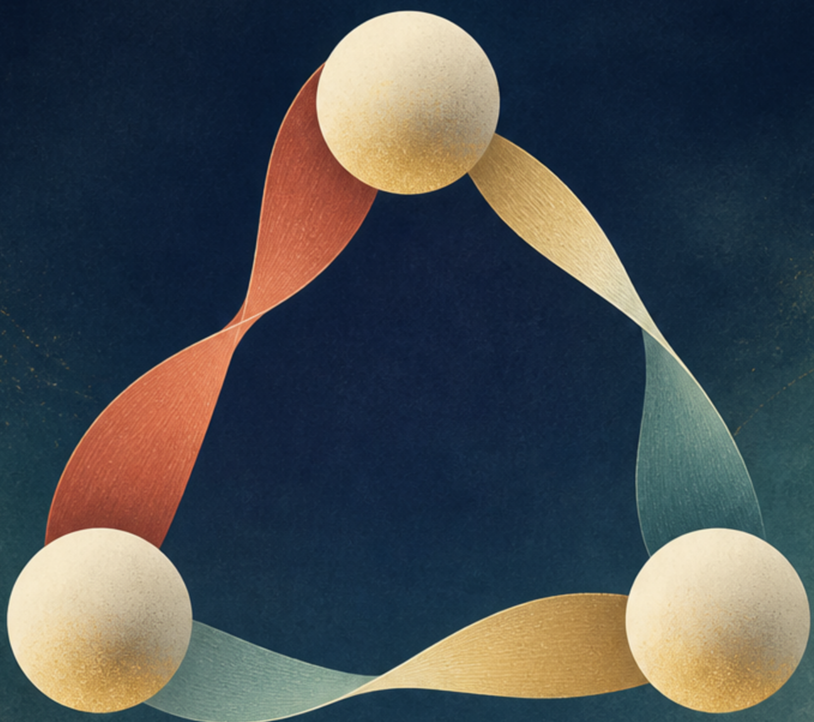
Quantum Entanglement, Part 3

Supplementary course, 2006 Fall



Lectures by Leonard Susskind

Companion edition by LazyingArt LLC with Video2Book

ABOUT THESE NOTES

These independently edited companion notes follow the third part of Leonard Susskind's 2006 lecture course on quantum entanglement. They were reconstructed from machine transcripts, subtitles, and selected blackboard frames, then checked against the available source material. They are not an original manuscript by Professor Susskind and have not been reviewed or endorsed by him.

The edition was prepared by LazyingArt LLC using the open-source Video2Book toolkit. Machine transcripts remain available separately and may contain recognition errors. Editorial clarifications and nontrivial reconstructions are identified in footnotes.

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CHAPTER 1

SPECIAL RELATIVITY: FRAMES, SIMULTANEITY, AND THE LORENTZ TRANSFORMATION

Relativity is the principle that the laws of physics do not depend on which inertial frame we use to describe them. Our first task is kinematics: the meaning of space, time, motion, synchronization, and the transformation from one inertial frame to another. Dynamics comes afterward, with relativistic energy, momentum, force, the analog of Newton's equation of motion, and the meaning of $E = mc^2$. Once that structure is secure, we can approach general relativity conceptually through curved spacetime, the motion of particles in a prescribed gravitational field, black holes, and the expansion of the universe, without beginning from the difficult problem of solving Einstein's field equations.

The distinction between conceptual and mathematical rigor is worth making. We will not use an approximation that is knowingly false. Rather, we will emphasize what a gravitational field does to clocks, particles, and geometry while postponing the machinery needed to construct arbitrary solutions of the field equations. Special relativity must be treated more explicitly because all of that later reasoning rests on its kinematics.

The right place to start is therefore not with a celebrated formula, but with a precise account of what an observer means by a place and a time.

1.1 A Frame Is a Physical Coordinate System

Imagine space filled with an arbitrarily fine lattice of meter sticks. Every lattice point is labeled by three spatial coordinates, and at every point there is a clock. Together the rods and clocks constitute a reference frame.

Definition 1.1. An *event* is something that happens at one place and one time. In one spatial dimension its coordinates are written (x, t); in three dimensions they are (x, y, z, t).

Newtonian physics adds a powerful assumption: all ideal clocks can be synchronized once and for all. Time is universal. A clock may be moved, and an observer may carry a watch, but everyone is supposed to agree on the same t everywhere. This is sometimes called absolute time; one may picture it as God's time, a single clock for the whole world.

The principle of relativity is older than Einstein. Its natural picture is a train moving so uniformly over perfectly smooth rails that nothing rattles or shakes. Inside the train we may toss and catch balls exactly as we do in a train standing at the station. Juggling is a surprisingly complete little mechanics experiment: it requires timing, gravity, alignment, and repeated control of trajectories. If the train moves at constant velocity, none of those rules changes. A ball thrown straight upward comes back to the same hand; the train does not slide out from under it. Galileo recognized this fact, and Newton expressed it systematically in his laws of motion.

Before Galileo, it was tempting to reason that the moving train ought to advance while the ball is in the air, forcing the juggler to reach backward or forward to recover it. The mistake is to forget that the ball already shares the train's horizontal velocity when it leaves the hand. No experiment confined to the smoothly moving carriage reveals a special state of absolute rest. Only relative motion between the two trains is observable.

The stationary train and the uniformly moving train are therefore two inertial frames. Before Einstein, they were related by Galilean transformations.

1.2 Drawing Spacetime

We suppress y and z for the moment and draw time vertically and position x horizontally. Every point in the drawing is an event. The line $x = 0$ is the history, or *worldline*, of the spatial origin. The neighboring vertical lines $x = 1, 2, 3, \ldots$ are the worldlines of the other lattice points. Horizontal lines are sets of events with a common time.

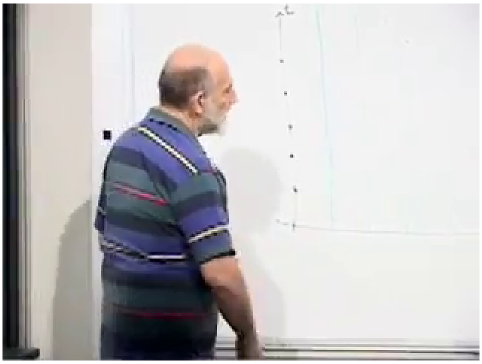

Figure 1.1: The rest-frame time axis and the first worldlines of the spatial lattice.

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The same construction is clearer in an idealized drawing:

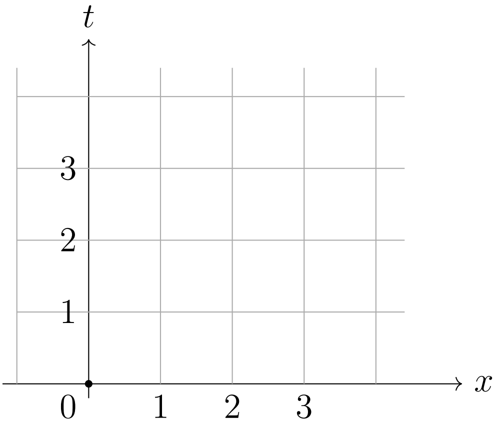

Figure 1.2: A stationary spacetime lattice. Vertical lines have constant position; horizontal lines have constant time.

The grid is imaginary, and its spacing is arbitrary. We can use meters, centimeters, microns, or any still smaller interval that makes physical sense. Its purpose is to assign a precise coordinate to every event.

The phrase "frame of reference" now has operational content. It is not merely a viewpoint or a person looking in some direction. It is an extended coordinate apparatus. Given an event, the local rods specify where it happened and the local clock specifies when. Other physical quantities can then be compared between frames: the position and velocity of a particle, its energy, perhaps the temperature of a body, and eventually its momentum and force. The simplest comparison is the one involving x and t, so we begin there.

Now place a second lattice on a train moving in the positive x-direction with constant velocity v. Choose the origins to coincide at $t = 0$. The moving origin then follows

$$x = vt. \tag{1.1}$$

Every other point fixed in the train follows a parallel tilted worldline. The tilt is simply the geometric statement that the train changes position as time passes.

Suppose the train's rear point is the primed origin. At $t = 0$ it coincides with the station origin. At a later time t, its stationary coordinate is vt. A nearby event at coordinate x is therefore only $x - vt$ from the train's origin. This is why the transformation is a subtraction rather than an arbitrary algebraic rule.

Let (x, t) be the coordinates of an event in the station frame and (x', t') its coordinates in the train frame. Galileo and Newton retain universal time,

$$t' = t. \tag{1.2}$$

During the time t, the moving origin advances by vt, so the distance from that origin to the event is

$$x' = x - vt. \tag{1.3}$$

Together,

$$t' = t, \qquad x' = x - vt. \tag{1.4}$$

Solving for the unprimed coordinates gives the inverse transformation,

$$t = t', \qquad x = x' + vt'. \tag{1.5}$$

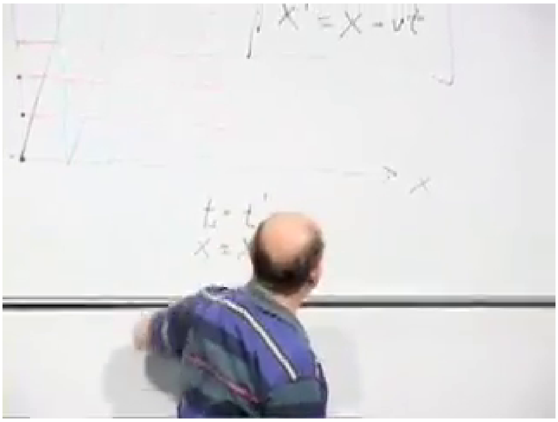

Figure 1.3: The Galilean transformation written beside the two spacetime lattices.

Lecture frame: 00:21:57.

The first pair tells the moving observer how to translate coordinates supplied by the stationary observer. The inverse pair translates back. This reciprocal conversion between descriptions is the mathematical content of Galilean relativity.

The replacement $t \to t'$ in the inverse spatial formula is harmless only because Galilean theory has already declared $t = t'$. It is useful to notice that assumption now. In Einstein's theory the two time coordinates will differ, and such substitutions will no longer be automatic.

Question from the audience. If a clock is carried away and then brought back, how can we know that its motion did not first speed it up and then slow it down, concealing the effect when the clocks are compared? [a]

Response. Galileo could not have established universal time with modern precision; it was a physical guess. We can test the guess in many ways: carry good clocks in opposite directions, repeat the trip north–south as well as east–west, move one clock around the other, or exchange timing signals without bringing the clocks together. At ordinary speeds all such tests agree to very high precision. With sufficiently accurate clocks or sufficiently large speeds, a small discrepancy does appear. That failure of universal time is not a defect in the experiment; it is one of the phenomena special relativity must explain.

[a] Lecture timestamp: 00:23:09.

1.3 Galilean Invariants and Newton's Law

An invariant is a quantity on which all the inertial observers under discussion agree. Coordinates themselves need not be invariant. What matters is whether the laws assembled from them retain the same form.

The word is used both as a noun and an adjective. A number can be an invariant, and an equation can be invariant under a transformation. The distinction between invariant quantities and invariant laws prevents a common confusion: relativity does not claim that every observer obtains the same coordinate for everything. It claims that the physical law connecting their measurements has the same form.

Take two simultaneous events at positions x_1 and x_2. Equation (1.4) gives

$$x_1' = x_1 - vt, \tag{1.6}$$
$$x_2' = x_2 - vt. \tag{1.7}$$

Their separation is therefore

$$x_2' - x_1' = (x_2 - vt) - (x_1 - vt) = x_2 - x_1. \tag{1.8}$$

In Galilean physics, the distance between events occurring at the same time is invariant even though the individual positions are not.

Question from the audience. Are the two occurrences of t in that subtraction really the same time? [a]

Response. Yes. We deliberately chose the two events to be (x_1, t) and (x_2, t). They lie on one horizontal constant-time line. The common term vt therefore cancels. If the events occurred at different times, this argument would not establish an invariant spatial separation.

[a] Lecture timestamp: 00:28:16.

Now describe a particle by a trajectory $x(t)$. The train observer assigns

$$x'(t) = x(t) - vt. \tag{1.9}$$

Differentiating once gives the familiar addition rule for velocities,

$$\frac{dx'}{dt} = \frac{dx}{dt} - v. \tag{1.10}$$

Position and velocity are frame-dependent. An observer moving alongside an arrow sees it travel more slowly than an observer standing beside the track.

Differentiate again. Since the relative velocity v is constant,

$$\frac{d^2x'}{dt^2} = \frac{d^2x}{dt^2}. \tag{1.11}$$

Acceleration is invariant under a Galilean transformation. This strongly suggests why Newton's law is written in terms of acceleration rather than position or velocity.

This is the acceleration of an observed object, not the acceleration of the train observer. Both frames are inertial, so neither frame accelerates relative to the other. Objects may accelerate left or right inside either description, and the two observers agree on the value of that acceleration even though they disagree on the object's position and velocity.

Suppose two particles exert a force on one another and the force depends only on their simultaneous separation. That separation is invariant, so the force is invariant. If mass is also an intrinsic, frame-independent property, then every inertial observer can write

$$F = ma \tag{1.12}$$

in exactly the same form. Newtonian mechanics is invariant under Galilean transformations. It is not merely plausible; throughout the domain of ordinary speeds it is extraordinarily accurate.

The argument applies to familiar central forces such as Newtonian gravity or electrostatics, where the magnitude is a function of distance. Since all Galilean observers agree on that distance at a common time, they agree on the force. With force and acceleration fixed, consistency then requires them to agree on mass as well: a one-kilogram object remains a one-kilogram object when carried past on a train. Every symbol in $F = ma$ fits the same invariant structure.

1.4 Light Does Not Fit the Galilean Rule

Electromagnetism creates the difficulty. Maxwell's equations predict waves that travel through empty space with a definite speed c. If Maxwell's equations are laws of nature and the laws of nature are the same in every inertial frame, then all inertial observers must obtain the same c.

Galilean kinematics predicts something else. A right-moving light pulse in the stationary frame satisfies

$$x = ct. \tag{1.13}$$

Applying the spatial transformation gives

$$x' = x - vt = (c - v)t. \tag{1.14}$$

Because $t' = t$, the train observer would measure the speed $c - v$.

The contradiction can be stated sharply. If Maxwell's equations retain their form, they predict c in the primed frame. If the Galilean transformation is retained, it predicts $c - v$. Both cannot be exact when $v \neq 0$.

This result looks perfectly natural if light behaves like an ordinary wave in a medium. A boat moving with a water wave measures a lower relative wave speed. A listener moving through still air measures a sound speed shifted by the listener's motion relative to the air. Nineteenth-century physicists were therefore led toward a preferred state of rest, an ether frame in which Maxwell's equations would have their simplest form and light would move at c. Experiments designed to detect motion relative to such a preferred frame found no corresponding effect.

There were several possible experimental arrangements, but their common aim was to compare light propagation for observers in different states or directions of motion. None revealed the expected Galilean shift. The failure was not a small technical nuisance. It forced a choice among the relativity principle, Maxwell's equations, and the inherited notions of space and time.

Einstein approached the conflict more directly. As a young man he imagined chasing a light wave. If he could catch up with it, its oscillating electromagnetic pattern would stand frozen in space, yet Maxwell's equations allow no such stationary light wave. He chose to treat Maxwell's theory and the relativity principle as fundamental, then reconsider the assumptions hidden in our coordinates.

The reconstruction rests on two postulates:

1. The laws of physics have the same form in every inertial frame.

2. Light in vacuum has the same speed c in every inertial frame.

The second postulate cannot coexist with the Galilean transformation. Something deeper than the rule for adding velocities must change. Einstein's economical choice is to keep the two postulates and reconsider the apparently innocent Galilean equation $t' = t$.

1.5 Synchronizing Distant Clocks

Spatial distances can be laid out with rods, but distant clocks require a synchronization convention. Put two clocks at equal distances from a midpoint observer. When each endpoint clock reads the appointed time, let it emit a flash. If the two flashes reach the midpoint together, call the endpoint clocks synchronized. If one flash arrives first, adjust the corresponding clock and repeat the procedure.

The word *midpoint* is defined using the rods already laid out in that frame. The test therefore contains two separate ingredients: equal spatial distance from the endpoints and equal arrival time of the signals. Neither ingredient may be dropped. Repeating the procedure throughout the lattice supplies the horizontal constant-time lines that Newton had simply assumed.

This definition uses light because the postulate assigns light the same speed in every inertial frame. Once all neighboring pairs have been treated in this way, the entire lattice carries a synchronized family of clocks.

Now let a second lattice move through the first. The midpoint of the moving apparatus remains halfway between its two moving endpoints in its own frame. Relative to the stationary apparatus, however, it moves toward one incoming flash and away from the other. Flashes emitted simultaneously according to the stationary clocks do not arrive together at the moving midpoint. Since the moving observer applies the same midpoint-and-light rule, that observer cannot call those emissions simultaneous. Each frame synchronizes its own clocks consistently, but the two synchronizations disagree.

This is not an optical illusion produced by the finite travel time of light. Each observer knows that light takes time to arrive and corrects for it using the same speed c. The disagreement remains after that correction because the observers do not select the same distant events as a simultaneous pair. Simultaneity itself, rather than merely the appearance of simultaneity, is relative.

Question from the audience. Could we avoid this conclusion by synchronizing clocks with sound instead of light? [a]

Response. Sound works as a synchronization signal in a frame where the air is at rest, but its measured speed is not the same in every inertial frame. An open train moving through still air is moving relative to the sound-carrying medium, so forward and backward sound signals do not behave symmetrically there. Light is different by postulate: there is no detectable optical medium whose rest frame selects one observer. We therefore use light and follow the consequences.

[a] Lecture timestamp: 00:49:03.

The qualitative disagreement is clear. We now calculate exactly which events the moving observer calls simultaneous.

1.6 The Light-Ray Construction

For the construction it is convenient to choose units in which

$$c = 1. \tag{1.15}$$

One unit of distance is then the distance light travels in one unit of time: light-seconds paired with seconds, or light-years paired with years. Space and time carry the same units, velocities are dimensionless, and light rays appear at 45°. A right-moving ray has

$$x = t + d, \tag{1.16}$$

while a left-moving ray has

$$x = -t + d'. \tag{1.17}$$

The two rays through the origin are simply $x = t$ and $x = -t$. Timelike trajectories, belonging to objects slower than light, lie closer to the vertical axis. A hypothetical superluminal trajectory would lie closer to the horizontal axis.

The angle in the drawing is meaningful only after this unit choice. In meters and seconds, a light ray would look almost horizontal because it covers roughly 3×10^8 meters in one second. Choosing light-seconds for distance rescales the horizontal axis and makes the causal geometry visible.

Consider three parallel worldlines in the moving lattice,

$$x = vt, \qquad x = vt + L, \qquad x = vt + 2L. \quad (1.18)$$

The middle worldline belongs to the observer who tests simultaneity between the two outer points. All three points move uniformly and preserve their relative arrangement in the primed frame.

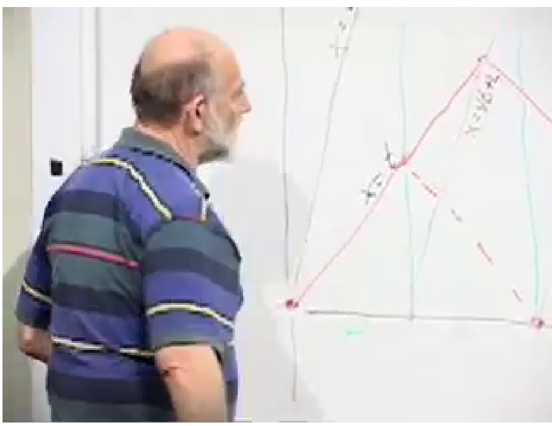

Figure 1.4: The forward light ray $x = t$ meeting the middle moving worldline.

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The complete geometry can be drawn without the obstruction of a person standing at the board:

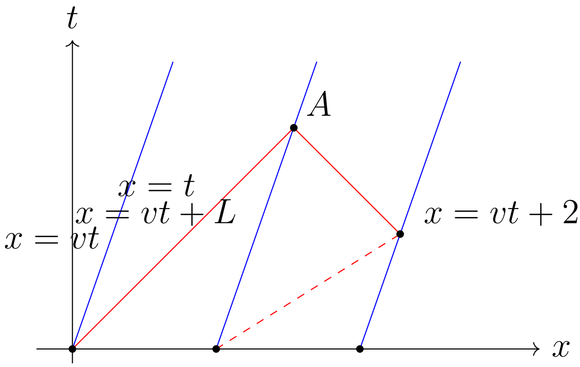

Figure 1.5: Light signals determine the moving frame's line of simultaneity.

Send a right-moving light ray from the origin. Let A be its intersection with the middle worldline. The two equations at A are

$$x = t, \tag{1.19}$$
$$x = vt + L. \tag{1.20}$$

Substitution gives

$$t = vt + L, \tag{1.21}$$

and hence

$$t_A = x_A = \frac{L}{1 - v}. \tag{1.22}$$

From A, trace a left-moving light ray backward to the rightmost moving worldline. Its equation has the form

$$x + t = b. \tag{1.23}$$

Because it passes through A, Equation (1.22) fixes the constant:

$$b = x_A + t_A = \frac{2L}{1-v}. \tag{1.24}$$

The backward ray therefore satisfies

$$x + t = \frac{2L}{1-v}. \tag{1.25}$$

Its intersection with the rightmost worldline follows from

$$x + t = \frac{2L}{1-v}, \tag{1.26}$$

$$x = vt + 2L. \tag{1.27}$$

Eliminating x,

$$(1+v)t = 2L\left(\frac{1}{1-v} - 1\right) \tag{1.28}$$

$$= \frac{2Lv}{1-v}, \tag{1.29}$$

so

$$t = \frac{2Lv}{1-v^2}. \tag{1.30}$$

Substitution into $x = vt + 2L$ gives

$$x = \frac{2L}{1-v^2}. \tag{1.31}$$

The coordinates themselves are less important than their ratio:

$$\frac{t}{x} = v. \tag{1.32}$$

Thus every event simultaneous with the origin according to the moving clocks lies on

$$t = vx, \tag{1.33}$$

whereas the moving origin follows

$$x = vt. \tag{1.34}$$

The two axes are tilted symmetrically around the 45° light ray.

All points on $x = vt$ have the same primed position, $x' = 0$, while their primed times vary. All points on $t = vx$ have the same primed time, $t' = 0$, while their primed positions vary. One line is therefore the primed time axis and the other the primed space axis. The long signal construction was needed to determine the second line; it could not be inferred from the moving worldline alone.

Restoring ordinary units in Equation (1.33) requires

$$t = \frac{v}{c^2}x. \tag{1.35}$$

The factor c^2 explains why the Newtonian picture works so well in daily life. For velocities tiny compared with c, the slope of the moving simultaneity line is extremely small.

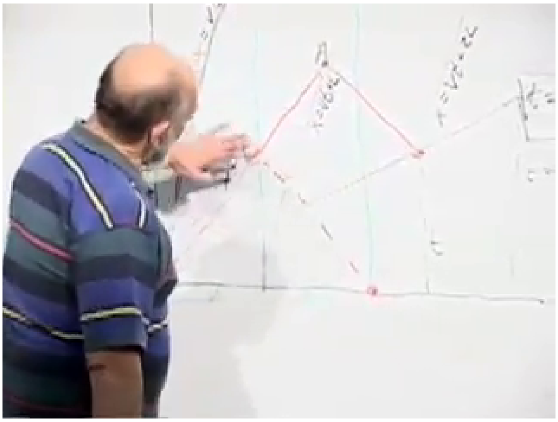

Figure 1.6: With $c = 1$, the primed time and simultaneity axes are symmetric about the light ray.

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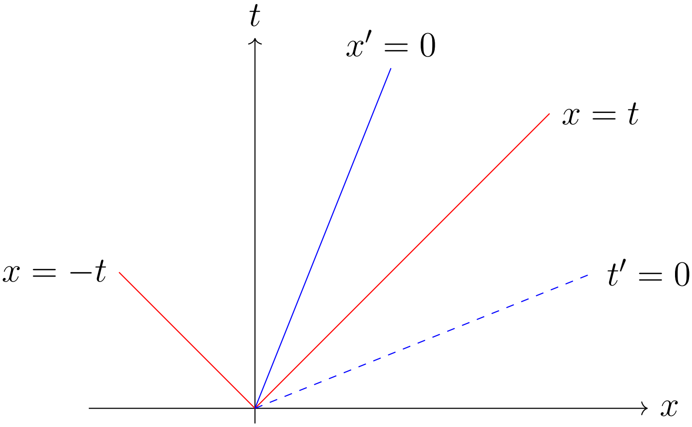

Figure 1.7: The primed axes and invariant light rays in $c = 1$ units.

1.7 Deriving the Lorentz Transformation

The geometry fixes where each primed coordinate vanishes. Along the moving time axis $x = vt$, we must have $x' = 0$. Along the moving simultaneity axis $t = vx$, we must have $t' = 0$. The most general linear relations with those zero-sets can be written

$$x' = \frac{x - vt}{F(v)}, \tag{1.36}$$

$$t' = \frac{t - vx}{G(v)}. \tag{1.37}$$

The factors F and G allow the primed coordinate scales to differ from the Galilean scales even though the axes have already been located. Linearity follows from the homogeneity of space and time: translating an experiment cannot change the transformation law.[1]

At this stage the geometry has told us a great deal but not everything. The numerators fix the tilt of the axes. They do not fix the unit spacing along those axes. That remaining scale information is exactly what $F(v)$ and $G(v)$ represent.

Now impose the invariance of light speed. A right-moving ray obeying $x = t$ must also obey $x' = t'$. Along that ray,

$$x' = \frac{(1 - v)t}{F(v)}, \tag{1.38}$$

[1] Editorial clarification: This short linearity argument makes explicit an assumption used implicitly in the blackboard derivation.

$$t' = \frac{(1-v)t}{G(v)}. \tag{1.39}$$

Equality for every point on the ray requires

$$F(v) = G(v). \tag{1.40}$$

We may therefore write

$$x' = \frac{x - vt}{F(v)}, \tag{1.41}$$

$$t' = \frac{t - vx}{F(v)}. \tag{1.42}$$

This step uses the first genuinely Einsteinian demand in the algebra. The same geometric line is a light ray for both observers, so the two primed coordinates must increase equally along it when $c = 1$. Different factors in the two equations would make the primed speed differ from unity.

One unknown function remains. Multiply Equation (1.42) by $F(v)$:

$$F(v)x' = x - vt, \tag{1.43}$$

$$F(v)t' = t - vx. \tag{1.44}$$

Multiplying the second equation by v and adding it to the first eliminates t, giving

$$(1 - v^2)x = F(v)(x' + vt'). \tag{1.45}$$

Similarly,

$$(1 - v^2)t = F(v)(t' + vx'). \tag{1.46}$$

The inverse transformation is therefore

$$x = \frac{F(v)}{1 - v^2}(x' + vt'), \tag{1.47}$$

$$t = \frac{F(v)}{1 - v^2}(t' + vx'). \tag{1.48}$$

The principle of relativity makes the two frames reciprocal. If the station sees the train move at $+v$, the train sees the station move at $-v$. The inverse equations must consequently have the same form as the forward equations with the sign of v reversed. Their prefactors must satisfy

$$\frac{F(v)}{1 - v^2} = \frac{1}{F(v)}. \tag{1.49}$$

Hence

$$F(v)^2 = 1 - v^2, \tag{1.50}$$

and continuity with the identity transformation at $v = 0$ selects the positive root,

$$F(v) = \sqrt{1 - v^2}. \tag{1.51}$$

Nothing has been fitted to a particular light pulse. The result follows from the axes fixed by synchronization, the common speed of light, and the demand that neither inertial frame be privileged over the other.

The Lorentz transformation in units with $c = 1$ is

$$x' = \frac{x - vt}{\sqrt{1 - v^2}}, \tag{1.52}$$

$$t' = \frac{t - vx}{\sqrt{1 - v^2}}. \tag{1.53}$$

Its inverse is

$$x = \frac{x' + vt'}{\sqrt{1 - v^2}}, \tag{1.54}$$

$$t = \frac{t' + vx'}{\sqrt{1 - v^2}}. \tag{1.55}$$

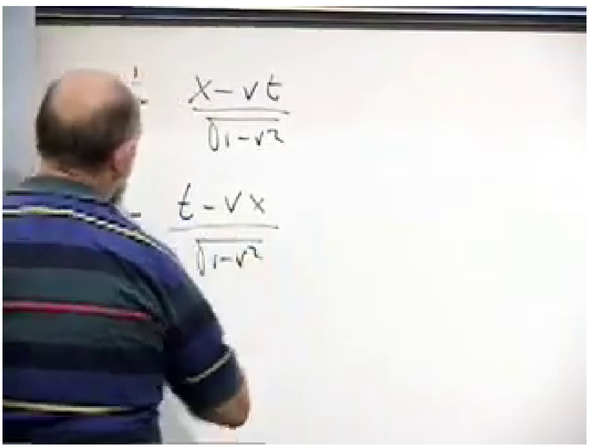

Figure 1.8: The completed $c = 1$ Lorentz transformation on the blackboard.

Lecture frame: 01:32:38.

The essential novelty is already visible in the time equation. Time is not simply carried unchanged from one frame to another. A primed time coordinate mixes unprimed time and position. Consequently t = constant and t' = constant describe different sets of events.

The mixing term is not an optional correction appended after the fact. Without it, the line $t' = 0$ would remain horizontal, distant simultaneity would stay universal, and the transformation could not preserve the speed of light.

1.8 Restoring the Speed of Light and Checking the Result

When $c \neq 1$, dimensional consistency restores the missing factors. The ratio v^2/c^2 is dimensionless, while $(v/c^2)x$ has units of time. Thus

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}, \tag{1.56}$$

$$t' = \frac{t - (v/c^2)x}{\sqrt{1 - v^2/c^2}}. \tag{1.57}$$

The inverse is obtained by reversing the sign of v:

$$x = \frac{x' + vt'}{\sqrt{1 - v^2/c^2}}, \tag{1.58}$$

$$t = \frac{t' + (v/c^2)x'}{\sqrt{1 - v^2/c^2}}. \tag{1.59}$$

Introducing

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \tag{1.60}$$

would put these in the familiar compact form, but the denominator form keeps the derivation visible.

Numerically, $c \simeq 3 \times 10^8$ m/s, so $c^2 \simeq 9 \times 10^{16}$ m^2/s^2. The large denominator makes both the simultaneity tilt and the factor v^2/c^2 imperceptible in most mechanical experiments. Their smallness explains Galileo's success; it does not restore absolute time.

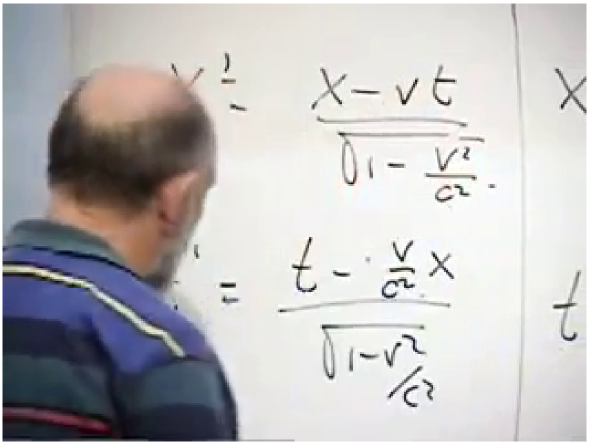

Figure 1.9: The Lorentz transformation after the factors of c have been restored.

Lecture frame: 01:35:58.

Two checks are immediate. First take $v \ll c$. Then v^2/c^2 is negligible, and for ordinary distances the relativity-of-simultaneity term vx/c^2 is also negligible. Equation (1.57) reduces to

$$t' \approx t, \qquad x' \approx x - vt. \tag{1.61}$$

Galilean relativity is recovered as the low-speed approximation.

Second, test a light ray. Once c has been restored, a right-moving ray obeys $x = ct$, not $x = t$. Substitution gives

$$x' = \frac{(c-v)t}{\sqrt{1 - v^2/c^2}}, \tag{1.62}$$

$$t' = \frac{(1-v/c)t}{\sqrt{1 - v^2/c^2}}. \tag{1.63}$$

Since $c - v = c(1 - v/c)$,

$$x' = ct'. \tag{1.64}$$

The primed observer measures the same speed c. For a left-moving ray, substitution of $x = -ct$ similarly gives

$$x' = -ct'. \tag{1.65}$$

The transformation treats light moving in either direction correctly.

The distinction between $x = t$ and $x = ct$ is an instructive dimensional check. The former is correct only in units where $c = 1$. Restoring ordinary units in the transformation while leaving the ray equation in natural units would mix two conventions and produce a false result. Dimensional consistency catches the mistake immediately.

1.9 The Midpoint Criterion

Question from the audience. Suppose an observer receives two flashes at the same instant but is not halfway between their sources. May that observer conclude that the flashes were emitted simultaneously? [a]

Response. No. Equal arrival times imply equal emission times only for the observer who is spatially midway between the two sources in the relevant frame. A receiver closer to one source can receive both flashes together even though the farther flash was emitted earlier. In the moving construction, the midpoint worldline $x = vt + L$ is therefore essential. The forward ray $x = t$ is followed until it meets that midpoint, and a backward light ray is then traced to $x = vt + 2L$. That second endpoint event, not an arbitrary equal-arrival event elsewhere, is simultaneous with the origin according to the moving frame.

[a] Lecture timestamp: 01:42:55.

We have reached the complete coordinate transformation but not yet its most familiar consequences. Length contraction, time dilation, and proper time follow next, followed by relativistic momentum and energy. The foundation is already in place: each inertial frame may use the same laws and measure the same speed of light only because space and time mix, and because distant simultaneity belongs to a frame rather than to the world absolutely.

CHAPTER 2

MAXWELL'S EQUATIONS, COVARIANCE, AND ELECTROMAGNETIC WAVES

Overview. Electric and magnetic fields act on charged matter through the Lorentz force, but matter must also create and rearrange the fields. We shall formulate that reciprocal half of electromagnetism, show that Maxwell's equations are Lorentz covariant, derive their wave solutions, and then restore charge and current in a form that makes local conservation manifest. Throughout the lecture we use units in which $c = 1$.

2.1 The Other Half of Electromagnetism

Begin with the part of the subject already in hand. If a particle of charge q moves with velocity $\mathbf{v}$ through electric and magnetic fields, the force is

$$\mathbf{F} = q\Big(\mathbf{E} + \mathbf{v} \times \mathbf{B}\Big). \tag{2.1}$$

The factor q is indispensable. An uncharged particle does not feel the electromagnetic force. The electric field contributes even when the particle is instantaneously at rest, whereas the magnetic contribution depends on its motion.

Equation (2.1) tells us what prescribed fields do to matter. It does not tell us what determines the fields. What produces an electric field around one object, a magnetic field around another, or a wave moving through empty space? The equations that answer those questions are Maxwell's equations.

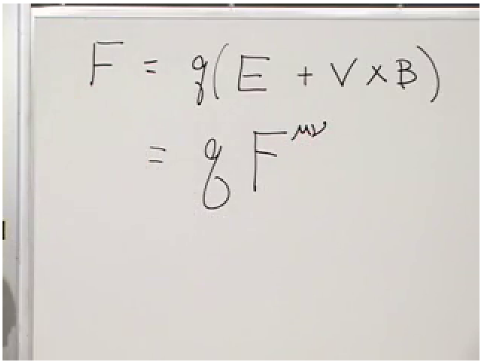

Figure 2.1: The three-vector Lorentz force and the beginning of its tensor reformulation.

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The relativistic form of the force law anticipates the notation we shall need:

$$f^\mu = qF^\mu{}_\nu u^\nu. \tag{2.2}$$

Here u^ν is the four-velocity, f^μ is the four-force, and $F^{\mu\nu}$ is an antisymmetric tensor assembled from $\mathbf{E}$ and $\mathbf{B}$. The board first displays only the schematic beginning of this expression; the contraction with u^ν supplies the missing vector needed to leave one free force index.

There is a physical reason that matter must act back on the field. An electromagnetic wave can strike a charge, accelerate it, and transfer energy to it. If the field could affect the charge while the charge could never affect the field, there would be no place from which that energy was withdrawn. Action and reaction are not quite the whole argument, but conservation of energy makes the reciprocal coupling unavoidable.

2.2 A Changing Source Sends Out a Disturbance

A point charge produces an electric field directed outward or inward from its position. A current in a wire–a flow of charged particles–produces a magnetic field that circulates around the wire. These familiar static pictures already show two different geometric patterns: electric flux emanates from charge, while magnetic field lines wind around current.

Now move the point charge abruptly. Near its new position the electric field must begin to rearrange, but a distant observer cannot learn of the motion instantly. The old field remains far away until a signal has had time to arrive. Between the near and far regions lies a moving front at which the field is changing. That moving rearrangement is an electromagnetic disturbance.

If the charge is driven back and forth, successive rearrangements follow one another and form an electromagnetic wave. This is the elementary function of an antenna: an oscillating electric current sends a field disturbance outward. Reversing a current in a wire gives the magnetic version of the same argument. The circulation of **B** must reverse, but the reversal propagates rather than appearing everywhere at once.

This causal picture sets two targets for Maxwell's equations. They must describe how charge and current source the fields, and they must permit changes in those fields to propagate no faster than light. The second target is also a test of relativity. If Maxwell's equations have the same form in every inertial frame and imply a definite wave speed, then every inertial frame must infer the same speed from them.

2.3 Reviewing the Relativistic Wave Equation

Before turning to vector fields, consider a scalar field $\phi(\mathbf{x}, t)$, one number at every spacetime point. A first-derivative equation

$$\partial_\mu \phi = 0 \tag{2.3}$$

would say that every spacetime derivative vanishes. The field would be constant and therefore unable to carry a wave. A nontrivial relativistic wave equation uses two derivatives:

$$\partial_\mu \partial^\mu \phi = 0. \tag{2.4}$$

With the sign convention used here,

$$\left(\partial_t^2 - \partial_x^2 - \partial_y^2 - \partial_z^2\right) \phi = 0, \tag{2.5}$$

or, more compactly,

$$\left(\partial_t^2 - \nabla^2\right) \phi = 0. \tag{2.6}$$

The choice $c = 1$ lets time and distance carry compatible units. Restoring ordinary units would place $1/c^2$ beside the second time derivative.

For a wave traveling along the z-axis, the field depends on z and t, not on x or y. Equation (2.5) reduces to

$$\left(\partial_t^2 - \partial_z^2\right)\phi = 0. \tag{2.7}$$

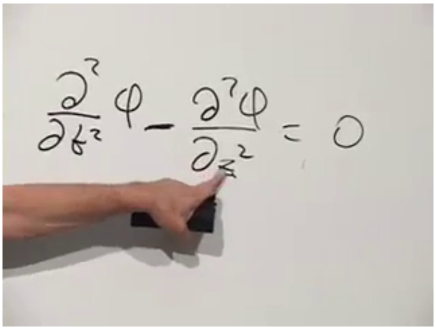

Figure 2.2: The one-dimensional scalar wave equation used as the model for electromagnetic waves.

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Its general traveling-wave form is

$$\phi(z,t) = f(z-t) + g(z+t). \tag{2.8}$$

The profile $f(z-t)$ moves toward increasing z, because a point of fixed phase satisfies $z - t =$ constant. The profile $g(z+t)$ moves in the opposite direction. Nothing requires either profile to be sinusoidal; any sufficiently smooth shape translates without changing form.

Sinusoids are nevertheless convenient:

$$\phi(z,t) = A\cos\big(k(z-t)\big). \tag{2.9}$$

The wave number k controls the wavelength,

$$\lambda = \frac{2\pi}{k}. \tag{2.10}$$

A large k produces rapid spatial oscillation and a short wavelength; a small k produces a long wavelength. These same profiles will soon become components of $\mathbf{E}$ and $\mathbf{B}$.

2.4 The Vector Calculus Maxwell Needs

Maxwell's equations are written with dot products, cross products, divergence, and curl. It is worth rebuilding the notation once, because the later component calculations use precisely these patterns.

For ordinary three-vectors $\mathbf{a}$ and $\mathbf{b}$, the dot product is the scalar

$$\mathbf{a}\cdot\mathbf{b} = a_x b_x + a_y b_y + a_z b_z. \tag{2.11}$$

The cross product is another vector,

$$(\mathbf{a}\times\mathbf{b})_x = a_y b_z - a_z b_y, \tag{2.12}$$
$$(\mathbf{a}\times\mathbf{b})_y = a_z b_x - a_x b_z, \tag{2.13}$$
$$(\mathbf{a}\times\mathbf{b})_z = a_x b_y - a_y b_x. \tag{2.14}$$

The remaining components follow from the cyclic order $x \to y \to z \to x$.

Introduce the differential operator

$$\boldsymbol{\nabla} = (\partial_x, \partial_y, \partial_z)\,. \tag{2.15}$$

Its entries are operators rather than ordinary numbers, but the dot- and cross-product patterns can still be used. For a vector field $\mathbf{A}$,

$$\boldsymbol{\nabla} \cdot \mathbf{A} = \partial_x A_x + \partial_y A_y + \partial_z A_z \tag{2.16}$$

is the divergence, while

$$(\boldsymbol{\nabla} \times \mathbf{A})_x = \partial_y A_z - \partial_z A_y, \tag{2.17}$$
$$(\boldsymbol{\nabla} \times \mathbf{A})_y = \partial_z A_x - \partial_x A_z, \tag{2.18}$$
$$(\boldsymbol{\nabla} \times \mathbf{A})_z = \partial_x A_y - \partial_y A_x \tag{2.19}$$

are the components of the curl.

The geometry is as important as the formulas. A vector field has positive divergence where it appears to flow outward from a source. It has curl where it circulates. The radial electric field of a point charge illustrates divergence; the magnetic field winding around a current-carrying wire illustrates curl. A spatially constant vector field has neither.

2.5 Maxwell's Equations in Vacuum

In the absence of charge and current, Maxwell's equations are

$$\boldsymbol{\nabla} \times \mathbf{E} = -\partial_t \mathbf{B}, \tag{2.20}$$
$$\boldsymbol{\nabla} \cdot \mathbf{B} = 0, \tag{2.21}$$
$$\boldsymbol{\nabla} \times \mathbf{B} = \partial_t \mathbf{E}, \tag{2.22}$$
$$\boldsymbol{\nabla} \cdot \mathbf{E} = 0. \tag{2.23}$$

The two curl equations nearly exchange **E** and **B**, but an essential minus sign distinguishes them. The two divergence equations say that neither field has a source in the vacuum region under consideration. Magnetic divergence remains zero even after electric sources are restored; ordinary electromagnetism contains no magnetic charge.

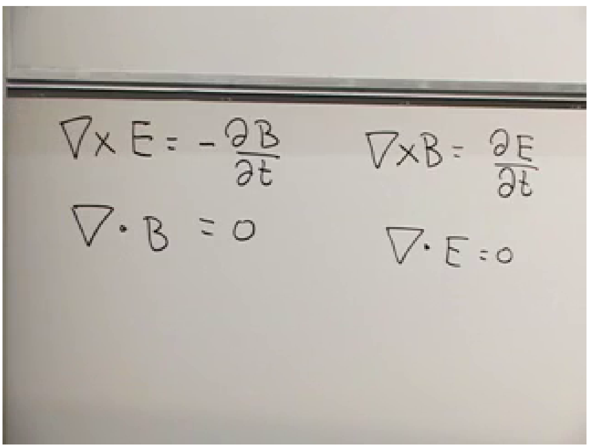

Figure 2.3: The four vacuum Maxwell equations, arranged to display the near symmetry between electric and magnetic fields.

Lecture frame: 00:27:30.

Charge density ρ will eventually replace the zero on the right of Equation (2.23), and current density **J** will enter Equation (2.22). We postpone those terms temporarily. The source-free equations expose the transformation structure and the wave mechanism with the least clutter.

They also sharpen Einstein's puzzle. A Galilean observer following a wave would expect to subtract velocities and measure a smaller wave speed. Yet if the same Maxwell equations hold in every inertial frame, each observer derives the same unit wave speed. The way out is not to alter the wave equation from frame to frame. It is to use the Lorentz transformation and to exhibit Maxwell's equations in objects whose transformation laws are already known.

2.6 Electric and Magnetic Fields as One Tensor

The electric and magnetic fields combine into the antisymmetric field tensor

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{pmatrix}, \tag{2.24}$$

where the row and column order is t, x, y, z. Antisymmetry,

$$F^{\mu\nu} = -F^{\nu\mu}, \tag{2.25}$$

forces the diagonal entries to vanish and determines the lower half of the matrix from the upper half. The time-space entries contain $\mathbf{E}$; the purely spatial entries contain $\mathbf{B}$.

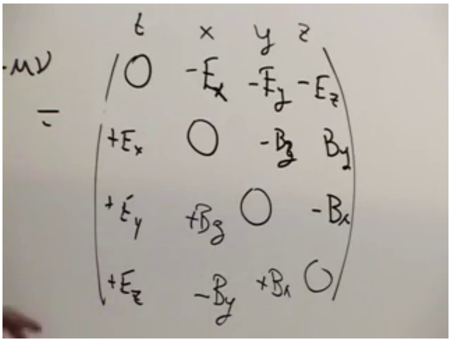

Figure 2.4: The completed electromagnetic field tensor, including its electric and magnetic entries.
Lecture frame: 00:32:05.

The importance of $F^{\mu\nu}$ is not merely that six field components fit into one matrix. Under a Lorentz transformation it transforms as a rank-two tensor. Electric and magnetic fields measured by different observers therefore mix in exactly the way required for the entire object to transform coherently.

There is a second antisymmetric tensor, the dual field tensor $\widetilde{F}^{\mu\nu}$. With the present convention it is obtained from Equation (2.24) by the replacement

$$\mathbf{E} \longmapsto -\mathbf{B}, \qquad \mathbf{B} \longmapsto \mathbf{E}. \tag{2.26}$$

Thus

$$\widetilde{F}^{\mu\nu} = \begin{pmatrix} 0 & B_x & B_y & B_z \\ -B_x & 0 & -E_z & E_y \\ -B_y & E_z & 0 & -E_x \\ -B_z & -E_y & E_x & 0 \end{pmatrix}. \tag{2.27}$$

The sign in Equation (2.26) matters. The lecture first tries the interchange verbally, catches the sign mismatch, and corrects it before completing the matrix.

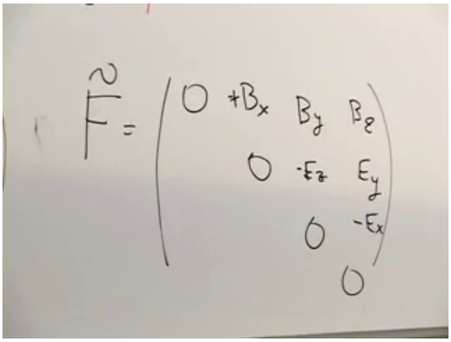

Figure 2.5: The dual tensor being assembled by exchanging electric and magnetic entries with the required sign.

Lecture frame: 00:35:45.

2.7 The Covariant Vacuum Equations

Each Maxwell equation contains one derivative. Starting from a rank-two tensor and contracting one derivative index leaves one free index, so the natural covariant candidates are

$$\partial_\mu F^{\mu\nu} = 0, \tag{2.28}$$

$$\partial_\mu \widetilde{F}^{\mu\nu} = 0. \tag{2.29}$$

For each fixed value of ν, Equation (2.28) is one equation. Since ν can be t, x, y, z, it packages four scalar equations. Equation (2.29) packages the other four. This matches the two vector curl equations and two scalar divergence equations in the original list.

Compact notation is useful only if it reproduces the familiar equations. Take $\nu = z$:

$$
\begin{aligned}
\partial_\mu F^{\mu z} &= \partial_t F^{tz} + \partial_x F^{xz} + \partial_y F^{yz} + \partial_z F^{zz} && (2.30)\\
&= -\partial_t E_z + \partial_x B_y - \partial_y B_x && (2.31)\\
&= -\partial_t E_z + \Big(\boldsymbol{\nabla} \times \mathbf{B}\Big)_z . && (2.32)
\end{aligned}
$$

The term F^{zz} vanishes by antisymmetry. Setting the result to zero gives the z-component of

$$\boldsymbol{\nabla} \times \mathbf{B} = \partial_t \mathbf{E}. \tag{2.33}$$

The x- and y-components follow by cycling the indices.

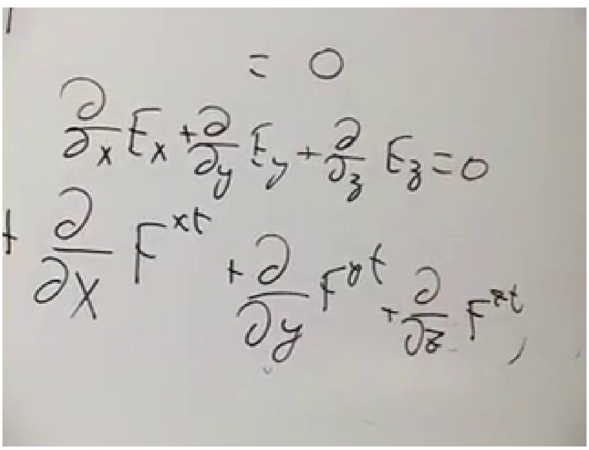

Figure 2.6: Component expansion of the covariant field equation; the diagonal tensor entry vanishes and the remaining terms reconstruct Maxwell's equations.

Lecture frame: 00:43:30.

Now take the time component:

$$\partial_\mu F^{\mu t} = \partial_t F^{tt} + \partial_x F^{xt} + \partial_y F^{yt} + \partial_z F^{zt} \tag{2.34}$$
$$= \partial_x E_x + \partial_y E_y + \partial_z E_z \tag{2.35}$$
$$= \boldsymbol{\nabla} \cdot \mathbf{E}. \tag{2.36}$$

It therefore gives Equation (2.23). Performing the same checks on the dual tensor yields $\boldsymbol{\nabla} \cdot \mathbf{B} = 0$ and $\boldsymbol{\nabla} \times \mathbf{E} = -\partial_t \mathbf{B}$.

The compact equations are now established rather than merely guessed. Their free-index structure makes them four-vector equations, so a Lorentz transformation changes their components but not their form. This is the precise sense in which the vacuum Maxwell equations are the same in every inertial frame.

2.8 From First-Order Equations to a Wave

Maxwell's equations contain only first derivatives, whereas a wave equation contains second derivatives. The apparent mismatch is resolved because Maxwell's theory couples two fields. A pair of first-order equations for $\mathbf{E}$ and $\mathbf{B}$ can be combined into a second-order equation for either field alone.

Differentiate Equation (2.22) with respect to time:

$$\partial_t^2 \mathbf{E} = \boldsymbol{\nabla} \times \partial_t \mathbf{B}. \tag{2.37}$$

Faraday's law supplies the time derivative of $\mathbf{B}$,

$$\partial_t \mathbf{B} = -\boldsymbol{\nabla} \times \mathbf{E}, \tag{2.38}$$

and therefore

$$\partial_t^2 \mathbf{E} = -\boldsymbol{\nabla} \times \left(\boldsymbol{\nabla} \times \mathbf{E}\right). \tag{2.39}$$

The board works through one component rather than invoking a memorized identity. For the x-component,

$$\left[\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \mathbf{E})\right]_x = \partial_y(\boldsymbol{\nabla} \times \mathbf{E})_z - \partial_z(\boldsymbol{\nabla} \times \mathbf{E})_y \tag{2.40}$$

$$= \partial_y(\partial_x E_y - \partial_y E_x) \tag{2.41}$$

$$- \partial_z(\partial_z E_x - \partial_x E_z) \tag{2.42}$$

$$= \partial_x(\partial_y E_y + \partial_z E_z) - \partial_y^2 E_x \tag{2.43}$$

$$- \partial_z^2 E_x. \tag{2.44}$$

Add and subtract $\partial_x^2 E_x$ to expose the divergence:

$$\left[\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \mathbf{E})\right]_x = \partial_x(\boldsymbol{\nabla} \cdot \mathbf{E}) - \nabla^2 E_x. \tag{2.45}$$

In vacuum $\boldsymbol{\nabla} \cdot \mathbf{E} = 0$. Combining Equations (2.39) and (2.45) gives

$$\left(\partial_t^2 - \nabla^2\right) E_x = 0. \tag{2.46}$$

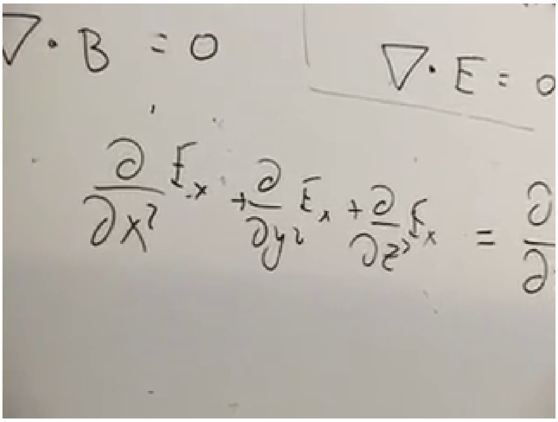

Figure 2.7: The component calculation reaches the wave equation for E_x.
Lecture frame: 00:57:10.

The same calculation applies to E_y and E_z, and an analogous elimination gives

$$\left(\partial_t^2 - \nabla^2\right)\mathbf{B} = 0. \tag{2.47}$$

Electric and magnetic fields can therefore propagate through empty space as waves. In the units used here their speed is one, which means the speed is c when ordinary units are restored.

2.9 A Plane Wave and Its Polarization

Construct a simple wave traveling in the positive z-direction:

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\,E_0\cos\Big(k(z-t)\Big). \tag{2.48}$$

Its profile translates along z without changing shape. Because the electric field points along the x-axis, the wave is linearly polarized in the x-direction. We have chosen $E_y = 0$ only to keep the example simple; a second transverse polarization could be included.

The longitudinal component is not equally free. Since the wave depends only on z and t,

$$\boldsymbol{\nabla} \cdot \mathbf{E} = \partial_z E_z = 0. \tag{2.49}$$

Thus E_z may be a constant, but it cannot contain the varying wave profile. A uniform background field can be superposed or removed independently; it is not part of the propagating radiation. The wave itself is transverse:

$$\mathbf{E}_{\text{wave}} \cdot \hat{\mathbf{z}} = 0. \tag{2.50}$$

Faraday's law fixes the magnetic field. For Equation (2.48), the only nonzero component of $\boldsymbol{\nabla} \times \mathbf{E}$ is the y-component,

$$(\boldsymbol{\nabla} \times \mathbf{E})_y = \partial_z E_x. \tag{2.51}$$

Consequently $\partial_t \mathbf{B}$ points along y, and a compatible choice is

$$\mathbf{B}(z,t) = \hat{\mathbf{y}}\, E_0 \cos\big(k(z-t)\big). \tag{2.52}$$

In these units the electric and magnetic amplitudes are equal. Their orientation obeys

$$\mathbf{E} \perp \mathbf{B}, \qquad \mathbf{E} \perp \hat{\mathbf{z}}, \qquad \mathbf{B} \perp \hat{\mathbf{z}}, \qquad \mathbf{E} \times \mathbf{B} \parallel \hat{\mathbf{z}}. \tag{2.53}$$

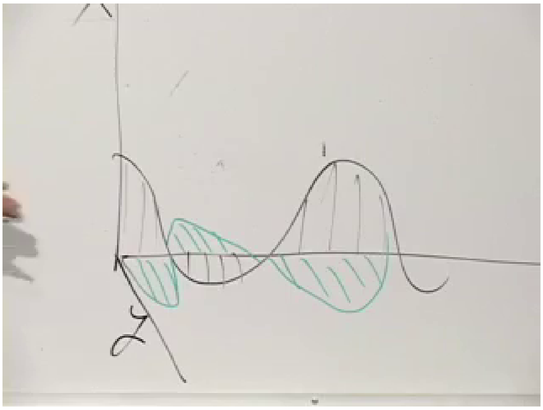

Figure 2.8: The completed board sketch of a transverse electromagnetic wave: the electric and magnetic oscillations are perpendicular to one another and to the direction of propagation.

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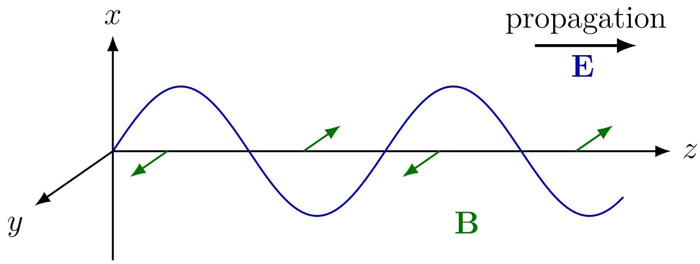

Figure 2.9: The fields **E** and **B** are perpendicular to each other and to the direction of propagation.

This is the character of light and of all electromagnetic radiation in vacuum. Since the equations that produced the wave are Lorentz covariant, another inertial observer again describes an electromagnetic wave moving at c, with transverse electric and magnetic fields.

2.10 Charge Density and Current Density

We now restore the sources that were temporarily set aside. When many charged particles occupy a region, it is useful to coarse-grain them into a continuous charge density

$$\rho(\mathbf{x}, t) = \frac{dQ}{dV}. \tag{2.54}$$

The approximation is like treating water as a continuous fluid even though it consists of molecules. Positive and negative charges contribute with opposite signs, so ρ is the net charge per unit volume.

The charge in a chosen region V is

$$Q_V(t) = \int_V \rho(\mathbf{x}, t)\, dV. \tag{2.55}$$

The density may vary with position and time. A pile of charge can move away from a point, changing the local density there. It cannot, however, vary in an arbitrary fashion. Nature conserves electric charge.

Global conservation alone would be too weak. It would allow charge to disappear from Earth and reappear on Mars at the same instant while the total remained unchanged. The stronger statement is local: charge can leave one place and reach another only through a flow in the intervening space.

To describe that flow, take an infinitesimal oriented surface element

$$d\boldsymbol{\sigma} = \hat{\mathbf{n}}\, d\sigma. \tag{2.56}$$

Its magnitude is the area, and its direction is normal to the surface. The current density **J** is defined so that the charge crossing the element per unit time is

$$\frac{dQ}{dt} = \mathbf{J} \cdot d\boldsymbol{\sigma}. \tag{2.57}$$

The magnitude of **J** is charge per unit area per unit time, and its direction is the direction of positive-charge flow.

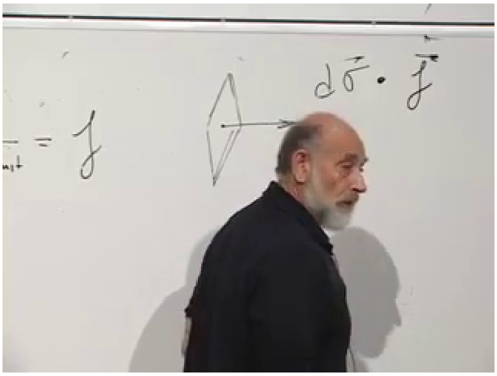

Figure 2.10: An oriented area element and the flux factor $d\boldsymbol{\sigma} \cdot \mathbf{J}$.

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The dot product selects the component normal to the surface. A current tangent to the surface crosses neither side and contributes zero. A current parallel to the normal gives the maximum flux for a fixed magnitude. A wire current is one special case, but **J** may also describe charge flowing through a sheet or throughout three-dimensional space.

2.11 Local Conservation and the Continuity Equation

Fix a region V in space and orient every element of its closed boundary ∂V outward. If there is a net outward current, the charge inside decreases:

$$\frac{dQ_V}{dt} = -\oint_{\partial V} \mathbf{J} \cdot d\boldsymbol{\sigma}. \tag{2.58}$$

The minus sign is physical rather than conventional: positive outward flux removes positive charge from the region. Using Equation (2.55),

$$\int_V \partial_t \rho \, dV = -\oint_{\partial V} \mathbf{J} \cdot d\boldsymbol{\sigma}. \tag{2.59}$$

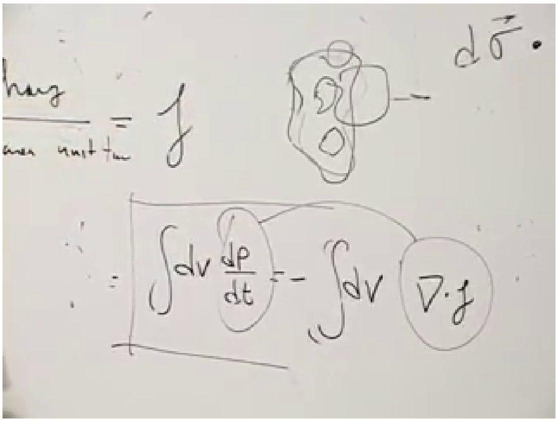


Figure 2.11: The fixed-region conservation law after the surface flux is converted into a volume statement. *Lecture frame: 01:29:00.*

Gauss's theorem, also called the divergence theorem, relates flux through a closed surface to divergence in its interior:

$$\oint_{\partial V} \mathbf{A} \cdot d\boldsymbol{\sigma} = \int_V \boldsymbol{\nabla} \cdot \mathbf{A} \, dV. \tag{2.60}$$

One can picture hidden pipes injecting water at points inside a vessel. The divergence measures the local rate at which water emerges from the pipe ends; the surface integral measures the total water leaving the vessel. Whatever is created as outward flow inside must eventually cross the boundary.

Applying Equation (2.60) to $\mathbf{J}$ gives

$$\int_V (\partial_t \rho + \boldsymbol{\nabla} \cdot \mathbf{J}) \, dV = 0. \tag{2.61}$$

This relation holds for every possible region V. The integrand must therefore vanish point by point:

$$\boxed{\partial_t \rho + \boldsymbol{\nabla} \cdot \mathbf{J} = 0.} \tag{2.62}$$

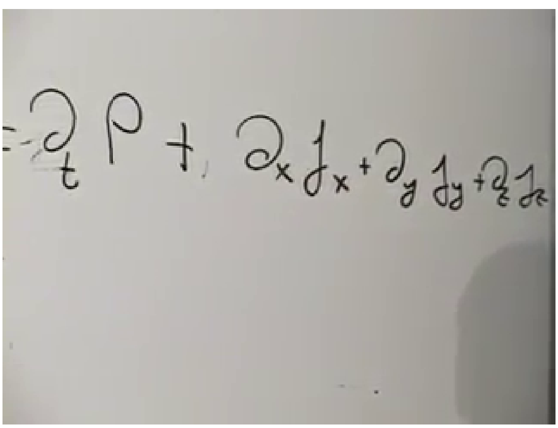


Figure 2.12: The local continuity equation written component by component.

Lecture frame: 01:29:50.

Equation (2.62) says exactly what local conservation requires. If charge density decreases at a point or within a small region, current must diverge outward from it. If current converges inward, charge density increases.

The density and current naturally form the four-current

$$J^\mu = (\rho, J_x, J_y, J_z) \tag{2.63}$$

in $c = 1$ units. The continuity equation becomes the Lorentz scalar equation

$$\partial_\mu J^\mu = 0. \tag{2.64}$$

The flow of a relativistic distribution should transform as a four-vector, just as the four-velocity of one particle does. Equation (2.64) makes local charge conservation valid in every inertial frame.

2.12 Maxwell's Equations with Sources

Restoring charge and current gives

$$\boldsymbol{\nabla} \cdot \mathbf{E} = \rho, \tag{2.65}$$
$$\boldsymbol{\nabla} \cdot \mathbf{B} = 0, \tag{2.66}$$
$$\boldsymbol{\nabla} \times \mathbf{E} = -\partial_t \mathbf{B}, \tag{2.67}$$
$$\boldsymbol{\nabla} \times \mathbf{B} = \partial_t \mathbf{E} + \mathbf{J}. \tag{2.68}$$

The equations involving electric sources collapse into

$$\boxed{\partial_\mu F^{\mu\nu} = J^\nu,} \tag{2.69}$$

while the source-free magnetic pair remains

$$\boxed{\partial_\mu \widetilde{F}^{\mu\nu} = 0.} \tag{2.70}$$

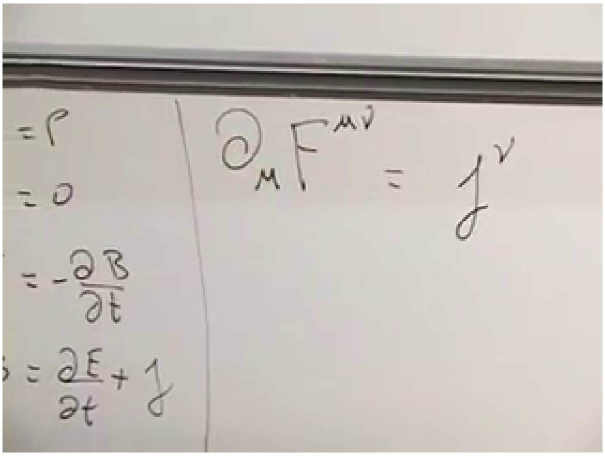


Figure 2.13: The sourced Maxwell equations beside their covariant form $\partial_\mu F^{\mu\nu} = J^\nu$.

Lecture frame: 01:34:00.

The time component of Equation (2.69) reads

$$\partial_\mu F^{\mu t} = J^t = \rho, \tag{2.71}$$

and the component calculation already performed turns its left side into $\boldsymbol{\nabla} \cdot \mathbf{E}$. The three spatial components reproduce $\boldsymbol{\nabla} \times \mathbf{B} = \partial_t \mathbf{E} + \mathbf{J}$.

There is also a useful consistency check. Applying ∂_ν to Equation (2.69) gives

$$\partial_\nu J^\nu = \partial_\nu \partial_\mu F^{\mu\nu} = 0, \tag{2.72}$$

because the two derivatives are symmetric under $\mu \leftrightarrow \nu$, whereas $F^{\mu\nu}$ is antisymmetric.[1]

[1] Editorial clarification: This short consistency check follows from the covariant equation and makes explicit why Maxwell's sourced equations automatically respect the continuity equation.

The reciprocal structure is now complete. The tensor force law $f^\mu = qF^\mu{}_\nu u^\nu$ describes how fields act on charge, and Equation (2.69) describes how charge and current generate fields. Both are tensor equations. In vacuum, the same field equations yield transverse waves moving at c. Lorentz covariance and the electromagnetic wave equation therefore meet at the point that motivated the construction: every inertial observer obtains the same laws of electromagnetism and the same speed of light.

CHAPTER 3

PROPER TIME, TIME DILATION, AND LENGTH CONTRACTION

We shall often choose units in which the speed of light is $c = 1$. In those units a year of time and a light-year of distance may be compared directly, and the formulas are much easier to read. Whenever c matters physically, we shall put it back. Dimensional analysis is enough to locate it.

3.1 The Galilean Spacetime Grid

Begin with the pre-Einstein relation between a frame at rest and a train moving at velocity v along the x-axis:

$$x' = x - vt, \qquad t' = t. \tag{3.1}$$

The second equation contains Newton's assumption of universal time. Every observer uses the same simultaneity slices, even though the spatial origins move relative to one another.

Suppose the unprimed observer stands beside the track and the primed observer rides the train. The train's origin has $x' = 0$, so its worldline in the unprimed coordinates is

$$x = vt. \tag{3.2}$$

That sloping line is the t'-axis. By contrast, $t' = t$ says that all lines of constant t' are the same horizontal lines as those of constant t.

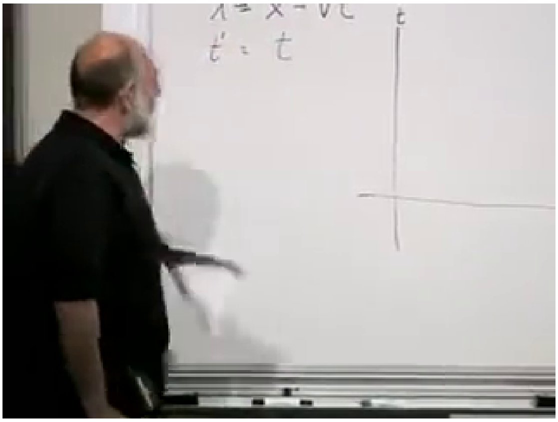


Figure 3.1: The Galilean transformation and the initial t-x grid. The moving origin will trace $x = vt$, while the two frames retain common constant-time lines.

Lecture frame: 00:03:06.

Solving Eq. (3.1) for the unprimed coordinates gives

$$x = x' + vt' = x' + vt, \qquad t = t'. \tag{3.3}$$

The inverse has the same form with v replaced by $-v$. If the train moves forward relative to the station, the station moves backward relative to the train. This reciprocity survives in special relativity, but universal time does not.

3.2 The Lorentz Correction

Galilean transformations cannot make the speed of light the same in every inertial frame. The corrected relations are, first in $c = 1$ units,

$$x' = \frac{x - vt}{\sqrt{1 - v^2}}, \qquad t' = \frac{t - vx}{\sqrt{1 - v^2}}. \tag{3.4}$$

Restoring ordinary units gives

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}, \qquad t' = \frac{t - vx/c^2}{\sqrt{1 - v^2/c^2}}. \tag{3.5}$$

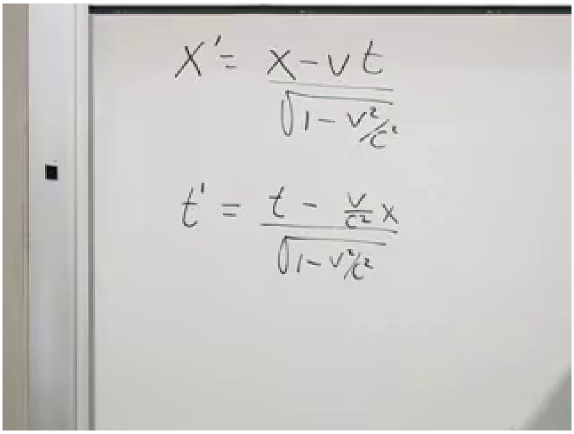


Figure 3.2: The Lorentz transformation with the factors of c restored.

Lecture frame: 00:09:10.

For $v \ll c$, both v^2/c^2 and vx/c^2 are tiny in the circumstances where Newtonian mechanics is used. Equation (3.5) then reduces to the Galilean result.

Relativity does not discard Newtonian kinematics; it identifies its range of validity.

At $v = c$, the denominator vanishes. An inertial frame cannot move at the speed of light relative to another inertial frame. Well below that limit the corrections are small, but near it the mixing of x and t becomes unavoidable.

Motion was chosen along x, so the transverse coordinates do not change:

$$y' = y, \qquad z' = z. \tag{3.6}$$

One way to see why is to place identical transverse meter sticks on the train and platform. An observer moving halfway between the two frames sees the rods approach with equal and opposite speeds. The setup has no feature that could make one transverse rod shorter than the other.

3.3 Lorentz Transformations Along the Light Cone

There is a geometric way to read Eq. (3.4). Add its two equations:

$$t' + x' = \frac{(t - vx) + (x - vt)}{\sqrt{1 - v^2}} \tag{3.7}$$

$$= (t + x)\frac{1 - v}{\sqrt{1 - v^2}} \tag{3.8}$$

$$= (t + x)\sqrt{\frac{1 - v}{1 + v}}. \tag{3.9}$$

Subtract them in the complementary order:

$$t' - x' = \frac{(t - vx) - (x - vt)}{\sqrt{1 - v^2}} \tag{3.10}$$

$$= (t - x)\frac{1 + v}{\sqrt{1 - v^2}} \tag{3.11}$$

$$= (t - x)\sqrt{\frac{1 + v}{1 - v}}. \tag{3.12}$$

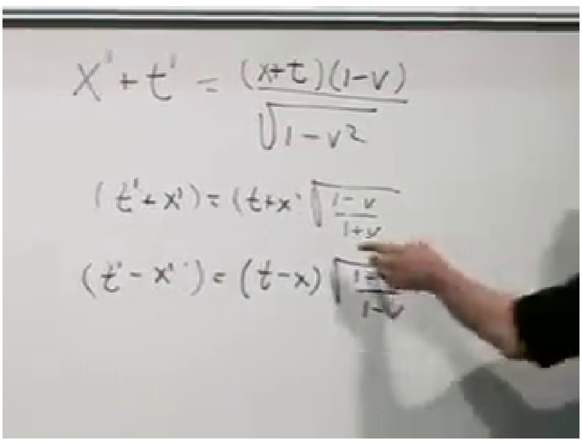


Figure 3.3: The Lorentz transformation rewritten in the null combinations $t + x$ and $t - x$. Their scale factors are reciprocal.

Lecture frame: 00:18:45.

The combinations $t+x$ and $t-x$ are coordinates along the two 45° lightlike directions. A boost squeezes one of these directions and stretches the other by the inverse factor. It does not rotate either one away from 45°. Thus light rays remain light rays, which is the spacetime picture of the invariant speed of light.

Question from the audience. The halfway observer sees a symmetric experiment, but how does that establish the result for the train and platform observers? [a]

Response. One additional fact is being used: a coincidence of events is invariant. If the ends of the rods meet, that meeting is one spacetime event. Every inertial observer may assign different coordinates to it, but none may split one coincidence into two different locations. Flashes emitted where the corresponding endpoints meet make those events operationally detectable. The symmetry of the halfway frame and the invariance of the two endpoint coincidences establish that the transverse lengths agree.

[a] Lecture timestamp: 00:20:48.

Question from the audience. Must the bicyclist who represents the halfway frame be located physically at the midpoint of the rods?[a]

Response. No. A reference frame is a whole network of observers and clocks, not one eye at one place. The bicyclist may receive the endpoint flashes later and infer where they originated. What matters is the spacetime location of each emission event, not where the light is eventually received.

[a] Lecture timestamp: 00:22:25.

The composition of successive velocities is another consequence of the same transformation, but it is deliberately left for the next lecture. The present task is to identify the invariant quantity that a moving clock measures.

3.4 The Invariant Interval

For a rotation of ordinary Cartesian axes, the quantity $x^2 + y^2$ is unchanged. Lorentz transformations preserve a different quadratic form. From Eq. (3.4),

$$
\begin{aligned}
t'^2 - x'^2 &= \frac{(t - vx)^2 - (x - vt)^2}{1 - v^2} && (3.13)\\
&= \frac{t^2 - 2vtx + v^2x^2 - x^2 + 2vtx - v^2t^2}{1 - v^2} && (3.14)\\
&= t^2 - x^2. && (3.15)
\end{aligned}
$$

The mixed terms cancel, and the remaining numerator contains a common factor of $1 - v^2$.

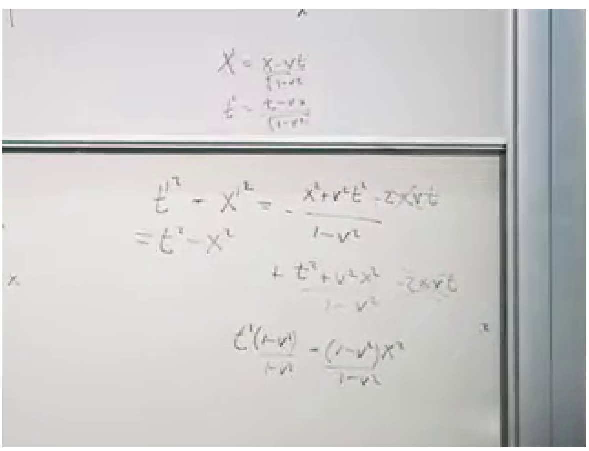


Figure 3.4: The board derivation of $t'^2 - x'^2 = t^2 - x^2$.
Lecture frame: 00:33:15.

With three spatial dimensions the invariant is

$$t'^2 - x'^2 - y'^2 - z'^2 = t^2 - x^2 - y^2 - z^2 \tag{3.16}$$

in $c = 1$ units. More generally, for two events we use coordinate differences:

$$\Delta s^2 = c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2. \tag{3.17}$$

Every inertial observer agrees on this number even though they disagree on its separate time and space terms.

3.5 Proper Time and Moving Clocks

Imagine a lattice of stationary observers whose clocks have been synchronized by exchanging light signals. Their readings define the coordinate time t. Now carry one wristwatch through that lattice. Its own reading follows the watch along its worldline; that reading is proper time.

For two timelike-separated events, define

$$c^2 \Delta\tau^2 = c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2. \tag{3.18}$$

In one spatial dimension with $c = 1$, this is simply

$$\Delta\tau = \sqrt{\Delta t^2 - \Delta x^2}. \tag{3.19}$$

The square root parallels the formula for an ordinary distance, but the minus sign changes the geometry.

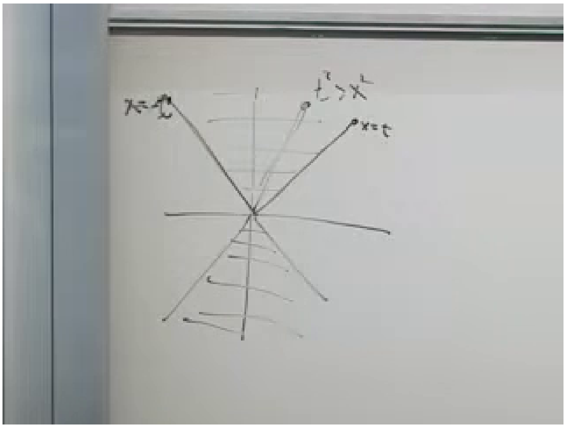
There are three cases:

$$\Delta t^2 - \Delta x^2 > 0 \qquad \text{timelike,} \tag{3.20}$$

$$\Delta t^2 - \Delta x^2 = 0 \qquad \text{lightlike,} \tag{3.21}$$

$$\Delta t^2 - \Delta x^2 < 0 \qquad \text{spacelike.} \tag{3.22}$$

Inside the light cone the proper time is real. On the cone, where $x = \pm t$, it is zero even though the events need not coincide. A light ray can connect precisely such events. Outside the cone the separation is spacelike; it is better to describe it by the real proper distance $\sqrt{\Delta x^2 - \Delta t^2}$ than by an imaginary clock time.



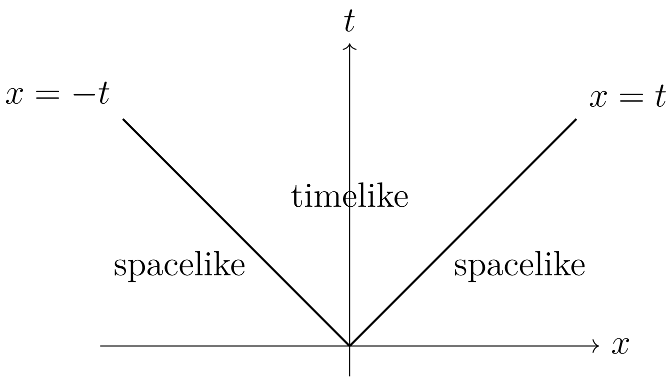


Figure 3.5: The board light cone and a clean classification of timelike, lightlike, and spacelike separation.

Lecture frame: 00:38:40.

Why does Eq. (3.18) equal a clock reading? For a clock at rest, $\Delta x = 0$, so $\Delta\tau = \Delta t$. For a uniformly moving clock, choose the primed frame in which that clock is at rest. There $\Delta x' = 0$, and invariance gives

$$\Delta\tau = \sqrt{\Delta t'^2 - \Delta x'^2} = \Delta t'. \tag{3.23}$$

Moving clocks therefore tick off proper time.

Question from the audience. Are “the interval between the events” and “the reading of the clock” being used as two names for the same thing? [a]

Response. They coincide when an ideal clock travels inertially from one event to the other. The invariant interval is a geometric quantity; the difference between the endpoint readings of the comoving clock is its physical measurement. A stationary coordinate clock generally records a different coordinate-time interval.

[a] Lecture timestamp: 00:44:49.

3.6 Time Dilation and Simultaneity

Consider the origin of the moving frame. Its worldline obeys

$$x' = 0, \qquad x = vt. \tag{3.24}$$

The same sloping line is both the path of the moving clock and the t'-axis.

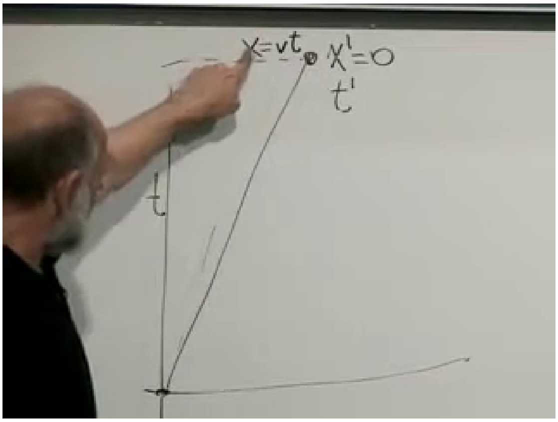


Figure 3.6: The moving origin: $x = vt$, $x' = 0$, and the t'-axis are three descriptions of the same worldline.

Lecture frame: 00:48:18.

Apply the invariant interval between the common origin and a later point on that worldline:

$$t'^2 - x'^2 = t^2 - x^2, \tag{3.25}$$

$$t'^2 = t^2 - v^2 t^2, \tag{3.26}$$

$$t' = t\sqrt{1 - v^2}. \tag{3.27}$$

With c restored,

$$\Delta\tau = \Delta t\sqrt{1 - \frac{v^2}{c^2}} = \frac{\Delta t}{\gamma}, \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}. \tag{3.28}$$

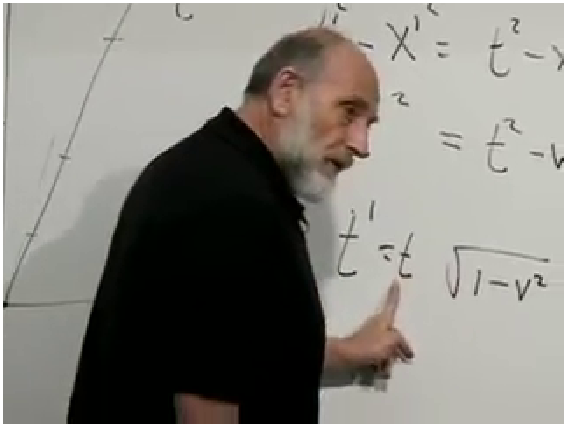


Figure 3.7: The time-dilation result on the board: the moving clock accumulates $t' = t\sqrt{1-v^2}$ in units with $c = 1$.

Lecture frame: 00:50:45.

The stationary observer says that the moving clock runs slow. Reciprocity requires the moving observer to say the same about stationary clocks. There is no contradiction because the two descriptions compare different pairs of distant events.

Indeed, a constant value of t' in Eq. (3.4) satisfies

$$t - vx = \text{constant}. \tag{3.29}$$

The moving observer's simultaneity lines are tilted relative to the horizontal lines $t =$ constant. Each observer calls a different collection of distant clock readings simultaneous. Once that difference is kept visible, the reciprocal slowing of clocks is consistent.

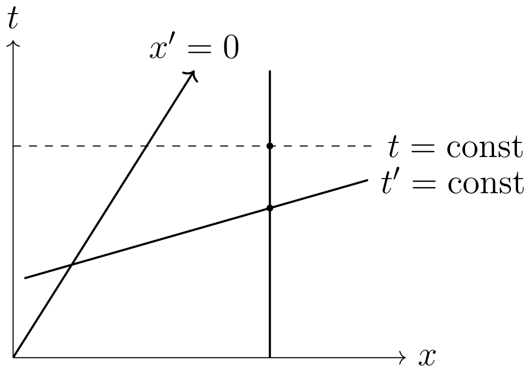


Figure 3.8: Reciprocal time dilation compares clock readings on different simultaneity slices.

3.7 The Twin Paths

Two twins begin at the same event. One remains at $x = 0$. The other travels outward at speed v, reverses direction, and returns at the same speed. When they reunite they compare every physical process that can serve as a clock: watches, heartbeats, metabolism, or aging itself.

Let each half of the trip take coordinate time T in the home frame. The home twin's proper time is

$$\tau_{\text{home}} = 2T. \tag{3.30}$$

The turn-around event has coordinates (vT, T). The first traveling leg therefore contributes

$$\tau_{\text{leg}} = \sqrt{T^2 - v^2T^2} = T\sqrt{1 - v^2}. \tag{3.31}$$

The return leg contributes the same amount, hence

$$\tau_{\text{travel}} = 2T\sqrt{1 - v^2} < 2T. \tag{3.32}$$

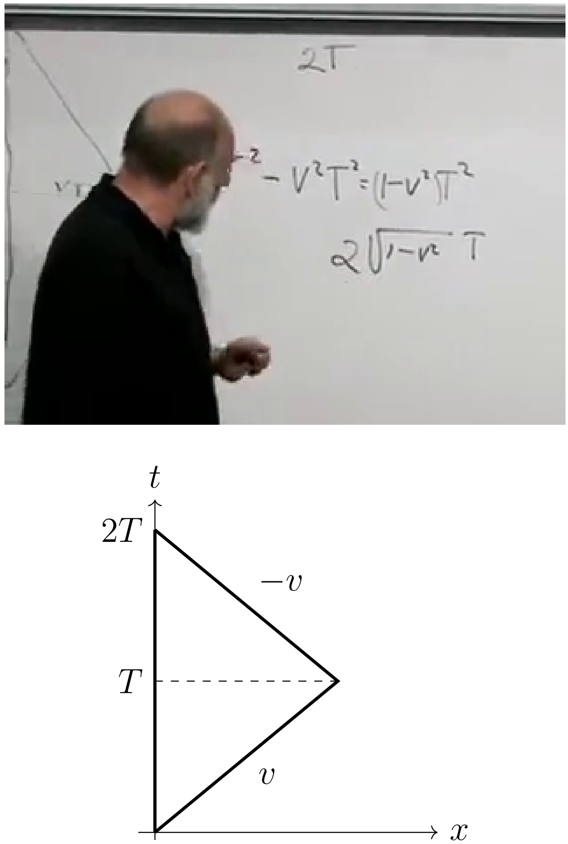


Figure 3.9: The home worldline and the two traveling legs. Their proper times are different because they are different paths through spacetime.

Lecture frame: 00:59:35.

There is no contradiction in the unequal ages. The traveling path contains a turn and is not related to the straight home path by a single inertial-frame symmetry. In ordinary Euclidean geometry a straight route and a broken route between two points also have different lengths. Minkowski geometry reverses which path has the larger interval, but the comparison is still geometric.

3.8 Acceleration and the Ideal Clock

Question from the audience. At the turnaround the physics is different and the traveler accelerates. Does that invalidate the clock calculation? [a]

Response. The acceleration is real and helps identify the asymmetry; it should not be argued away. The sharp corner is only an idealized drawing and may be rounded into a finite turn. The further physical assumption is that a sufficiently small, well-built clock is insensitive to modest acceleration and continues to record proper time. A large or fragile clock may deform or break, but that is a defect of the mechanism rather than of the spacetime interval.

[a] Lecture timestamp: 01:02:19.

This ideal-clock assumption is not derived from the Lorentz transformation alone. It is tested through the behavior of actual clock mechanisms. An atom is a much better approximation than a clock hundreds of miles across because the compact mechanism is harder to distort under acceleration or gravity. Invoking general relativity by name would not eliminate this physical issue; one would still have to know how the chosen mechanism responds to its environment.

Question from the audience. Can an accelerated trajectory be reduced to short intervals on which the inertial analysis remains valid?[a]

Response. Yes. Divide the curve into short, nearly straight segments, calculate the proper time on each segment, and add the results. This is the spacetime analogue of finding the length of an ordinary curve from many small line segments. In the continuum limit,

$$\tau = \int \sqrt{1 - \frac{v(t)^2}{c^2}}\, dt,$$

for motion in flat spacetime described in an inertial coordinate system.

[a]Lecture timestamp: 01:09:08.

3.9 What the Twins See

The final clock readings can be made operational by sending light pulses. Mark one-second intervals of proper time on both worldlines and let each twin emit a pulse at every mark. During the outward leg, pulses received from the other twin are spread apart. During the return, they arrive bunched together. This is the relativistic Doppler effect expressed directly on the spacetime diagram.

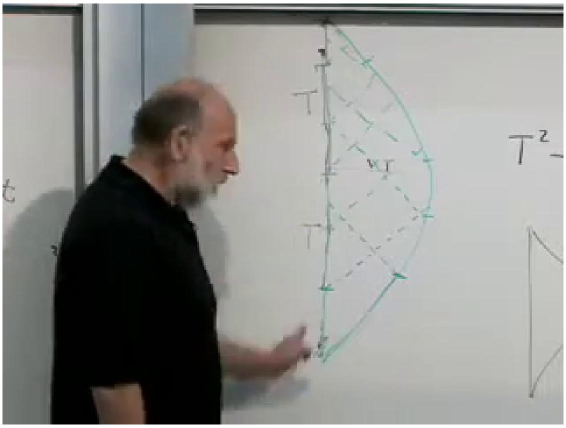


Figure 3.10: Light signals exchanged between the home and traveling worldlines. Sparse arrivals on the outward leg and compressed arrivals on the return leg account for every emitted pulse.

Lecture frame: 01:12:55.

No frames of the story disappear. Each observer eventually receives all the pulses emitted by the other, but their reception rates depend on the relative motion. The outbound redshift and inbound blueshift are not equal-duration effects for the two paths, so counting all received pulses agrees with the proper-time comparison.

Question from the audience. If the returning traveler receives blueshifted light, could the traveler collect more energy than the source emitted? [a]

Response. That question requires the relativistic definition and frame dependence of energy, which have not yet been developed. It is reasonable, but coordinates and pulse counts alone are not enough to settle the energy bookkeeping, especially for an accelerated observer. The answer should therefore be postponed rather than guessed.

[a] Lecture timestamp: 01:13:50.

3.10 A Closed Spatial Direction

Question from the audience. Suppose one spatial direction is compact. Could a twin move uniformly around it, return to the starting place, and avoid acceleration altogether? What then distinguishes the twins?[a]

Response. The space should be imagined intrinsically as a cylinder, not as a sheet bent inside a larger space; there need be no radial acceleration. The returning observer nevertheless encounters the global fact that the same places recur. That indicates an asymmetry, but a complete answer requires a careful treatment of special relativity with the spatial identification. The question is left open here rather than answered improvisationally.

[a] Lecture timestamp: 01:15:14.

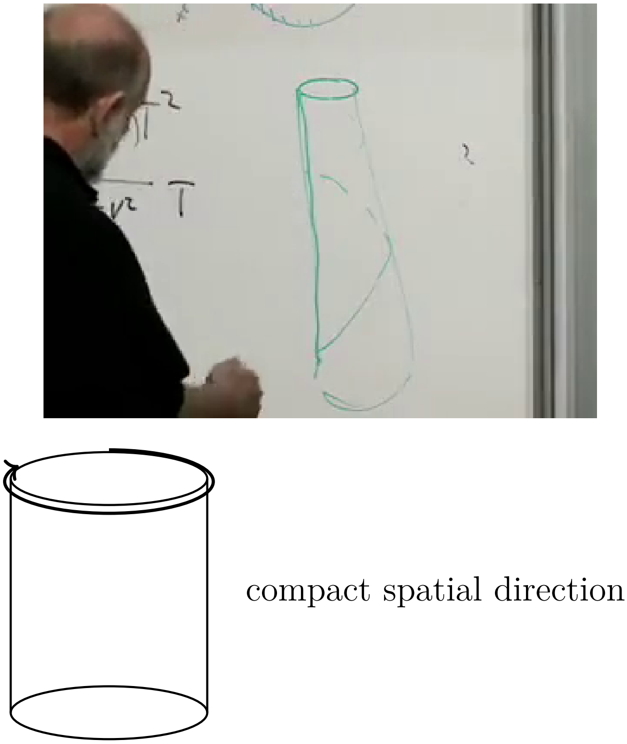


Figure 3.11: The compact-space question as drawn on the board, with a clean intrinsic cylinder. It is a question about global identification, not motion around a cylinder embedded in an extra radial dimension.

Lecture frame: 01:17:56.

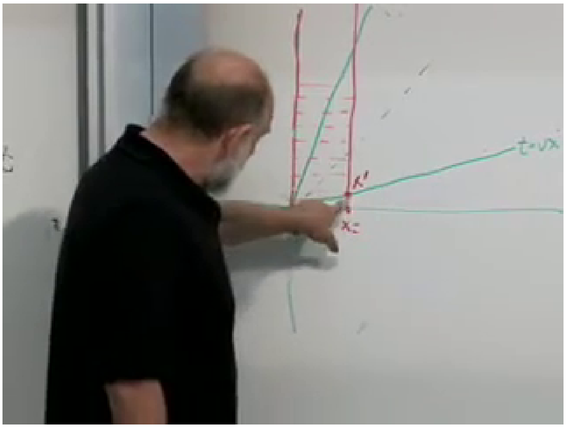
3.11 Length Is Measured on a Simultaneity Slice

For a special-relativity problem, first draw the spacetime picture. Take a meter stick at rest in the unprimed frame, with endpoints at $x = 0$ and $x = 1$. As time passes, the two endpoints trace vertical worldlines. The stick sweeps out a ribbon between them.

The moving observer must measure both endpoints at one value of the moving time t'. From Eq. (3.29), the slice through the origin with $t' = 0$ satisfies

$$t = vx \tag{3.33}$$

in $c = 1$ units. It intersects the right endpoint $x = 1$ at $t = v$.



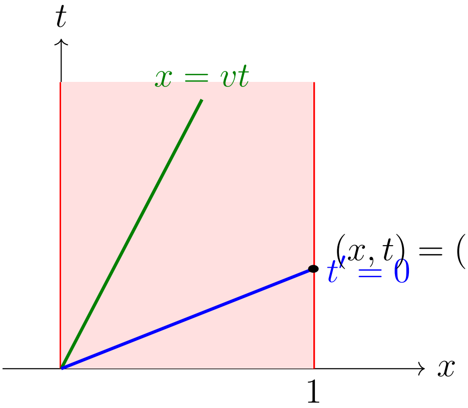


Figure 3.12: A rod at rest sweeps out a spacetime ribbon. The moving observer's length is the separation between its endpoints on one $t' =$ const slice.

Lecture frame: 01:24:20.

The selected endpoint separation is spacelike, so use the positive spatial form of the invariant:

$$x'^2 - t'^2 = x^2 - t^2. \tag{3.34}$$

At the chosen endpoint, $t' = 0$, $x = 1$, and $t = v$. Hence

$$x'^2 = 1 - v^2, \tag{3.35}$$

$$x' = \sqrt{1 - v^2}. \tag{3.36}$$

Thus a rest length L is measured in the moving frame as

$$L' = L\sqrt{1 - v^2/c^2} = \frac{L}{\gamma}. \tag{3.37}$$

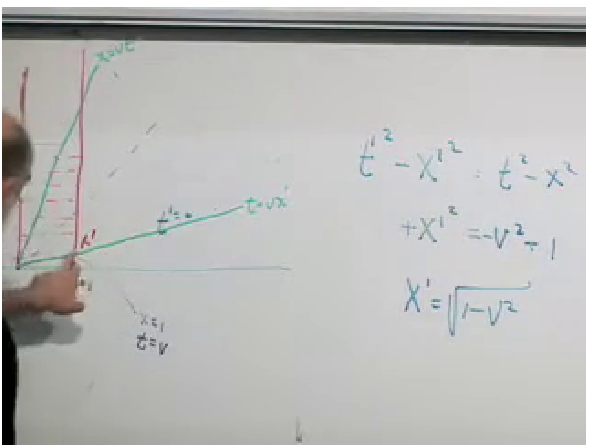


Figure 3.13: The completed length-contraction construction and the board result $x' = \sqrt{1 - v^2}$ for a unit rest length.

Lecture frame: 01:28:10.

The reciprocal statement is also true: the platform observer assigns a shorter length to the train's longitudinal meter stick. Once again the two observers choose different simultaneous endpoint events, so reciprocity is not contradictory.

3.12 Questions at the End

Question from the audience. Does anything besides light propagate at the speed of light? In particular, what about gravity?[a]

Response. General relativity predicts gravitational waves that propagate at c. At the time of this lecture, the evidence came indirectly from systems such as binary pulsars; gravitational waves had not yet been detected directly.[b]

[a] Lecture timestamp: 01:29:29.

[b] Editorial clarification: The experimental situation changed after this 2006 lecture. LIGO directly detected GW150914 in 2015, and the joint observation of GW170817 and its electromagnetic counterpart in 2017 showed that the propagation speeds of gravity and light agree to within a few parts in 10^{15}. See https://ligo.org/detections/gw150914/ and https://doi.org/10.1103/PhysRevLett.119.251302.

Question from the audience. Can an astronaut cross a distance of many light-years in less proper time than the distance expressed in light-years? Would that mean moving faster than light?[a]

Response. The astronaut's proper time is $\sqrt{\Delta t^2 - \Delta x^2}$, which is smaller than the coordinate time Δt. In the astronaut's inertial frame, the distance between the stars is Lorentz-contracted. The stars move with speed $-v$, not faster than light; there is simply less contracted distance to cross.

[a] Lecture timestamp: 01:30:38.

If the astronaut begins at rest and accelerates, the description changes between a succession of instantaneous inertial frames. The distance assigned to the stars can change during that transition. One cannot apply one fixed Lorentz transformation to the entire accelerated history, but the total proper time still follows by adding the contributions along the worldline. The same statement applies to a realistic twin who launches, turns, and decelerates rather than being born at cruising speed.

Question from the audience. Are these puzzles caused by treating time as special? Must we call it a fourth dimension?[a]

Response. It is not necessary to begin with the slogan that time is a fourth dimension. The Lorentz transformation itself forces space and time to mix, while the interval $t^2 - x^2 - y^2 - z^2$ treats them as parts of one geometry with a different sign for time and space. At low velocity the mixing becomes negligible and familiar Newtonian intuition returns. At relativistic velocity, however, space and time cannot be regarded as fundamentally decoupled.

[a] Lecture timestamp: 01:36:43.

Time dilation, length contraction, the twin comparison, and shortened travel time are therefore not separate tricks. They all follow from three habits: identify the events being compared, draw their spacetime geometry, and use the invariant interval.

CHAPTER 4

FIREBALLS, FRAMES, AND FOUR-VECTORS

We begin with a question left over from the discussion of high-energy collisions. A collision of two heavy nuclei creates a violently changing object, not a quiet astronomical black hole. What, then, can it mean to describe the collision by a horizon? The answer leads through a five-dimensional picture of strongly coupled matter, a set of practical questions about accelerators, and finally back to the geometry of special relativity. By the end, velocity and momentum have become four-vectors, and energy has taken its place beside the three components of momentum.

4.1 A Fireball in a Fifth Dimension

Question from the audience. The object produced in the collision is unstable and changes shape. How can such a complicated object be called a black hole?

Response. Picture a fifth spatial direction running between a floor and a ceiling. Gravity in that direction pulls excitations toward the floor. The collision deposits a hot lump of energy there; the lump spreads and changes shape, but it can still possess a horizon, temperature, and entropy.

Imagine particles, or more pictorially little strings, moving in this five-dimensional world. Excitations that fall to the floor appear in four dimensions as hadrons: protons, neutrons, mesons, and other extended composite objects. Excitations attached near the ceiling appear more elementary. The vertical location is therefore a geometric way of encoding scale: the floor represents the long-distance hadronic regime, while the ceiling represents short-distance degrees of freedom.

When two hadrons collide, two objects near the floor strike one another and make a large splash of energy. The result is not a rigid body. It is a hot horizon-bearing lump, a kind of puddle in the five-dimensional gravitational field. It is not an astronomical black hole in ordinary four-dimensional space, nor a Planck-scale black hole attached to the ceiling. It is the gravitational description of the strongly interacting fireball near the floor.

At high energy the incoming nuclei are Lorentz-contracted into thin pancakes. They collide, deposit their energy in a small region, and produce a fluid that spreads toward equilibrium. Its small viscosity was one of the surprising features of the quark–gluon plasma. In the five-dimensional description the same evolution is the relaxation of a distorted horizon.

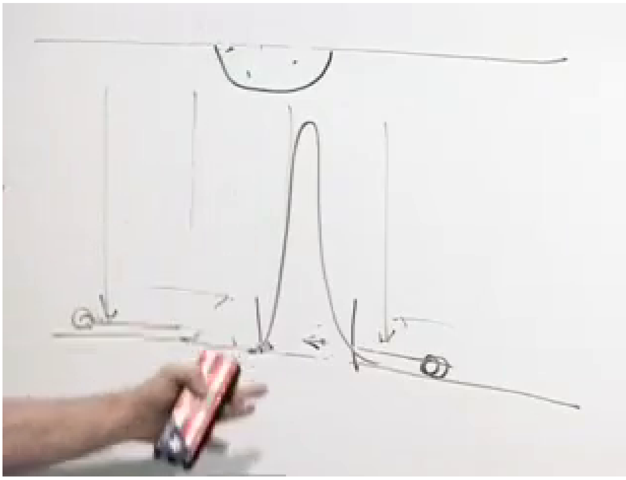


Figure 4.1: The floor-and-ceiling picture of the fifth dimension. The central horizon-bearing lump spreads along the floor after the collision.

Lecture frame: 00:04:22.

A horizon does not remain forever. Hawking radiation carries energy away, and in the four-dimensional description that radiation is the shower of ordinary hadrons emerging from the collision. The thousands of tracks in a detector event are, in this language, the particles boiled off the dual black hole.

4.1.1 Is the extra dimension real?

Question from the audience. Is the fifth dimension a spatial dimension, and is the line drawn between floor and ceiling literally extending in that direction?

Response. Yes. It is spacelike, and the surface drawn in that five-dimensional geometry is a horizon. Whether one calls the extra direction "real" or regards it as an equivalent mathematical description is a deeper question.

The useful statement is not that gravity replaces quantum chromodynamics. Rather, two languages may describe the same physics. In the four-dimensional language there is a hot fireball of quarks and gluons, with no ordinary gravitational field. In the five-dimensional language there is gravity and a horizon. If the correspondence is correct, an honest QCD calculation and the dual gravitational calculation must agree wherever both can be performed.

Question from the audience. QCD has neither gravity nor horizons. Is the black-hole picture adding new physics, or does it merely redescribe the fireball?

Response. In four dimensions it is a plasma of quarks and gluons; in the five-dimensional incarnation it has a gravitational description with a horizon. The advantage is calculational. Strongly coupled QCD is extremely difficult, while some horizon properties—the viscosity in particular—are comparatively easy to calculate.

The distinction between "real dimension" and "mathematical analogy" need not be settled before the description can be useful. A structure earns a physical interpretation when it joins the other variables into a simpler, more coherent set of laws. Still, one should not overstate the case: applications of gauge/gravity duality to the observed quark–gluon plasma are models of strongly coupled dynamics, not a proof that real-world QCD has a known exact gravitational dual.[1]

4.2 Why Colliders Collide Beams

The questions now become practical. Why use two moving beams instead of a single beam and a stationary target? Why are circular accelerators so large? Why are electrons and protons handled

[1]Editorial clarification: This qualification makes explicit the modern status of the QCD application; it does not alter the five-dimensional mechanism developed here.

differently? All of these questions are questions about relativistic energy.

Question from the audience. Why collide two beams rather than fire one energetic beam into a stationary target?

Response. What creates new particles is the energy available in the center-of-mass frame. Two equal and opposite relativistic beams have no net momentum, so essentially all their energy is available to the collision. A fixed target forces much of the incident energy to remain as motion of the center of mass.

For two ultrarelativistic particles with opposite momenta of magnitude p, the lecture-level estimate in $c = 1$ units is

$$E_{\mathrm{cm}}^{\mathrm{collider}} \simeq 2p. \tag{4.1}$$

The corresponding fixed-target estimate grows only as the square root of the beam momentum,

$$E_{\mathrm{cm}}^{\mathrm{fixed}} \sim \sqrt{mp}. \tag{4.2}$$

The scaling, rather than the omitted numerical factor, is the point: doubling the beam energy doubles the collider energy, but it increases fixed-target center-of-mass energy only by a square root.

The invariant calculation makes the comparison precise. If P_1 and P_2 are the incoming four-momenta, then

$$s = (P_1 + P_2)^2, \qquad E_{\mathrm{cm}} = \sqrt{s}. \tag{4.3}$$

For equal counter-propagating beams of energy E, the spatial momenta cancel and $\sqrt{s} = 2E$. For a projectile of energy $E \gg m$ striking an equal-mass target at rest,

$$s = 2m^2 + 2mE, \qquad \sqrt{s} \simeq \sqrt{2mE} \qquad (c = 1). \tag{4.4}$$

This short invariant derivation supplies the factor suppressed in the spoken estimate.[2]

Question from the audience. Is there a fundamental reason a circular accelerator must be so large? Could the same beam simply circulate through a smaller machine more times?

Response. Fields and materials impose limits. A particle receives a finite increase in energy per metre, so a higher-energy machine needs more effective accelerating length. A tight circle also demands greater transverse acceleration, and an accelerated charge radiates. If the ring is too small, the beam radiates energy as fast as the machine supplies it.

The bending relation is approximately

$$p = qBR, \tag{4.5}$$

so for fixed magnetic field B, a larger momentum p requires a larger radius R. The radio-frequency cavities do the work that raises the beam energy; the magnets bend and focus the beam. The broad lesson

[2] Editorial clarification: The derivation is included to make the lecture's scaling argument reproducible.

remains the one given at the board: field strength and radiative losses prevent us from compressing an arbitrarily energetic accelerator into a small ring.

Circular motion has acceleration even at constant speed because the velocity changes direction. An accelerated charge emits synchrotron radiation. A linear accelerator avoids the continuous bending loss, although the short accelerating pushes themselves are not literally radiation-free. Circular machines accept the bending cost in exchange for reusing the same apparatus on every orbit.

Question from the audience. Why were electrons associated with the straight machine at SLAC while circular high-energy colliders use protons or other hadrons?

Response. Electrons are much lighter. At a given high energy their Lorentz factor is much larger, and circular acceleration makes them radiate far more strongly. Heavier protons can circle repeatedly, gaining energy on each pass, before synchrotron losses become prohibitive.

For fixed energy and ring radius, the radiated power scales very steeply with the particle mass,

$$P_{\text{sync}} \propto \frac{E^4}{m^4 R^2}. \tag{4.6}$$

This is why the electron–proton distinction is much sharper than the phrase "the lighter particle moves faster" might suggest: both beams can already be extremely close to c, but their Lorentz factors and radiation losses are very different.[3]

Question from the audience. Why not accelerate hadrons in a straight line as well?

Response. One can. The difficulty is length. In a ring the same particle passes the accelerating structures many times; an equivalent linear hadron machine would have to be enormously long. The useful compromise is a large ring, large enough that proton radiation remains tolerable.

These accelerator questions expose exactly why Newtonian rules are insufficient. We now return to special relativity and build the language that makes such comparisons systematic.

4.3 The Garage and the Limousine

Consider a garage about twelve feet long and a stretch limousine about twenty feet long. The driver accelerates until the limousine is Lorentz-contracted, hoping that it will fit. In the garage frame the moving limousine is shorter, so it appears possible. In the limousine frame, however, the moving garage is shorter, so it appears impossible. Can the whole limousine be inside the garage?

Open both doors so that no collision with a wall distracts us. To say that the limousine is inside means that its front and rear are inside *simultaneously*. That final word contains the entire puzzle.

[3] Editorial clarification: The bending relation and synchrotron scaling make the lecture's qualitative accelerator argument precise; they were not worked out on the board.

Draw a spacetime diagram with t vertical and x horizontal. Light rays form the 45° light cone in $c = 1$ units. The garage is at rest, so the worldlines of its two doors are vertical. The front and rear of the limousine trace two parallel sloping worldlines. The moving frame's t'-axis is tilted toward the light cone, and its lines of simultaneity are parallel to the tilted x'-axis.

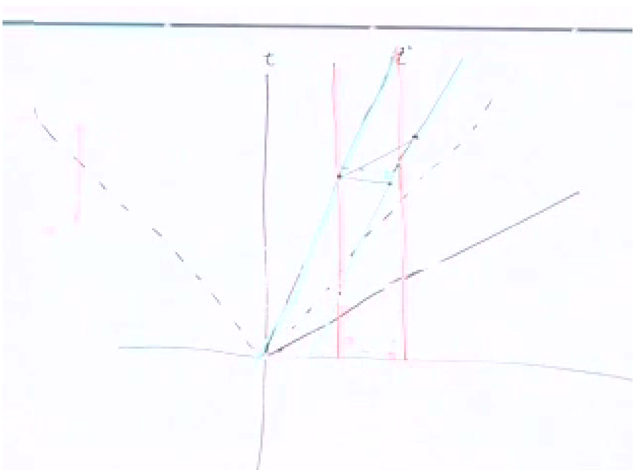


Figure 4.2: The completed garage–limousine spacetime diagram. Vertical lines bound the garage; sloping lines trace the two ends of the limousine. Horizontal and tilted cuts represent the two frames' different simultaneity slices.

Lecture frame: 00:31:05.

In the garage frame, a horizontal line can cross both limousine worldlines between the two garage worldlines. At that garage time the rear has entered and the front has not yet left. The contracted limousine is wholly inside.

In the limousine frame, the relevant simultaneous events lie on a tilted line. Draw such a line through the event at which the rear enters. It meets the front worldline after the front has already passed the far door. The limousine is never wholly inside according to its own simultaneity. Both statements are true because they compare different pairs of events.

Question from the audience. Do the two green worldlines and the horizontal segment between them form a parallelogram? Is that black segment the length of the vehicle?

Response. The horizontal segment is the Lorentz-contracted length, because its endpoints are simultaneous in the garage frame. The proper length is obtained from endpoints simultaneous in the limousine frame, on a line parallel to x'. The two measurements use different cuts through the same pair of worldlines.

If L_0 is the proper length, the horizontal garage-frame cut has length

$$L = \frac{L_0}{\gamma}, \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}. \tag{4.7}$$

The formula alone does not resolve the paradox. The diagram does, because it shows which events were compared. Whenever a relativity puzzle seems to produce two incompatible answers, look for an unspoken use of “simultaneously.”

4.4 Four Coordinates and Two Index Positions

To move from puzzles to laws of motion, we need notation that remembers the geometry. Package the coordinates as

$$(X^1, X^2, X^3, X^0) = (x, y, z, t), \qquad X^\mu = (x, y, z, t), \tag{4.8}$$

where μ runs through $1, 2, 3, 0$. The ordering puts time last on the page even though its index is 0, matching the convention used at the board.

Question from the audience. Are we setting $c = 1$? If ordinary units are restored, should the time coordinate be ct?

Response. Yes. We set $c = 1$ while developing the geometry and put the factors of c back when dimensions or the nonrelativistic limit matter. With conventional length units, the time coordinate is ct.

The same vector may be written with a lower index. With the sign convention used here,

$$X_\mu = (-x, -y, -z, t). \tag{4.9}$$

Thus lowering an index reverses the signs of the three spatial components and leaves the time component unchanged. For example,

$$X^\mu = (3, 2, 4, 10) \quad \Longrightarrow \quad X_\mu = (-3, -2, -4, 10). \tag{4.10}$$

These are not two spacetime points; they are two component forms of one geometric object.

Question from the audience. Which components are covariant and contravariant, and are the labels meant to be superscripts or subscripts?

Response. Upper-index components are contravariant; lower-index components are covariant. A useful memory aid from the room is "co is low, contra is high." Physicists often simply say upstairs and downstairs.

The notation earns its keep in the invariant contraction

$$X_\mu X^\mu = t^2 - x^2 - y^2 - z^2. \tag{4.11}$$

In differential form this is

$$d\tau^2 = dt^2 - dx^2 - dy^2 - dz^2 \qquad (c = 1). \tag{4.12}$$

For timelike motion, $d\tau$ is the time recorded by a clock moving along the worldline.

Whenever the same index appears once upstairs and once downstairs, it is understood to be summed. Einstein's convention turns

$$\sum_\mu X_\mu X^\mu \tag{4.13}$$

into the shorter expression in Eq. (4.11). This is not merely typographical economy. It prevents the alternating signs of Minkowski geometry from being rewritten by hand in every equation.

Question from the audience. Is τ the standard symbol for proper time?

Response. Yes. The symbol s is also sometimes used, but τ is a standard choice for proper time.

4.5 Rotations as the Easy Symmetry

A Lorentz boost mixes space and time. Before studying that matrix, consider the more familiar operation of rotating spatial axes. If the relative motion points in an arbitrary direction, we can rotate the axes until that direction becomes the new x-axis and then apply the known boost. Rotations are therefore part of the construction, not an ornamental detour.

For a rotation through angle θ about the z-axis,

$$\begin{aligned} x' &= x\cos\theta + y\sin\theta, \\ y' &= -x\sin\theta + y\cos\theta, \\ z' &= z. \end{aligned} \tag{4.14}$$

At $\theta = 0$ these equations return $x' = x$ and $y' = y$; at $\theta = \pi/2$ they exchange the axes with the expected signs.

The transformation preserves Euclidean distance. Writing $c_\theta = \cos\theta$ and $s_\theta = \sin\theta$,

$$\begin{aligned} x'^2 + y'^2 &= (xc_\theta + ys_\theta)^2 + (-xs_\theta + yc_\theta)^2 \\ &= x^2(c_\theta^2 + s_\theta^2) + y^2(s_\theta^2 + c_\theta^2) \\ &\quad + 2xyc_\theta s_\theta - 2xyc_\theta s_\theta \\ &= x^2 + y^2. \end{aligned} \tag{4.15}$$

The cross terms cancel and $c_\theta^2 + s_\theta^2 = 1$. Since $z' = z$, the full length $x^2 + y^2 + z^2$ is invariant.

The same equations become one matrix equation:

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \underbrace{\begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{M} \begin{pmatrix} x \\ y \\ z \end{pmatrix}. \tag{4.16}$$

In index notation,

$$x'^i = M^i{}_j x^j, \qquad i, j = 1, 2, 3. \tag{4.17}$$

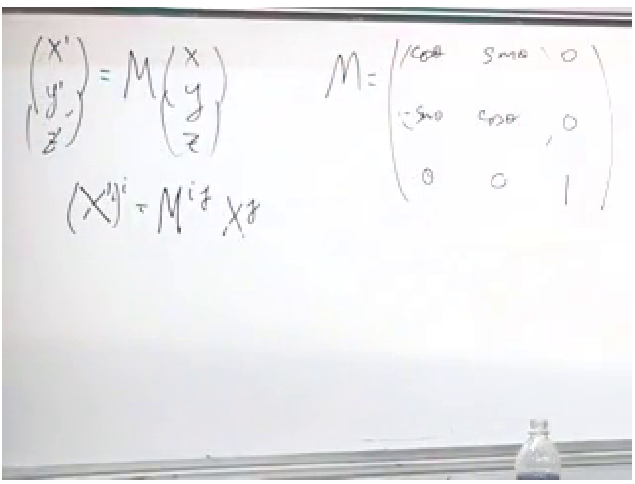


Figure 4.3: The spatial rotation written both as a matrix equation and in index notation.

Lecture frame: 01:00:40.

4.6 Lorentz Matrices and Composed Velocities

For motion along x, the Lorentz transformation in $c = 1$ units is

$$\begin{aligned} x' &= \gamma_v(x - vt), & y' &= y, \\ z' &= z, & t' &= \gamma_v(t - vx), \end{aligned} \tag{4.18}$$

where

$$\gamma_v = \frac{1}{\sqrt{1-v^2}}. \tag{4.19}$$

With the component order (x, y, z, t), this becomes

$$\begin{pmatrix} x' \\ y' \\ z' \\ t' \end{pmatrix} = \begin{pmatrix} \gamma_v & 0 & 0 & -\gamma_v v \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\gamma_v v & 0 & 0 & \gamma_v \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix}. \tag{4.20}$$

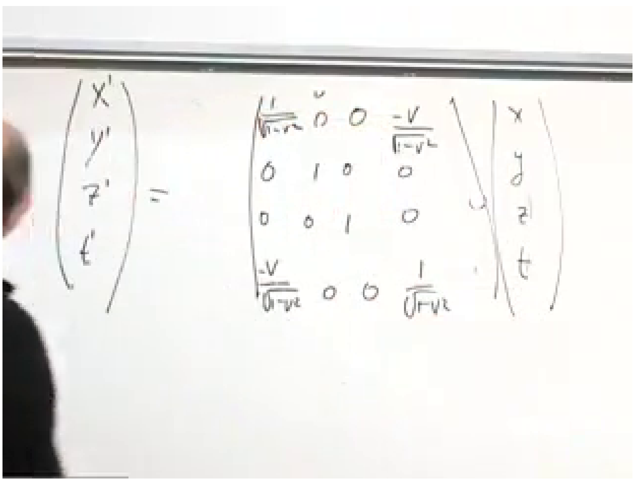


Figure 4.4: The 4×4 Lorentz matrix. Only the x and t components mix; y and z remain unchanged.

Lecture frame: 01:04:40.

Matrix multiplication now answers a physical question. Suppose one observer moves with velocity v relative to us and a third moves with velocity u relative to that observer, all along the same axis. Newtonian kinematics would give $w = u + v$, which would allow two velocities of $3c/4$ to combine into $3c/2$. Relativity must give a different rule.

The active part of the boost is the 2×2 matrix

$$B(v) = \gamma_v \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix}. \tag{4.21}$$

Composing the two boosts gives

$$\begin{aligned} B(u)B(v) &= \gamma_u\gamma_v \begin{pmatrix} 1 & -u \\ -u & 1 \end{pmatrix} \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix} \\ &= \gamma_u\gamma_v \begin{pmatrix} 1+uv & -(u+v) \\ -(u+v) & 1+uv \end{pmatrix}. \end{aligned} \tag{4.22}$$

The ratio of an off-diagonal entry to a diagonal entry identifies the velocity of the resulting boost:

$$w = \frac{u+v}{1+uv} \qquad (c=1). \tag{4.23}$$

Restoring dimensions,

$$w = \frac{u+v}{1+uv/c^2}. \tag{4.24}$$

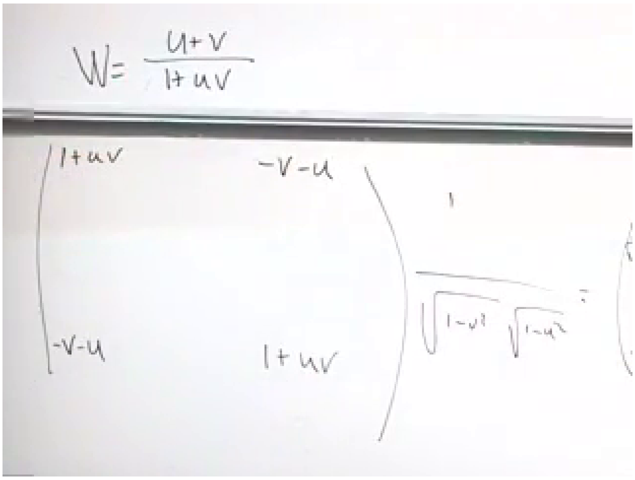


Figure 4.5: The composed-boost matrix and the velocity $w = (u+v)/(1+uv)$ in natural units.
Lecture frame: 01:15:35.

If u and v are each close to c, the numerator approaches $2c$ while the denominator approaches 2, so w approaches c, not $2c$. More generally, Eq. (4.24) maps subluminal inputs to a subluminal result. Matrix multiplication has encoded the speed limit automatically.

4.7 Four-Vectors and the Failure of Ordinary Velocity

A four-vector is any object whose four components transform under rotations and Lorentz boosts in the same way as the spacetime coordinates. If V^μ is a four-vector, then with our sign convention

$$V_\mu = (-V^1, -V^2, -V^3, V^0), \tag{4.25}$$

and its invariant norm is

$$V_\mu V^\mu = (V^0)^2 - (V^1)^2 - (V^2)^2 - (V^3)^2. \quad (4.26)$$

For two four-vectors V^μ and W^μ, the contraction

$$W_\mu V^\mu = W^0V^0 - W^1V^1 - W^2V^2 - W^3V^3 \quad (4.27)$$

is likewise invariant. The components change between frames; the contraction does not.

Ordinary velocity has only three components,

$$\mathbf{v} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}\right). \quad (4.28)$$

Appending $dt/dt = 1$ does not make it a useful four-vector. The fourth entry would be trivial, and the resulting object would not transform linearly under Lorentz transformations. Coordinate time t changes from one frame to another, so differentiating by t spoils the desired transformation law.

Proper time provides the cure. Along a worldline define

$$u^\mu = \frac{dX^\mu}{d\tau}. \quad (4.29)$$

The differential dX^μ is a four-vector and $d\tau$ is invariant, so their ratio is a four-vector. The adjective "proper" refers to the clock in the denominator.

The chain rule relates this object to ordinary velocity:

$$\frac{dx}{d\tau} = \frac{dx}{dt}\frac{dt}{d\tau}, \quad (4.30)$$

and similarly for y and z. From Eq. (4.12),

$$\left(\frac{d\tau}{dt}\right)^2 = 1 - \left(\frac{dx}{dt}\right)^2 - \left(\frac{dy}{dt}\right)^2 - \left(\frac{dz}{dt}\right)^2$$
$$= 1 - v^2 \qquad (c = 1), \tag{4.31}$$

and therefore

$$\frac{dt}{d\tau} = \gamma_v = \frac{1}{\sqrt{1-v^2}}. \tag{4.32}$$

The four components are

$$u^i = \gamma_v v^i, \qquad u^0 = \gamma_v \qquad (c = 1). \tag{4.33}$$

At low speed, $u^i \simeq v^i$ and $u^0 \simeq 1$. Near the speed of light, γ_v grows without bound and proper velocity differs sharply from ordinary velocity.

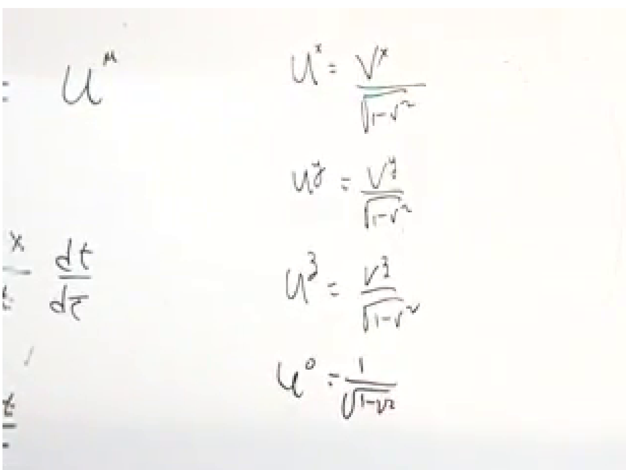


Figure 4.6: The four components of proper velocity expressed through ordinary velocity and the Lorentz factor.

Lecture frame: 01:34:40.

Restoring conventional units and using $X^0 = ct$ gives the standard form

$$U^\mu = (\gamma_v \mathbf{v}, \gamma_v c) \tag{4.34}$$

in the spatial-first ordering used here. Its invariant norm is

$$U_\mu U^\mu = c^2. \tag{4.35}$$

Thus every massive particle has a four-velocity of the same invariant magnitude even though its spatial velocity depends on the frame.

4.8 The Tangent to a Worldline

There is a familiar Euclidean model for Eq. (4.29). Along a curve in ordinary three-dimensional space,

$$ds^2 = dx^2 + dy^2 + dz^2. \tag{4.36}$$

The differential displacement (dx, dy, dz) points along the curve but is not a unit vector. Dividing by arc length gives

$$\mathbf{T} = \left(\frac{dx}{ds}, \frac{dy}{ds}, \frac{dz}{ds}\right), \qquad \mathbf{T} \cdot \mathbf{T} = 1. \tag{4.37}$$

The relativistic construction is the same idea with proper time in place of Euclidean arc length. The vector $u^\mu = dX^\mu / d\tau$ is tangent to the worldline.

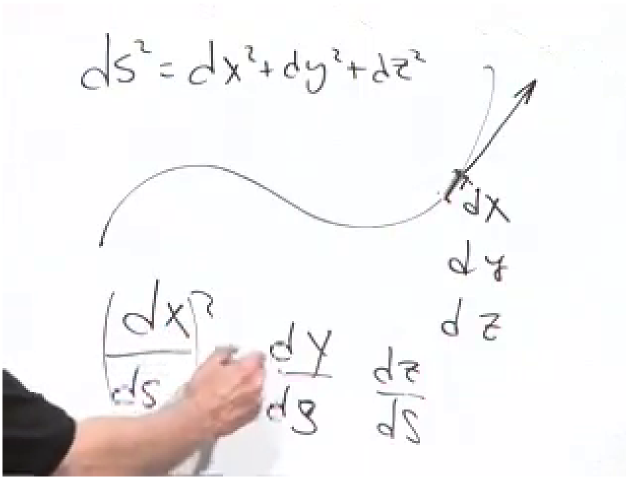


Figure 4.7: The Euclidean tangent-vector construction: divide the differential displacement by the arc-length element ds. Proper time plays the analogous role along a timelike worldline.

Lecture frame: 01:39:30.

Question from the audience. If $c = 1$, are proper time and proper distance the same?

Response. Setting $c = 1$ puts time and distance in the same units; it does not erase the minus signs of spacetime geometry. In the Euclidean analogy, s is ordinary distance measured along a spatial curve. Along a timelike worldline, τ is the elapsed time measured by a clock. The constructions are analogous, but the geometries are different.

Differentiating once more defines proper acceleration,

$$a^\mu = \frac{du^\mu}{d\tau}. \tag{4.38}$$

This is the quantity that must replace $d^2\mathbf{x}/dt^2$ when Newton's second law is made relativistic. Since $u_\mu u^\mu$ is constant,

$$\frac{d}{d\tau}(u_\mu u^\mu) = 2u_\mu a^\mu = 0, \tag{4.39}$$

so four-acceleration is Minkowski-orthogonal to four-velocity.[4]

4.9 Four-Momentum and Energy

Momentum is our final example because it is conserved in collisions and decays. The particle has a fixed intrinsic mass m, often called its rest mass. It does not vary with velocity. Relativistic momentum is that fixed mass times four-velocity:

$$p^\mu = mu^\mu. \tag{4.40}$$

In $c = 1$ units,

$$p^i = \gamma_v m v^i, \qquad p^0 = \gamma_v m. \tag{4.41}$$

At small velocity $\gamma_v \simeq 1$, so the spatial components reduce to the Newtonian momentum $m\mathbf{v}$. At high velocity they grow more rapidly than $m\mathbf{v}$.

The time component is energy. With conventional units and $X^0 = ct$,

$$P^\mu = (\mathbf{p}, E/c) = (\gamma_v m\mathbf{v}, \gamma_v mc), \tag{4.42}$$

[4]Editorial clarification: The orthogonality follows immediately from the invariant norm and is included as the next mathematical consequence of the definition.

and hence

$$E = \gamma_v mc^2 = mc^2 \left(1 - \frac{v^2}{c^2}\right)^{-1/2}. \tag{4.43}$$

To recover ordinary mechanics, let $\epsilon = v^2/c^2 \ll 1$. The binomial expansion gives

$$(1 - \epsilon)^p = 1 - p\epsilon + O(\epsilon^2). \tag{4.44}$$

Taking $p = -1/2$,

$$\begin{aligned} E &= mc^2 \left(1 - \frac{v^2}{c^2}\right)^{-1/2} \\ &= mc^2 \left[1 + \frac{1}{2}\frac{v^2}{c^2} + O\left(\frac{v^4}{c^4}\right)\right] \\ &= mc^2 + \frac{1}{2}mv^2 + O\left(\frac{mv^4}{c^2}\right). \end{aligned} \tag{4.45}$$

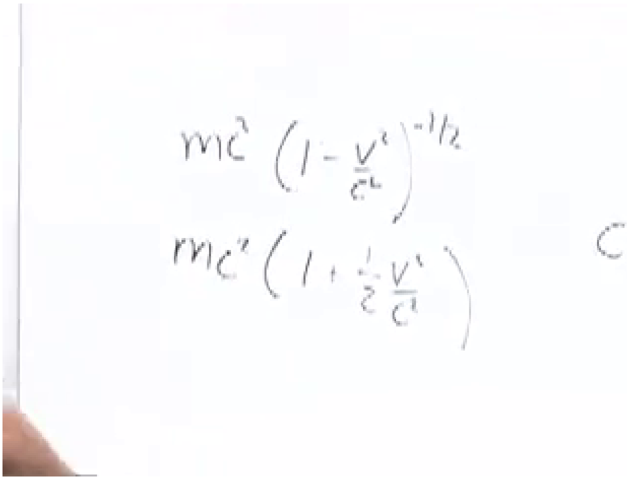


Figure 4.8: The low-velocity expansion of relativistic energy, revealing rest energy and Newtonian kinetic energy.

Lecture frame: 01:51:30.

The first term, mc^2, is present even at rest. In a Newtonian problem with a fixed collection of particles and unchanging masses, that term adds the same constant before and after the motion, so only $\frac{1}{2}mv^2$ affects the familiar bookkeeping. Relativity reveals that the discarded constant was physical energy all along.

Energy and momentum are therefore not unrelated conservation laws. They are the four components of one conserved vector. A Lorentz boost mixes them: what one observer calls energy contributes partly to another observer's momentum, and conversely. Nevertheless, for an isolated system the total four-momentum is the same before and after every collision or decay in every inertial frame.

This is the threshold of relativistic mechanics. The next tasks are to express force through proper acceleration and to understand relativistic wave equations, especially the equations governing electromagnetism and light.

CHAPTER 5

INVARIANT LAWS, FOUR-VECTORS, AND THE FIELD TENSOR

We shall begin with a law that nature could have obeyed but does not. Its failure will tell us how a physical law ought to be written. The same test first leads from coordinate components to ordinary vectors, then from ordinary vectors to four-vectors, and finally from four-vectors to tensors. By the end, the electric and magnetic fields will no longer be separate objects: together they will occupy the six independent slots of one antisymmetric tensor.

Unless stated otherwise, we use units in which $c = 1$, order spacetime components as $(1, 2, 3, 0) = (x, y, z, t)$, and use the metric signature $(-, -, -, +)$.

5.1 A possible law with the wrong symmetry

Fix an x- and a y-axis permanently in space. Now decree that two particles interact whenever their x-coordinates agree. The interaction might reverse their motion or send them away in new directions; its detailed dynamics does not matter. The proposed trigger is simply

$$x_1 - x_2 = 0, \tag{5.1}$$

where the subscripts label particles rather than coordinate components. There is no mathematical contradiction in this rule. The question is what makes it physically implausible.

Question from the audience. Does the rule violate the homogeneity of space?

Response. No. Translating the origin changes both coordinates by the same amount and leaves $x_1 - x_2$ unchanged. The rule specifies no preferred place. Its defect is isotropy: it singles out one preferred direction.

Rotate the axes through 90°. The old x-direction is now called y, so the same physical events would be described by equality of the y-coordinates. At a general angle the condition would involve a linear combination of x and y. The form of the law therefore depends on how the blackboard was oriented. It is translation invariant but not rotation invariant.

We usually give physical laws their simplest form in orthogonal coordinates. One may use more general coordinates, but then the metric and the formulas become less simple. Within the class of orthogonal Cartesian frames, no orientation should be privileged.

The repair is immediate. In two dimensions require both coordinates to agree; in three dimensions require all three:

$$x_1 = x_2, \qquad y_1 = y_2, \qquad z_1 = z_2. \tag{5.2}$$

The particles can then interact only when they reach the same point, even if they approach it along different lines.

Question from the audience. Must the z-coordinates and the times agree as well?

Response. In three-dimensional space the z-coordinates must also agree. Time is being suppressed for the moment; it will be restored as soon as the spatial argument is complete.

Let $\boldsymbol{r}_a$ be the position of particle a. The three component conditions become one vector equation,

$$\boldsymbol{r}_1 - \boldsymbol{r}_2 = \mathbf{0}. \tag{5.3}$$

A component can vanish in one rotated frame and fail to vanish in another. The whole vector is different: if every component is zero in one frame, every component is zero in every rotated frame. Likewise, if two vectors are equal in one frame, they remain equal after both are transformed.

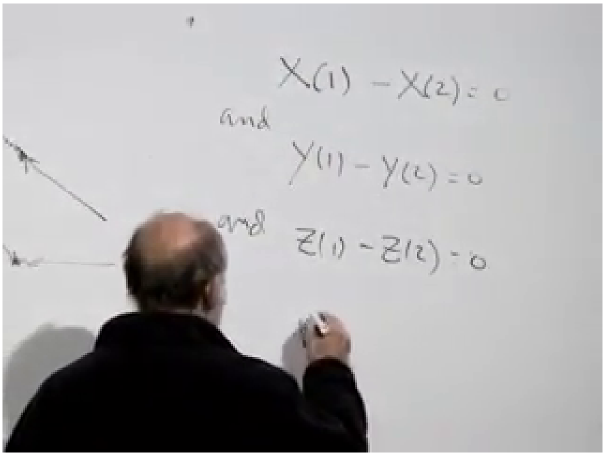


Figure 5.1: The three coordinate equalities that make collision mean spatial coincidence rather than coincidence along one preferred axis.

Lecture frame: 00:09:00.

This is why vector equations are useful. Their components change, but the object transforms coherently. A law written for the complete object can retain its form when the coordinates are rotated.

5.2 Coincidence in spacetime

Now replace the vertical spatial axis by time. A Newtonian rule might say that two changes of motion occur only at equal times. In Newtonian mechanics simultaneity is shared by all inertial observers, so such a rule is possible. In special relativity it selects a preferred frame. Events simultaneous in one frame need not be simultaneous in another.

Demanding equality only of the spatial coordinates is no better. A spacetime displacement can have vanishing spatial components and a nonzero time component,

$$\Delta X^{\mu} = (0, 0, 0, \Delta t), \tag{5.4}$$

and a boost mixes Δt with a spatial component. The invariant local statement is that the two changes of motion occur at the same event. Thus a simple collision requires

$$X_1^{\mu} = X_2^{\mu}, \qquad \text{or} \qquad X_1^{\mu} - X_2^{\mu} = 0. \tag{5.5}$$

All four coordinates agree: the particles occupy the same place at the same time.

The collision example gives a general rule for writing physics. We may put everything on one side and set a complete four-vector equal to zero, or set one complete four-vector equal to another. Equating disconnected pieces of a four-vector does not, by itself, give a Lorentz-invariant statement.

5.3 Boosts, rotations, and matrix notation

A four-vector is an object that transforms as a spacetime displacement does. To make that definition useful, first recall the transformations of the coordinates themselves. For a boost of speed v in the x-direction,

$$x' = \gamma(x - vt), \quad y' = y, \quad z' = z, \quad t' = \gamma(t - vx), \tag{5.6}$$

where

$$\gamma(v) = \frac{1}{\sqrt{1-v^2}}. \tag{5.7}$$

Question from the audience. Has the speed of light been set equal to one in these formulas?

Response. Yes. With units restored, take $X^0 = ct$, replace $1-v^2$ by $1-v^2/c^2$, and restore the corresponding factors of c in the mixed space–time terms.

A rotation about the z-axis mixes x and y, leaving z and t alone:

$$\begin{pmatrix} x' \\ y' \\ z' \\ t' \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta & 0 & 0 \\ -\sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix}. \tag{5.8}$$

The zeros are worth displaying: they show explicitly which components do not enter a given transformed coordinate. In the same ordering, the boost in Eq. (5.6) is

$$\begin{pmatrix} x' \\ y' \\ z' \\ t' \end{pmatrix} = \begin{pmatrix} \gamma & 0 & 0 & -\gamma v \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\gamma v & 0 & 0 & \gamma \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix}. \tag{5.9}$$

Rotations and boosts can be composed by matrix multiplication. To boost in a direction not aligned with the original x-axis, one may rotate that direction into the x-axis, apply the boost, and rotate as needed. Products of these elementary matrices generate more general Lorentz transformations.

Question from the audience. Does it matter whether one rotates first and then boosts, or boosts first and then rotates?

Response. Yes. After the first operation the axes on which the second one acts have changed. The two matrix products need not agree, so the order of operations matters.

All these linear transformations can be written at once as

$$X'^{\mu} = \Lambda^{\mu}{}_{\nu} X^{\nu}. \tag{5.10}$$

The repeated index ν is summed over $1, 2, 3, 0$. This is simply the usual row-by-column sum in matrix multiplication, expressed in index notation.

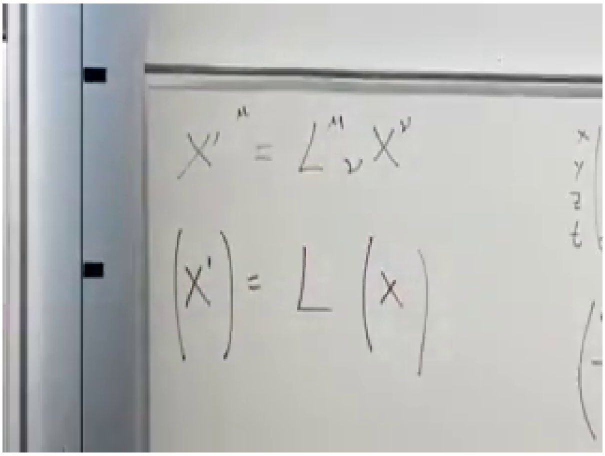


Figure 5.2: The indexed Lorentz transformation above the beginning of its column-vector form.

Lecture frame: 00:31:55.

Equation (5.10) is also the definition of a general contravariant four-vector. If A^μ is such a vector, then

$$A'^\mu = \Lambda^\mu{}_\nu A^\nu. \tag{5.11}$$

Its four components need not represent position. What makes the object a four-vector is its transformation law.

5.4 Scalars, the metric, and invariant products

Before using more four-vectors, it is helpful to step back to the simpler notion of a scalar. Under spatial rotations, the temperature at a specified point is a scalar, and so is the distance between two specified points. Each is one number, independent of the orientation of the axes. One could even make the deliberately silly numerical claim that a distance measured in centimetres equals the room temperature measured in degrees Fahrenheit. The claim has no physical depth, but because both sides are rotational scalars, its truth or falsity does not depend on how the axes are turned.

The components of a vector are not scalars. We therefore need a systematic way to combine them into scalars. Write a general contravariant four-vector as

$$A^\mu = (A^1, A^2, A^3, A^0). \tag{5.12}$$

The covariant components describe the same geometric object with a lower index. With our signature they are obtained using the Minkowski metric,

$$A_\mu = \eta_{\mu\nu} A^\nu, \qquad \eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{5.13}$$

so that

$$A_\mu = (-A^1, -A^2, -A^3, A^0). \tag{5.14}$$

Question from the audience. Are two indices and signs on the board inconsistent?

Response. Yes. The spatial component in the displayed column needs the corresponding minus sign, and the components on the right of the index-lowering equation carry upper indices before multiplication by $\eta_{\mu\nu}$. The time label is 0, not 4.

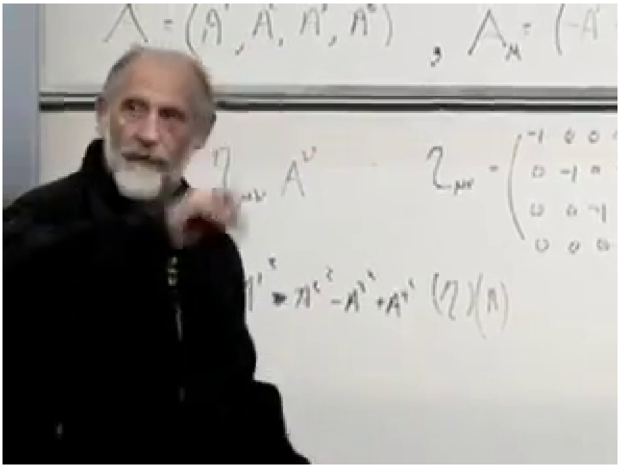


Figure 5.3: Contravariant and covariant components, the metric matrix, and the beginning of the invariant contraction.

Lecture frame: 00:45:00.

Index lowering is not decoration. It packages the signs required for the Lorentz-invariant norm:

$$A_\mu A^\mu = -(A^1)^2 - (A^2)^2 - (A^3)^2 + (A^0)^2, \quad (5.15)$$

$$X_\mu X^\mu = t^2 - x^2 - y^2 - z^2. \quad (5.16)$$

The second expression is the invariant spacetime interval from the origin. Because A^μ transforms in the same way as X^μ, its contraction has the same invariance.

The notation also teaches a useful bookkeeping rule. A repeated summation index appears once upstairs and once downstairs. For example,

$$A_\mu A^\mu = \eta_{\mu\nu} A^\mu A^\nu. \quad (5.17)$$

The names of summed, or dummy, indices may be changed without changing the quantity.

Question from the audience. Should one of these vectors be regarded as a row and the other as a column, and does their order matter?

Response. For the present calculation, regard the indexed quantities as components and let the metric perform the contraction; there is no need to turn the discussion into row-versus-column operator algebra. These are ordinary numbers, so their order does not matter. Noncommuting operators would require more care, but that is not the subject here.

The same construction gives an inner product between two different four-vectors,

$$A \cdot B = A_\mu B^\mu = A^\mu B_\mu. \tag{5.18}$$

Its invariance follows without writing an explicit Lorentz matrix. Since $A + B$ and $A - B$ are four-vectors, both of their squared norms are invariant:

$$(A + B)^2 = A^2 + B^2 + 2A \cdot B, \tag{5.19}$$

$$(A - B)^2 = A^2 + B^2 - 2A \cdot B. \tag{5.20}$$

Subtracting gives

$$(A + B)^2 - (A - B)^2 = 4A \cdot B, \tag{5.21}$$

so $A \cdot B$ is invariant as well. The factor of four follows directly from the expansion.[1]

[1]Editorial clarification: The board argument supplies this polarization identity; writing out the factor makes the spoken cancellation unambiguous.

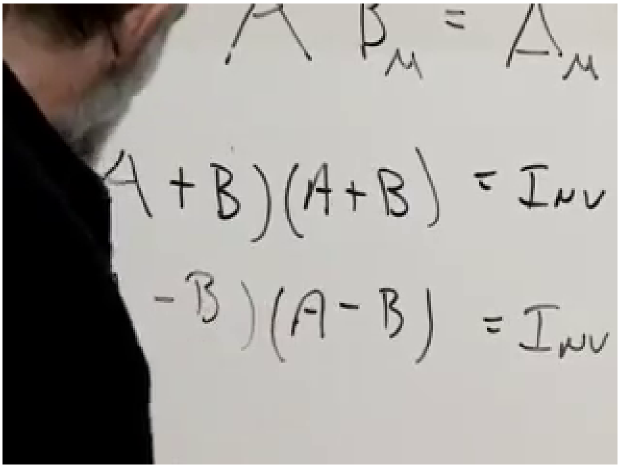


Figure 5.4: The two invariant squares used to isolate the inner product of different four-vectors.

Lecture frame: 00:52:20.

Invariant quantities are natural ingredients of physical laws: unlike an isolated component, they carry the same value from frame to frame.

5.5 Proper velocity and four-momentum

Consider two neighboring events on a particle's worldline. Their separation is the infinitesimal four-vector

$$dX^\mu = (dx, dy, dz, dt). \tag{5.22}$$

For a timelike worldline, proper time is defined by

$$d\tau^2 = dX_\mu dX^\mu = dt^2 - dx^2 - dy^2 - dz^2. \tag{5.23}$$

The proper four-velocity is

$$U^\mu = \frac{dX^\mu}{d\tau}. \tag{5.24}$$

It measures the rate of change of spacetime position with respect to the particle's own proper time, not with respect to the coordinate time of a chosen frame.

Although U^μ has four components, only three are independent. Dividing Eq. (5.23) by $d\tau^2$ gives

$$U_\mu U^\mu = 1. \tag{5.25}$$

Thus the four-velocity is a unit timelike vector. The future-directed choice fixes the sign of its time component.

Question from the audience. At speeds much smaller than c, does four-velocity reduce to ordinary velocity?

Response. Its three spatial components approach ordinary velocity, while its time component approaches one in $c = 1$ units. The relation follows by applying the chain rule and computing $dt/d\tau$.

For each spatial component,

$$U^i = \frac{dx^i}{d\tau} = \frac{dx^i}{dt}\frac{dt}{d\tau} = v^i\frac{dt}{d\tau}. \tag{5.26}$$

Divide Eq. (5.23) by dt^2:

$$\left(\frac{d\tau}{dt}\right)^2 = 1 - v_x^2 - v_y^2 - v_z^2 = 1 - \boldsymbol{v}^2. \tag{5.27}$$

Therefore

$$\frac{dt}{d\tau} = \frac{1}{\sqrt{1-\boldsymbol{v}^2}} = \gamma(|\boldsymbol{v}|), \tag{5.28}$$

and

$$U^\mu = (\gamma v_x, \gamma v_y, \gamma v_z, \gamma). \tag{5.29}$$

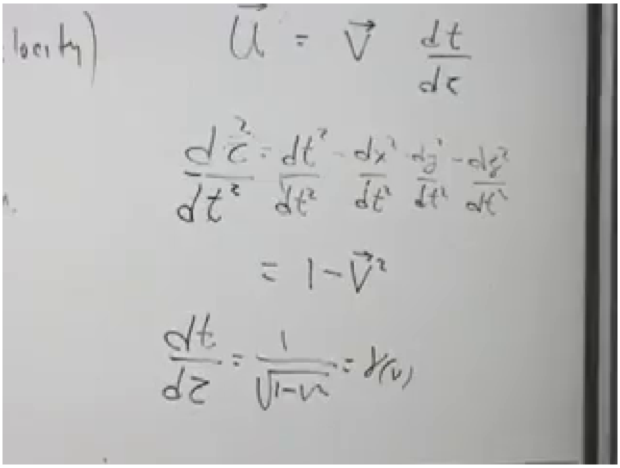


Figure 5.5: The proper-time calculation that produces $dt/d\tau = \gamma$ and relates proper velocity to ordinary velocity.

Lecture frame: 01:02:00.

Question from the audience. Does dx^2/dt^2 mean a second derivative, and should squaring a vector produce cross terms?

Response. Here it means $(dx/dt)^2$, the square of a first derivative. Moreover, $\boldsymbol{v}^2$ means the Euclidean dot product $v_x^2 + v_y^2 + v_z^2$, not the square of the formal sum $v_x + v_y + v_z$; there are no cross terms.

Question from the audience. Is the v in $\gamma(v)$ the relative speed of two coordinate frames or the speed of the particle?

Response. Here it is the particle's speed in the chosen frame. The function $\gamma(v) = 1/\sqrt{1 - v^2}$ can be evaluated for any speed: in a boost it is evaluated at the relative frame speed; in four-velocity it is evaluated at the particle speed.

For $v \ll 1$, $\gamma \simeq 1$, so the spatial components of U^μ approach $\boldsymbol{v}$ and U^0 approaches one. As the ordinary speed approaches the speed of light, γ and all nonzero components of the four-velocity grow without bound.

Question from the audience. Should not U^0 have the units of a velocity rather than being close to the number one?

Response. The units disappeared when c was set to one. With $X^0 = ct$, one has $U^\mu = (\gamma \boldsymbol{v}, \gamma c)$, so the time component approaches c and $U_\mu U^\mu = c^2$.

For a particle of rest mass m, define four-momentum by

$$P^\mu = m U^\mu. \tag{5.30}$$

In natural units,

$$P^\mu = (\gamma m \boldsymbol{v}, \gamma m). \tag{5.31}$$

The first three components approach the Newtonian momentum $m\boldsymbol{v}$ at low speed. Restoring units gives

$$P^\mu = (\gamma m \boldsymbol{v}, E/c), \qquad E = \gamma m c^2, \tag{5.32}$$

and hence

$$E = mc^2 + \frac{1}{2}m\boldsymbol{v}^2 + O(mv^4/c^2). \qquad (5.33)$$

The time component is energy in $c = 1$ units, or energy divided by c when all four components are assigned momentum units.

Question from the audience. Is ordinary $\boldsymbol{v}$, with a fourth entry appended, itself a four-vector?

Response. No. Ordinary three-velocity does not transform as a four-vector. The factor γ and the proper-time definition are essential; the four-vector is U^μ, not $\boldsymbol{v}$.

5.6 Why electric and magnetic fields must mix

We now take what was called a short vacation from relativity. Start with the pre-Einstein equation of motion and ask how it must be reorganized before it can be valid in every inertial frame. Ordinary acceleration is the time derivative of ordinary velocity. Its four-dimensional analogue is naturally

$$A^\mu = \frac{dU^\mu}{d\tau}. \qquad (5.34)$$

At low speed its spatial components approach ordinary acceleration. The full relativistic equation will be postponed; first inspect the old force law.

For a charge q in electric and magnetic fields,

$$\boldsymbol{F} = m\boldsymbol{a} = q\left(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}\right). \qquad (5.35)$$

The electric term is independent of the particle velocity, whereas the magnetic term is not.

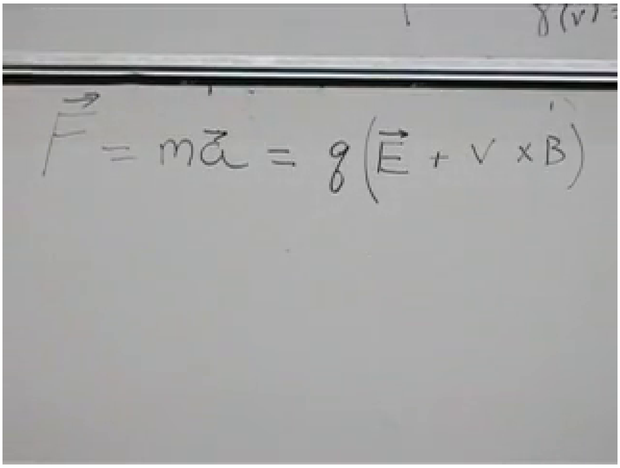


Figure 5.6: The pre-relativistic form of the Lorentz force law that motivates combining $\boldsymbol{E}$ and $\boldsymbol{B}$.

Lecture frame: 01:18:20.

Suppose an observer sees no electric field but does see a magnetic field and a moving charge. The charge feels a magnetic force. In the instantaneous rest frame of the charge, however, $\boldsymbol{v} = 0$, so the magnetic term in Eq. (5.35) vanishes. The physical worldline has not become inertial merely because the coordinates changed; the force in the new frame must therefore include an electric contribution. More carefully, a nonzero four-acceleration cannot be transformed to zero by changing inertial frames, even though the three-acceleration and its components do transform. [2]

[2]Editorial clarification: This four-acceleration statement

Electric and magnetic fields must consequently transform into one another. Together they have six components, so they are neither one scalar nor one four-vector. We need a new transforming object.

5.7 Tensors: one transformation per index

A scalar may be called a tensor of rank zero; it has no indices. A four-vector is a tensor of rank one; it has one index with four possible values. The simplest rank-two example is a product

$$A^{\mu}B^{\nu}. \tag{5.36}$$

Each index has four values, so the object has $4^2 = 16$ components, which may be displayed as a 4×4 array. A product of three vectors,

$$A^{\mu}B^{\nu}C^{\sigma}, \tag{5.37}$$

has $4^3 = 64$ components and is a rank-three tensor. Higher ranks continue the pattern. The name is less important than the rule that defines the object.

If

$$A'^{\mu} = \Lambda^{\mu}{}_{\nu}A^{\nu}, \qquad B'^{\sigma} = \Lambda^{\sigma}{}_{\tau}B^{\tau}, \tag{5.38}$$

then their product transforms as

$$A'^{\mu}B'^{\sigma} = \Lambda^{\mu}{}_{\nu}\Lambda^{\sigma}{}_{\tau}A^{\nu}B^{\tau}. \tag{5.39}$$

This gives the defining law for a contravariant rank-two tensor:

$$T'^{\mu\sigma} = \Lambda^{\mu}{}_{\nu}\Lambda^{\sigma}{}_{\tau}T^{\nu\tau}. \tag{5.40}$$

---

is the invariant form of the classroom argument and avoids treating ordinary three-acceleration itself as frame invariant.

There is one Lorentz matrix for each index. The repeated ν and τ are summed; the open μ and σ label the component being calculated in the primed frame.

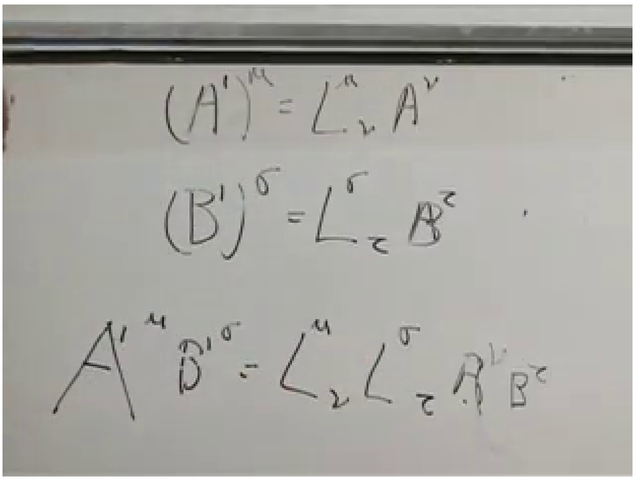


Figure 5.7: The transformation of two vectors and the resulting two-matrix transformation law for a rank-two tensor.

Lecture frame: 01:27:15.

A tensor need not be a single product of vectors. Both $A^\mu B^\nu$ and $A^\nu B^\mu$ obey the tensor transformation law, as do their sum and any sum such as

$$A^\mu B^\nu + C^\mu D^\nu. \tag{5.41}$$

The transformation law, not a particular factorization, is the definition. A third-rank tensor acquires a third Lorentz matrix, and so on.

5.8 Symmetric and antisymmetric tensors

Two classes of rank-two tensors will be especially useful. A symmetric tensor satisfies

$$T^{\mu\nu} = T^{\nu\mu}, \tag{5.42}$$

so entries reflected across the diagonal agree. Its four diagonal entries and six entries on one side of the diagonal determine the rest, giving

$$4 + 6 = 10 \tag{5.43}$$

independent components.

An antisymmetric tensor satisfies

$$T^{\mu\nu} = -T^{\nu\mu}. \tag{5.44}$$

Putting $\mu = \nu$ in Eq. (5.44) gives $T^{\mu\mu} = -T^{\mu\mu}$, so every diagonal entry vanishes. The six entries in one triangular half determine the other six by a sign change. There are therefore exactly six independent components.

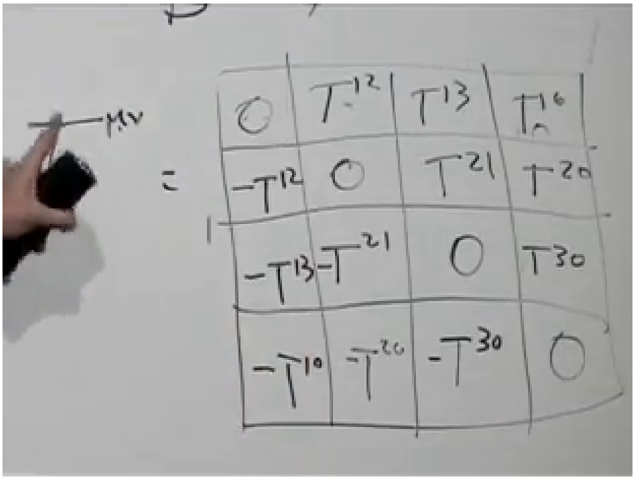


Figure 5.8: A completed antisymmetric 4 × 4 array: zeros on the diagonal and opposite signs across it.
Lecture frame: 01:34:25.

That count is the clue. The three electric components and three magnetic components can fit into one antisymmetric rank-two tensor.

5.9 The electromagnetic field tensor

Let the spacetime labels run across rows and columns in the order $(1, 2, 3, 0) = (x, y, z, t)$. The arrangement written on the board is

$$F_{\mu\nu} = \begin{pmatrix} 0 & -B_3 & B_2 & E_1 \\ B_3 & 0 & -B_1 & E_2 \\ -B_2 & B_1 & 0 & E_3 \\ -E_1 & -E_2 & -E_3 & 0 \end{pmatrix}. \tag{5.45}$$

The purely spatial slots contain the magnetic field; the slots carrying one spatial and one time index contain the electric field. Antisymmetry supplies the lower triangular entries once the upper triangular entries are known.

Question from the audience. Are the magnetic entries all negative, and do the labels correspond to x, y, z?

Response. The signs in the upper spatial triangle alternate as shown: $-B_3, +B_2, -B_1$. The labels $1, 2, 3$ correspond to x, y, z, while 0 labels time. Reflection across the diagonal reverses every sign.

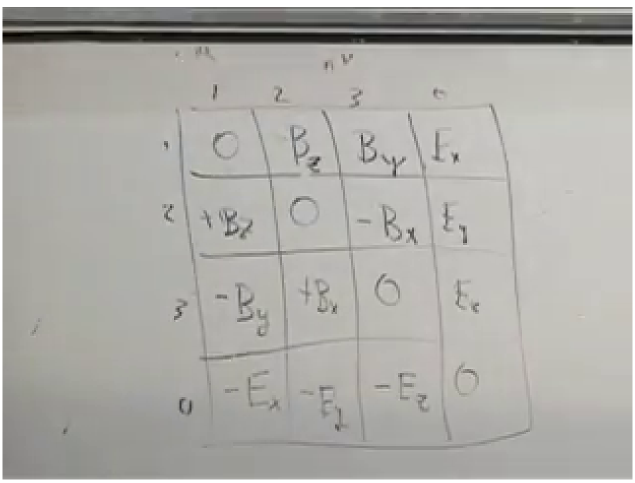


Figure 5.9: The completed electromagnetic field tensor in the lecture's spatial-first component ordering.

Lecture frame: 01:41:05.

The notation is conventionally $F_{\mu\nu}$, rather than a generic $T_{\mu\nu}$. The lecturer conjectures that the letter might refer to Faraday, while explicitly saying that he does not know the history of the notation. What matters here is its antisymmetry and its six field components, not that conjecture about the letter.

Question from the audience. Does each matrix entry represent the strength of a field in the indicated coordinate plane?

Response. It is a component of the electromagnetic field tensor. The entries involving time are electric components, and the spatial–spatial entries encode magnetic components.

Fence off the row and column carrying the time index. The remaining 3×3 antisymmetric block is magnetic. Its independent entries may be remembered cyclically:

$$F_{12} = -B_3, \qquad F_{23} = -B_1, \qquad F_{31} = -B_2. \tag{5.46}$$

Starting from the first relation, advance every label around $1 \to 2 \to 3 \to 1$. Equivalently,

$$F_{ij} = -\epsilon_{ijk} B_k, \tag{5.47}$$

with $\epsilon_{123} = +1$.[3]

[3]Editorial clarification: The Levi–Civita formula is the compact standard form of the cyclic rule developed explicitly on the board.

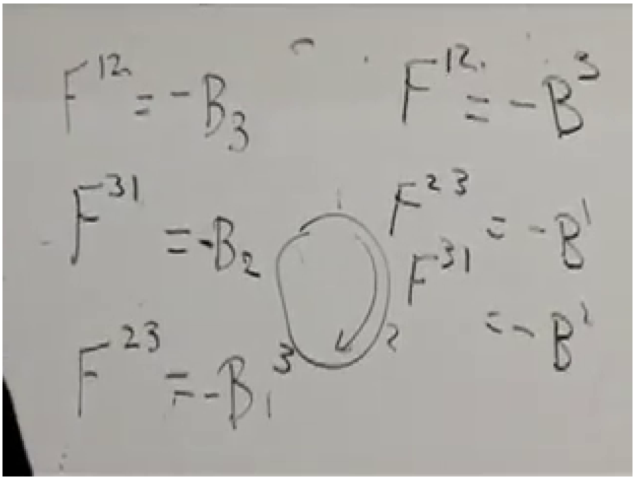


Figure 5.10: The cyclic spatial components that encode the magnetic field in the antisymmetric tensor.

Lecture frame: 01:46:20.

The electric identifications are more direct:

$$E_1 = F_{10}, \qquad E_2 = F_{20}, \qquad E_3 = F_{30}. \tag{5.48}$$

The magnetic field consequently has two useful identities. Within three-dimensional space, a three-component axial vector can be paired with a 3×3 antisymmetric tensor. In spacetime, however, it is only the spatial part of the six-component tensor $F_{\mu\nu}$. This pairing is special to three dimensions: an antisymmetric 4×4 tensor has six independent components and cannot be identified with a four-vector. [4]

[4]Editorial clarification: Under proper rotations the magnetic

The purpose of this construction is not component counting for its own sake. Because $F_{\mu\nu}$ transforms as a tensor, a Lorentz boost automatically mixes its electric and magnetic entries. In the next lecture it can be combined with four-velocity and four-acceleration to replace Eq. (5.35) by a covariant equation of motion.

5.10 Questions about Einstein and Minkowski

The final minutes turn from the formalism to how it arose historically. The answers are offered as recollections rather than a settled history, and that qualification is part of the exchange.

field behaves as a vector; under reflections it is more precisely an axial vector, or pseudovector.

Question from the audience. Was Einstein seeking a Lorentz-invariant formulation on conceptual grounds, or was he driven by particular observations?

Response. The principle that the laws of nature should have the same form in every inertial frame was central to his reasoning. There was also experimental evidence concerning light and the failure to detect motion through an ether, although the lecturer is explicitly uncertain about which results Einstein knew or how strongly they influenced him. The conceptual pressure was that Maxwell's equations and the Lorentz force should obey the relativity principle, which requires electric fields, magnetic fields, and velocity components to transform together.

Einstein's 1905 presentation did not use the compact four-vector notation used here. It worked component by component, with the necessary factors of γ and velocity written explicitly. Minkowski subsequently organized the same structure geometrically as four-dimensional spacetime.

Question from the audience. Did Einstein resist Minkowski's spacetime formulation, and how soon did Minkowski introduce it?

Response. The recollection offered in class is that Einstein did not initially welcome the four-dimensional geometric language but adopted its value quickly. The timing is given tentatively as roughly two years after the 1905 paper, perhaps 1907; the speaker stresses that he is recalling rather than establishing the history.

Question from the audience. Was Minkowski a mathematician or a physicist, what else did he contribute, and was he influenced by Lorentz or FitzGerald?

Response. No confident answer is claimed. The distinction between mathematician and physicist was not pursued, an audience correction interrupts a broad remark about Minkowski's other work, and the possible lines of influence are left expressly uncertain. The secure point for this discussion is Minkowski's decisive organization of relativity in four-dimensional geometric notation.

Question from the audience. Were the transformations that mix $\boldsymbol{E}$ and $\boldsymbol{B}$ already present in Einstein's 1905 work?

Response. Yes, but written component by component rather than as one field-tensor equation. The comparison made in class is with Maxwell's component equations: the content can be present before the compact notation that reveals its common structure. Minkowski's contribution was to make that structure systematic in spacetime language.

That returns us to the mathematical point with which we began. Components are indispensable for calculation, but the law becomes intelligible when those components are recognized as parts of one transforming object. The next task is to use $F_{\mu\nu}$ to write the charged particle's equation of motion in a form that is the same in every inertial frame.

CHAPTER 6

CHARGED PARTICLES, FIELD TRANSFORMATIONS, AND RELATIVISTIC WAVES

Last time we began asking how a charged particle moves in electric and magnetic fields once the principles of relativity are imposed. The familiar Lorentz force law already contains the clue: its magnetic term depends on the particle's velocity, so different observers cannot agree separately about what is electric and what is magnetic. Our task is to formulate the law so that every inertial observer uses the same equation, even though the fields and the particle velocity change from frame to frame.

Unless stated otherwise, we use units in which $c = 1$ and the metric signature $(-, -, -, +)$. Spacetime components are ordered with the three spatial entries first:

$$\mu = (1, 2, 3, 0) = (x, y, z, t). \tag{6.1}$$

Latin indices m, n run only over the three spatial directions.

6.1 What survives of Newton's force law

In Newtonian mechanics,

$$\boldsymbol{F} = m\boldsymbol{a} \tag{6.2}$$

has the same form in every inertial frame because Galilean observers agree on acceleration. They also assign the same mass and force to the particle. Once the kinematics changes, there is no reason to expect this particular relation between force and ordinary acceleration to survive unchanged.

Electromagnetism and gravitation already warn us that relativistic extensions need not resemble one another. Coulomb's law and Newton's law of gravity are both inverse-square laws, but gravitation cannot be incorporated naturally into special relativity without a deeper change in spacetime geometry. Electromagnetism can. Indeed, magnetism is itself a relativistic effect: when factors of c are restored, a magnetic force carries a factor of velocity divided by c, and it disappears in the formal limit $c \to \infty$.

The durable part of Newton's law is the statement that force is the rate of change of momentum. For constant rest mass,

$$\boldsymbol{F} = m\frac{d\boldsymbol{v}}{dt} = \frac{d}{dt}(m\boldsymbol{v}) = \frac{d\boldsymbol{p}}{dt}. \tag{6.3}$$

Relativity will alter the relation between momentum and velocity, but it will preserve this way of organizing the equation.

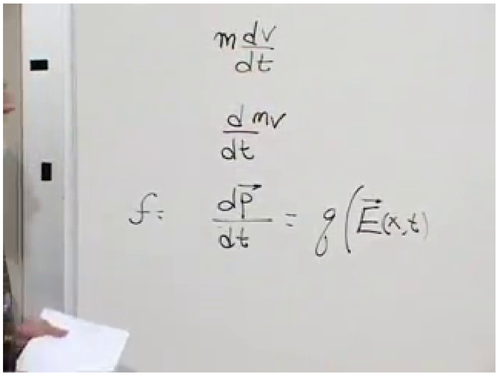


Figure 6.1: Force rewritten as $d\boldsymbol{p}/dt$ while the electric part of the Lorentz law is being placed on the board.

Lecture frame: 00:08:20.

In units $c = 1$, the pre-relativistic Lorentz force law is

$$\frac{d\boldsymbol{p}}{dt} = q\Big(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}\Big). \tag{6.4}$$

Both fields may depend on position and time. The nonrelativistic limit is now the limit $v \ll 1$, not a limit in which the numerical constant $c = 1$ somehow changes. The electric term in (6.4) is already an ordinary vector. To understand the magnetic term in a form that can be carried into spacetime, we must look again at the cross product.

6.2 The cross product as an antisymmetric tensor

Take two three-dimensional vectors $\boldsymbol{A}$ and $\boldsymbol{B}$. Their outer product

$$T_{mn} = A_m B_n \tag{6.5}$$

is a rank-two tensor with nine components. It has no symmetry in general. Antisymmetrizing gives

$$K_{mn} = A_m B_n - A_n B_m, \qquad K_{mn} = -K_{nm}. \quad (6.6)$$

Its diagonal entries vanish, while every entry below the diagonal is fixed by the corresponding entry above it. A 3×3 antisymmetric tensor therefore has only three independent components.

That count is suggestive but not sufficient by itself. Three quantities do not automatically transform as a vector. In three dimensions, however, the three independent entries of (6.6) can be identified with the components of the cross product $\boldsymbol{C} = \boldsymbol{A} \times \boldsymbol{B}$. The board first makes a temporary placement of C_2 in the upper-right position:

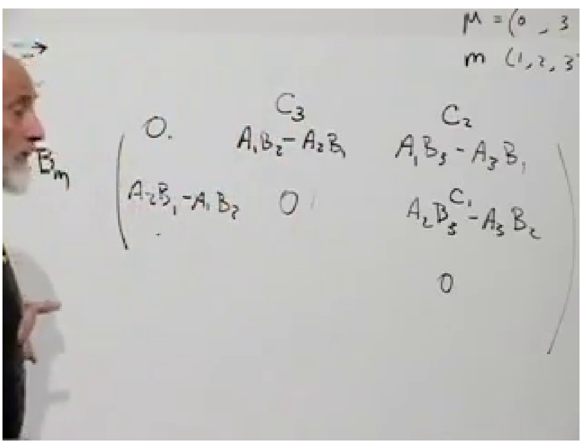


Figure 6.2: The antisymmetric product while its three independent entries are being matched to the components of $\boldsymbol{C}$. The upper-right label is still provisional here.

Lecture frame: 00:17:33.

Cycling $1 \to 2 \to 3 \to 1$ gives the component formulas

$$C_3 = A_1 B_2 - A_2 B_1, \tag{6.7}$$
$$C_1 = A_2 B_3 - A_3 B_2, \tag{6.8}$$
$$C_2 = A_3 B_1 - A_1 B_3. \tag{6.9}$$

The last line shows that the upper-right entry is $-C_2$, not C_2. The correction is part of the argument rather than a detail to hide.

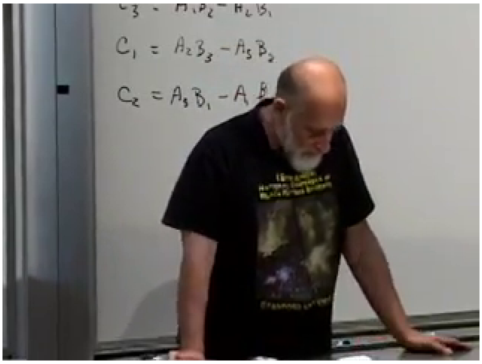


Figure 6.3: The completed cyclic component identities that fix the temporary sign in the antisymmetric array.

Lecture frame: 00:21:54.

The resulting matrix is

$$K_{mn} = \begin{pmatrix} 0 & C_3 & -C_2 \\ -C_3 & 0 & C_1 \\ C_2 & -C_1 & 0 \end{pmatrix}. \tag{6.10}$$

There is a useful geometry behind this notation. A component with indices $1, 2$ belongs naturally to the xy-plane and is identified with a vector component along the perpendicular z-axis. One may describe the same oriented object either by a plane or by an arrow normal to that plane. The arrow description is special to three dimensions; in higher dimensions an antisymmetric rank-two tensor still exists, but it no longer has the same number of components as a vector. Thus the tensor, not the perpendicular arrow, is the form of the cross product that generalizes.

Equivalently, after choosing an orientation,

$$C_k = \frac{1}{2}\epsilon_{kmn}K_{mn}. \tag{6.11}$$

The need to choose an orientation is the source of the right-hand rule and of the familiar distinction between polar and axial vectors. [1]

6.3 The magnetic field in tensor form

The same dual description applies to the magnetic field. We define its antisymmetric spatial tensor by

$$B_{mn} = \begin{pmatrix} 0 & -B_3 & B_2 \\ B_3 & 0 & -B_1 \\ -B_2 & B_1 & 0 \end{pmatrix}. \tag{6.12}$$

The extra overall sign relative to (6.10) is a convention. Once it has been chosen, it must be used

[1] Editorial clarification: The Levi–Civita formula and the axial-vector terminology are standard notation for the plane-versus-normal geometry developed explicitly in the lecture.

consistently; changing the sign of all three vector components would describe the same tensor-to-vector identification with the opposite orientation convention.

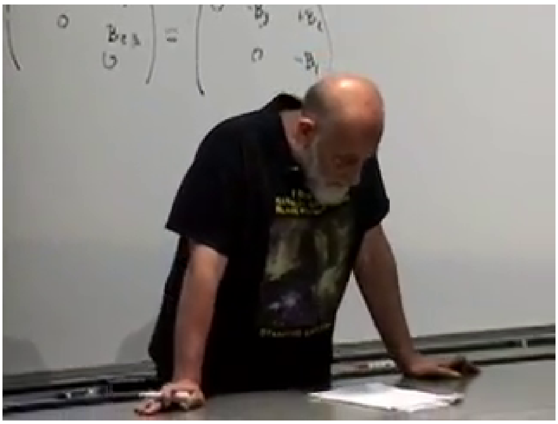


Figure 6.4: The magnetic field written both as an antisymmetric spatial array and in terms of its three vector components.

Lecture frame: 00:24:40.

Now take the third component of the cross product:

$$
\begin{aligned}
(\boldsymbol{v} \times \boldsymbol{B})_3 &= v_1 B_2 - v_2 B_1 && (6.13)\\
&= v_1 B_{13} + v_2 B_{23} + v_3 B_{33}. && (6.14)
\end{aligned}
$$

The last term may be added because $B_{33} = 0$. Cycling the free component gives the compact rule

$$(\boldsymbol{v} \times \boldsymbol{B})_m = v_n B_{nm}, \tag{6.15}$$

where the repeated index n is summed. The magnetic force is therefore

$$F_m^{(B)} = q v_n B_{nm}. \tag{6.16}$$

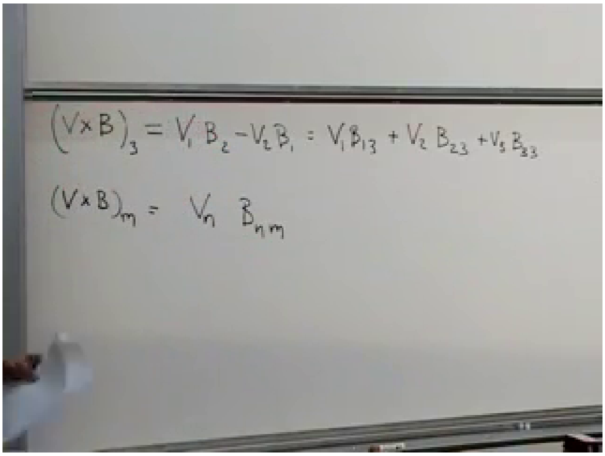


Figure 6.5: The third component of $\boldsymbol{v} \times \boldsymbol{B}$ and its general expression as a contraction with B_{nm}.
Lecture frame: 00:28:30.

Question from the audience. Does the free index m label a separate magnetic component of the force?

Response. No. It labels whichever spatial component of the force is being calculated. The electric field has simply been set to zero for the present argument, and every index here runs only over the three spatial directions.

The tensor notation makes the no-work property of a magnetic field nearly automatic. Differentiate the squared speed:

$$\frac{d}{dt}(v_m v_m) = 2\dot{v}_m v_m \tag{6.17}$$
$$\propto 2(\boldsymbol{v} \times \boldsymbol{B})_m v_m \tag{6.18}$$
$$= 2v_n B_{nm} v_m. \tag{6.19}$$

Terms with $m = n$ vanish because the diagonal of B_{nm} is zero. Every off-diagonal term has a partner with m and n interchanged; the two cancel because $v_n v_m$ is symmetric while B_{nm} is antisymmetric. Hence

$$\frac{d}{dt}(v_m v_m) = 0. \tag{6.20}$$

A pure magnetic force bends a trajectory without changing its speed or kinetic energy.

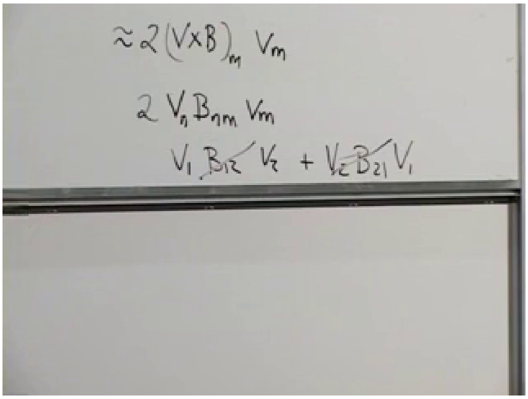


Figure 6.6: The paired terms in $v_n B_{nm} v_m$ cancel because the magnetic tensor changes sign when its indices are interchanged.

Lecture frame: 00:35:40.

This proof is intentionally more elaborate than saying that a cross product is perpendicular to its first factor. The latter statement is peculiar to the three-dimensional vector representation. The contraction of a symmetric pair with an antisymmetric tensor is the argument that will survive in four dimensions.

Question from the audience. Must the magnetic field be time independent for the speed to remain constant?

Response. The algebra above never differentiates the field and does not require $\partial_t \boldsymbol{B} = 0$. A magnetic term by itself still does no work. In electrodynamics, however, a changing magnetic field produces an electric field, and that electric field can change the particle's speed.

6.4 Why electric and magnetic fields must mix

The mixing of $\boldsymbol{E}$ and $\boldsymbol{B}$ can be seen before any four-dimensional calculation. Consider slow motion, so that ordinary Newtonian statements about acceleration are adequate. Let a magnetic field point along the y-axis and let a charged particle move along the x-axis. The magnetic force points along the z-axis:

$$\boldsymbol{F} = q\boldsymbol{v} \times \boldsymbol{B}. \tag{6.21}$$

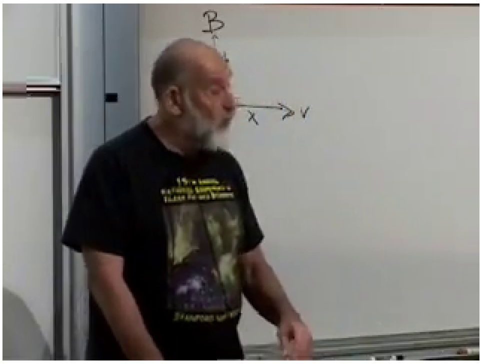


Figure 6.7: The board sketch for a particle moving across a magnetic field; the three directions are used to compare the laboratory and co-moving descriptions.

Lecture frame: 00:42:20.

The geometry may be redrawn without the perspective of the classroom:

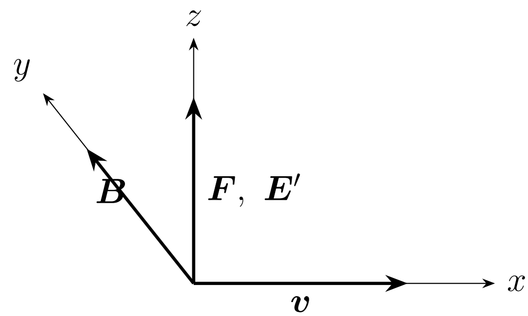


Figure 6.8: In the laboratory the z-directed force is magnetic; in the instantaneous co-moving frame the same force must be electric.

Now introduce an observer moving with the particle at that instant. The particle's velocity in the new frame is zero, so the magnetic term $q\boldsymbol{v}' \times \boldsymbol{B}'$ vanishes. Yet in the slow-motion limit both observers must agree that the particle has a z-acceleration. If the Lorentz force law has the same form in both frames, the co-moving observer must attribute the force to an electric field $\boldsymbol{E}'$ along z. A field that was purely magnetic in the laboratory is therefore not purely magnetic in the moving frame.

This is the physical reason to seek one object whose components include both fields. A change of frame does not merely rotate $\boldsymbol{E}$ among its own three components and $\boldsymbol{B}$ among its own three components; it mixes the two sets.

6.5 The electromagnetic field tensor

Together the electric and magnetic fields have six components. A four-dimensional antisymmetric tensor also has six independent components, since

$$\frac{4(4-1)}{2} = 6. \tag{6.22}$$

This does not mean that an antisymmetric tensor is literally the only collection of six quantities one could invent: six scalars, or a vector and two scalars, would also contain six numbers. It is the natural geometrical object whose components can mix in the required way.

To keep the board convention explicit, we display the contravariant tensor in the spatial-first order

$(1, 2, 3, 0)$:

$$F^{\mu\nu} = \begin{pmatrix} 0 & -B_3 & B_2 & E_1 \\ B_3 & 0 & -B_1 & E_2 \\ -B_2 & B_1 & 0 & E_3 \\ -E_1 & -E_2 & -E_3 & 0 \end{pmatrix}. \tag{6.23}$$

The upper-left 3×3 block is precisely the magnetic tensor (6.12); components with one spatial and one time index contain the electric field. Antisymmetry supplies the remaining entries.

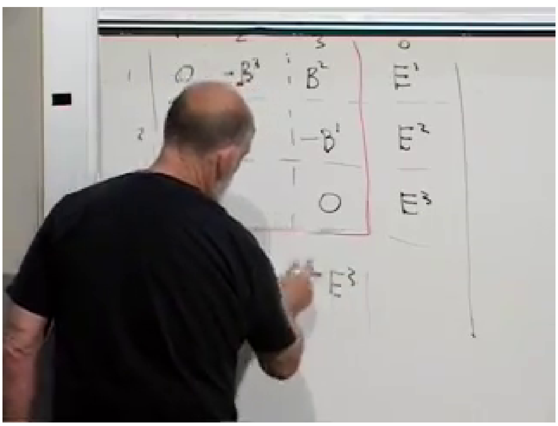


Figure 6.9: The magnetic spatial block and electric time-space column being assembled into the four-dimensional field tensor.

Lecture frame: 00:46:35.

Question from the audience. When should these field components carry upper rather than lower indices?

Response. For the three spatial labels alone, the board sometimes suppresses the distinction while identifying the familiar Maxwell–Faraday fields. In a covariant calculation the distinction must be restored: raising or lowering a spatial index changes a sign in the chosen metric convention.

What is invariant is not a particular arrangement of signs on the board, but the antisymmetric spacetime tensor together with a consistent metric and index convention. The notation F may suggest Faraday, although the lecture does not claim that historical attribution with confidence. Einstein worked out the transformation of electric and magnetic components in 1905; the compact four-dimensional tensor language came with the later spacetime formulation associated with Minkowski.

6.6 Four-velocity and four-force

The particle side of the equation must also be made covariant. Proper time τ is invariant, so differentiating the position four-vector with respect to it gives another four-vector:

$$u^\mu = \frac{dx^\mu}{d\tau} = \gamma(v_x, v_y, v_z, 1), \qquad \gamma = \frac{1}{\sqrt{1 - v^2}}. \tag{6.24}$$

The last component follows from $dt/d\tau = \gamma$. The four-momentum is

$$p^\mu = mu^\mu, \tag{6.25}$$

whose spatial part is the relativistic momentum $\boldsymbol{p} = m\gamma\boldsymbol{v}$.

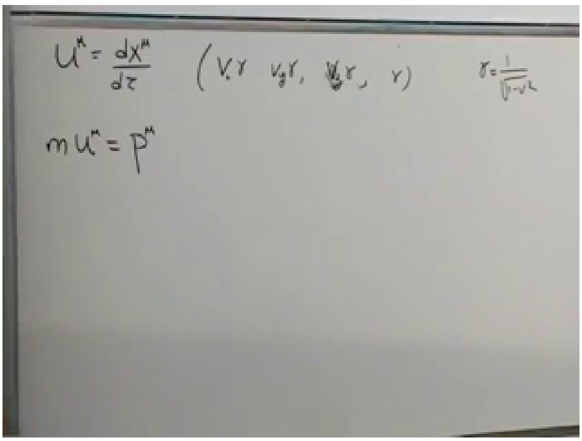


Figure 6.10: Four-velocity, its components, the Lorentz factor, and four-momentum.
Lecture frame: 00:50:30.

Because p^μ is a four-vector and τ a scalar, the natural relativistic left-hand side is

$$f^\mu \equiv \frac{dp^\mu}{d\tau}. \tag{6.26}$$

The four-force cannot be chosen arbitrarily. Four-velocity has fixed norm,

$$u_\mu u^\mu = 1. \tag{6.27}$$

Differentiating along the world line gives

$$0 = \frac{d}{d\tau}(u_\mu u^\mu) \tag{6.28}$$
$$= \frac{du_\mu}{d\tau}u^\mu + u_\mu\frac{du^\mu}{d\tau} \tag{6.29}$$
$$= 2u_\mu\frac{du^\mu}{d\tau}. \tag{6.30}$$

For constant mass this implies

$$f^\mu u_\mu = 0. \tag{6.31}$$

Thus four-force is Minkowski-orthogonal to four-velocity. This is a purely kinematic constraint following from the normalization of u^μ, not a special property of electromagnetism.

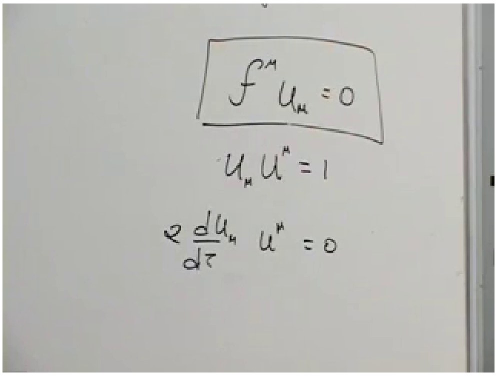


Figure 6.11: The unit norm of four-velocity and the resulting orthogonality of four-force and four-velocity.

Lecture frame: 00:56:50.

The three-dimensional magnetic calculation has already supplied the trick for satisfying (6.31).

Contract an antisymmetric tensor with one copy of the velocity:

$$\boxed{\frac{dp^\mu}{d\tau} = qF^{\mu\nu}u_\nu.} \tag{6.32}$$

Indeed,

$$f^\mu u_\mu = qF^{\mu\nu}u_\nu u_\mu = 0, \tag{6.33}$$

because $F^{\mu\nu}$ is antisymmetric and $u_\nu u_\mu$ is symmetric. This is not the only conceivable relativistic force law, but it is the simplest one linear in both the electromagnetic field and the velocity.

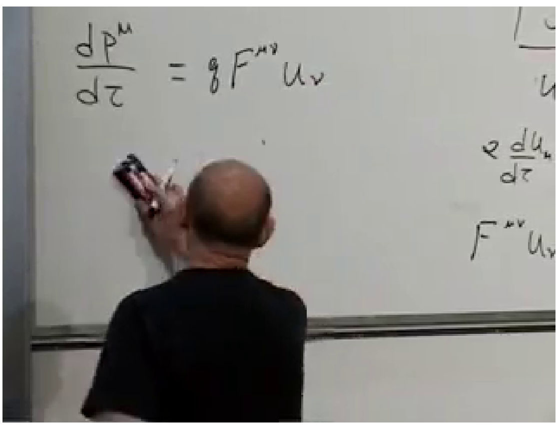


Figure 6.12: The covariant Lorentz-force equation on the left and its vanishing contraction with a second four-velocity on the right.

Lecture frame: 01:00:00.

6.7 Recovering the ordinary Lorentz force

A covariant-looking equation must still reproduce the known force law. With the metric convention stated at the beginning,

$$u_\mu = \gamma(-v_x, -v_y, -v_z, 1). \tag{6.34}$$

For the x-component, (6.23) and (6.32) give

$$\frac{dp^1}{d\tau} = q\Big(F^{12}u_2 + F^{13}u_3 + F^{10}u_0\Big) \tag{6.35}$$
$$= q\gamma\Big(B_3 v_y - B_2 v_z + E_1\Big) \tag{6.36}$$
$$= q\gamma\Big[\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}\Big]_x. \tag{6.37}$$

The other spatial components follow by cycling the indices.

Question from the audience. Where is the qE_x term in the component expansion?

Response. It is the term $qF^{10}u_0$. A field-tensor component with one spatial and one time index is an electric field component, and $u_0 = \gamma$.

The factor γ on the right is not an extra force. Since

$$\frac{dp^i}{d\tau} = \frac{dt}{d\tau}\frac{dp^i}{dt} = \gamma\frac{dp^i}{dt}, \tag{6.38}$$

it cancels from both sides of (6.37). The exact spatial equation is therefore again

$$\frac{d\boldsymbol{p}}{dt} = q\Big(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}\Big), \qquad \boldsymbol{p} = m\gamma\boldsymbol{v}. \tag{6.39}$$

The familiar form survives, but the relativity is hidden in the momentum.

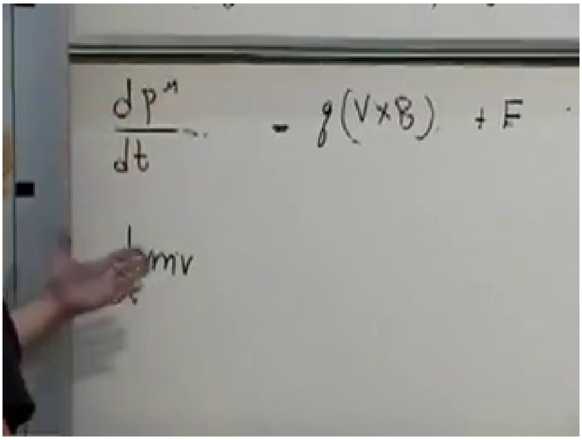


Figure 6.13: The three-dimensional Lorentz law recovered after the common proper-time factor is removed; the momentum retains its factor of γ.

Lecture frame: 01:06:30.

The time component carries the corresponding energy statement,

$$\frac{dp^0}{dt} = q\boldsymbol{E} \cdot \boldsymbol{v}, \tag{6.40}$$

so a magnetic field performs no work while an electric field may change the particle energy. [2]

[2] Editorial clarification: The time-component check is the direct companion of the lecture's spatial calculation; it is included to expose the same no-work property in four-dimensional notation.

Question from the audience. Is this derivation mathematics or physics?

Response. It is both. Tensor algebra supplies the compact form, but substantial physics entered through the principle that all inertial frames obey the same laws and through the assumption that the equations of motion are second order. A second-order law predicts the future from position and velocity; a third-order law would require acceleration as additional initial data. Experiment, not notation, tells us which kind of law nature uses.

6.8 How the fields transform

Equation (6.32) has the same form in every frame because both sides are four-vectors. The field tensor itself transforms with one Lorentz matrix for each index:

$$F'^{\mu\nu} = \Lambda^{\mu}{}_{\rho}\Lambda^{\nu}{}_{\sigma}F^{\rho\sigma}. \tag{6.41}$$

The board derives this rule by first transforming a simple product $A^\mu B^\nu$. Once the rule is known for a product of vectors, linearity gives the rule for any rank-two tensor.

For a boost along the 1-axis, the coordinates transform as

$$x'^1 = \gamma(x^1 - vx^0), \qquad x'^0 = \gamma(x^0 - vx^1), \tag{6.42}$$

$$x'^2 = x^2, \qquad x'^3 = x^3. \tag{6.43}$$

Apply the same transformation separately to both tensor indices. For the component parallel to the boost, the board begins with a product:

$$(A'^0B'^1) = \gamma^2\Big(A^0B^1 - vA^0B^0 - vA^1B^1 + v^2A^1B^0\Big). \tag{6.44}$$

Replacing each product by the corresponding tensor component and then using antisymmetry, $F^{00} = F^{11} = 0$ and $F^{10} = -F^{01}$, gives

$$F'^{01} = \gamma^2(1 - v^2)F^{01} = F^{01}. \tag{6.45}$$

Thus the electric field component parallel to the motion is unchanged. The parallel magnetic component behaves similarly.

The transverse components are different. Following the primed-frame and sign convention used on the board,

$$F'^{02} = \gamma\Big(F^{02} - vF^{12}\Big), \tag{6.46}$$

$$E'_2 = \gamma(E_2 + vB_3). \tag{6.47}$$

A transverse electric component acquires part of a magnetic component. The remaining transformation laws follow from the same two-index rule and are left as the lecture's exercise.

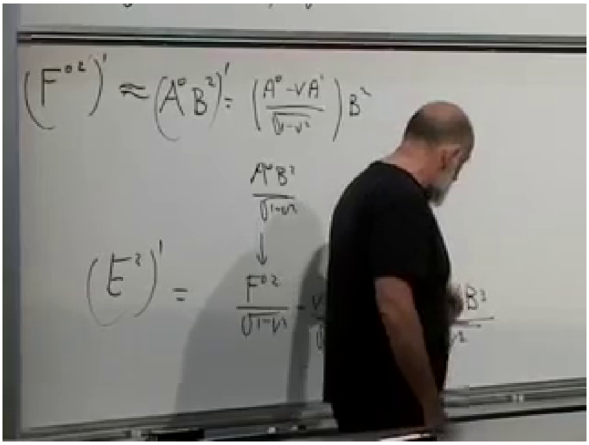


Figure 6.14: The transverse tensor transformation at the point where the E_2' component is seen to contain a term proportional to vB_3.

Lecture frame: 01:23:40.

The sign in (6.47) depends on which frame is declared to move with $+v$, as well as on index and metric conventions. Its invariant content does not: transverse electric and magnetic components mix, and the covariant force equation remains valid after the transformation.

Question from the audience. For a current in a wire, can one move at some velocity that makes the magnetic field vanish?

Response. Moving with the conduction electrons is not enough because the positive lattice charges then move in the new frame. A neutral current-carrying wire is therefore subtler than a single moving lump of charge. A purely magnetic field cannot be transformed completely away. If electric and magnetic fields are both present, their transformed pieces can cancel in special circumstances; for one uniformly moving charge, the charge's rest frame is the simple frame in which its magnetic field vanishes.

The precise invariant test uses

$$\boldsymbol{B}^2 - \boldsymbol{E}^2, \qquad \boldsymbol{E} \cdot \boldsymbol{B}. \tag{6.48}$$

A frame with $\boldsymbol{B}' = 0$ can exist only for an electric-dominated field with $\boldsymbol{E} \cdot \boldsymbol{B} = 0$; a field that is purely magnetic in one frame is magnetic-dominated and cannot meet that condition. [3]

Question from the audience. Do the electric scalar potential and magnetic vector potential form a four-vector?

Response. Yes. With convention-dependent signs they combine into the four-potential A^μ. Maxwell's equations will make systematic use of that object.

[3]Editorial clarification: These invariant criteria state precisely the qualification reached in the classroom discussion; their derivation is deferred.

6.9 Derivatives of fields

We now leave the trajectory of one particle and consider a field throughout spacetime. Let

$$\phi = \phi(x^1, x^2, x^3, x^0) \tag{6.49}$$

be a scalar field. Its four derivatives form the covariant components of a four-vector:

$$\partial_\mu \phi \equiv \frac{\partial \phi}{\partial x^\mu} = \left(\frac{\partial \phi}{\partial x}, \frac{\partial \phi}{\partial y}, \frac{\partial \phi}{\partial z}, \frac{\partial \phi}{\partial t} \right). \tag{6.50}$$

The chain rule applied to the Lorentz coordinate transformation proves this covariant transformation law. Raising the index reverses the spatial signs but leaves the time sign unchanged:

$$\partial^\mu \phi = \left(-\frac{\partial \phi}{\partial x}, -\frac{\partial \phi}{\partial y}, -\frac{\partial \phi}{\partial z}, \frac{\partial \phi}{\partial t} \right). \tag{6.51}$$

Every differentiation contributes a lower index. Thus $\partial_\nu V^\mu$ is a rank-two tensor with one upper and one lower index, and differentiating a tensor produces an object with one additional lower index. This is the basic rule needed to write field equations without choosing a preferred inertial frame.

6.10 The invariant scalar wave equation

The simplest wave equations contain two derivatives. If the equation is to have the same form in every inertial frame, its left-hand side should be a scalar.

Contracting the derivative indices gives the unique simplest choice:

$$\boxed{\partial^\mu \partial_\mu \phi = 0.} \tag{6.52}$$

Writing out the components in the present signature,

$$\frac{\partial^2 \phi}{\partial t^2} - \frac{\partial^2 \phi}{\partial x^2} - \frac{\partial^2 \phi}{\partial y^2} - \frac{\partial^2 \phi}{\partial z^2} = 0. \tag{6.53}$$

The contraction in (6.52) makes Lorentz invariance manifest; equation (6.53) is the same statement written component by component.

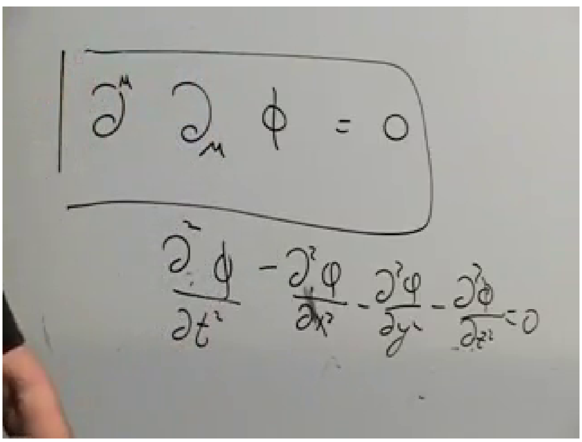


Figure 6.15: The contracted derivative form of the scalar wave equation and its time-minus-space component expansion.

Lecture frame: 01:39:40.

Question from the audience. Must the right-hand side of the wave equation be zero?

Response. No. It may be any scalar compatible with the problem. A constant times ϕ, or a nonlinear scalar function such as one proportional to ϕ^2, still preserves covariance. Zero gives the simplest free wave equation.

The point of this scalar example is preparation. Maxwell's equations mix electric and magnetic fields more intricately, but they too can be written so that Lorentz invariance is visible from their index structure. That is the problem to which we return next time.

CHAPTER 7

METRICS, EXPANSION, AND A FIRST COSMOLOGY

At the end of the preceding lecture we had reached a Lorentz-invariant wave equation and were preparing to write Maxwell's equations in spacetime form. An audience question now supplies a useful warning before we turn to gravitation: how much of a physical theory can symmetry determine by itself? The answer is emphatic: a great deal, but not everything. Lorentz invariance is a powerful filter on possible laws. It is not a substitute for the remaining physical assumptions or for experiment.

7.1 What symmetry does not determine

Question from the audience. Once the electric and magnetic fields are known to transform together under Lorentz transformations, do Maxwell's equations follow from relativity alone?

Response. No. Relativity constrains the form of the equations, but the construction also uses their linearity, the low order of their derivatives, the way the fields transform, and experimental information. Without those additional ingredients many Lorentz-covariant field equations would still be possible.

The most important of those extra assumptions for the opening discussion is linearity. If $A(x, t)$ is a solution of a linear field equation, then $\lambda A(x,t)$ is also a solution for every constant λ. If A and B are solutions, then so is

$$C(x,t) = A(x,t) + B(x,t). \tag{7.1}$$

This is the superposition principle. Two source-free electromagnetic waves may cross one another and emerge unchanged. They do not collide directly in the way material objects do. They can interact indirectly through charged matter, because each wave moves the charges and the moving charges radiate, but the vacuum field equations themselves are linear.

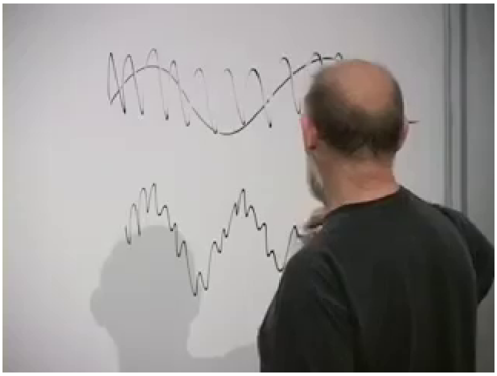


Figure 7.1: Two wave profiles on the board during the discussion of superposition. Their sum is again an allowed source-free wave.

Lecture frame: 00:05:30.

The harmonic oscillator makes the algebra transparent. Suppose

$$m\ddot{x} = -kx. \tag{7.2}$$

If x_1 and x_2 solve (7.2), then

$$m(\ddot{x}_1 + \ddot{x}_2) = -k(x_1 + x_2), \tag{7.3}$$

so $x_1 + x_2$ is a solution. Add a nonlinear term, however,

$$m\ddot{x} = -kx + px^2, \tag{7.4}$$

and the sum no longer works:

$$(x_1 + x_2)^2 = x_1^2 + x_2^2 + 2x_1x_2. \tag{7.5}$$

The cross term $2x_1x_2$ has no counterpart in the two separate equations. The two motions now influence one another.

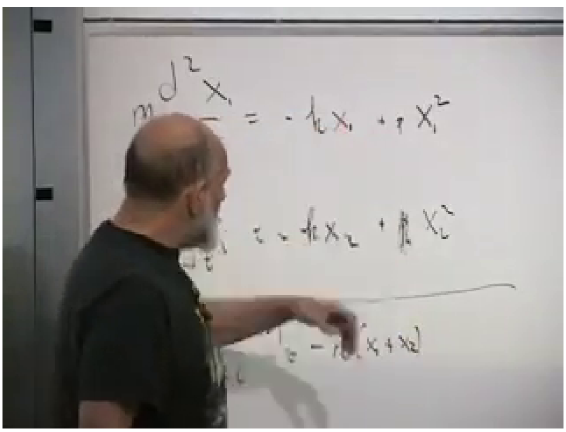


Figure 7.2: Separate oscillator equations compared with the equation for their sum. The quadratic term exposes the extra cross term that breaks superposition.

Lecture frame: 00:10:10.

This example fixes the methodological point that will remain in force throughout the cosmology discussion.

Symmetry tells us which quantities may be combined and which equations are allowed. Dynamics and observation tell us which allowed equation nature uses.

7.2 The interval as geometry

We now leave electromagnetism and take a short route into general relativity and cosmology. The starting object is the interval between neighboring events. Label an event by four coordinates x^μ and a nearby event by $x^\mu + dx^\mu$. In inertial Cartesian coordinates, with $c = 1$ and the $(+,-,-,-)$ convention,

$$d\tau^2 = dt^2 - dx^2 - dy^2 - dz^2 = dx^\mu dx_\mu = \eta_{\mu\nu}\, dx^\mu dx^\nu. \tag{7.6}$$

The quantity $d\tau^2$, not any one coordinate difference, is invariant. Lorentz transformations change dt, dx, dy, dz while preserving this quadratic combination. Special relativity may therefore be read as the geometry of Minkowski spacetime.

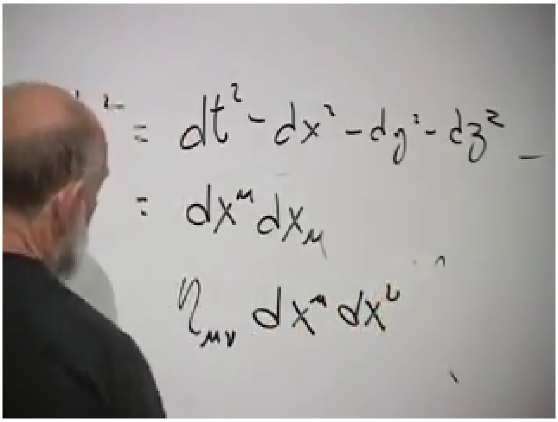


Figure 7.3: The Minkowski interval written successively in component, contracted-index, and metric-tensor notation.

Lecture frame: 00:14:50.

The simplicity of (7.6) depends on the coordinates. Even a flat plane can acquire unfamiliar coefficients if we label it unwisely. Let x be measured in centimeters while y is measured in meters. For a physical displacement,

$$ds^2 = dx^2 + \left(\frac{dy}{100}\right)^2 = dx^2 + 10^{-4}dy^2. \qquad (7.7)$$

The factor 10^{-4} says nothing about curvature. It merely translates between units.

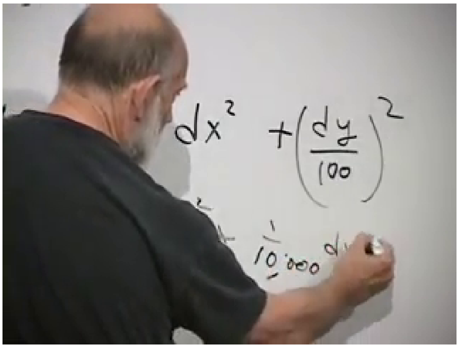


Figure 7.4: A flat plane described with centimeters on one axis and meters on the other. The unequal metric coefficients encode units, not curvature.

Lecture frame: 00:20:10.

Oblique coordinates add a mixed term. In two dimensions the most general local quadratic form can be written

$$\begin{aligned} ds^2 = a_{11}\,dx^2 &+ a_{12}\,dx\,dy \\ &+ a_{21}\,dy\,dx + a_{22}\,dy^2, \qquad a_{12} = a_{21}. \end{aligned} \tag{7.8}$$

Because ordinary differentials commute, $dx\,dy = dy\,dx$, only the symmetric part of the coefficient array contributes. We collect the coefficients into the matrix

$$g = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}. \tag{7.9}$$

This matrix is the *metric*. It is the local rule that converts coordinate differences into physical intervals.

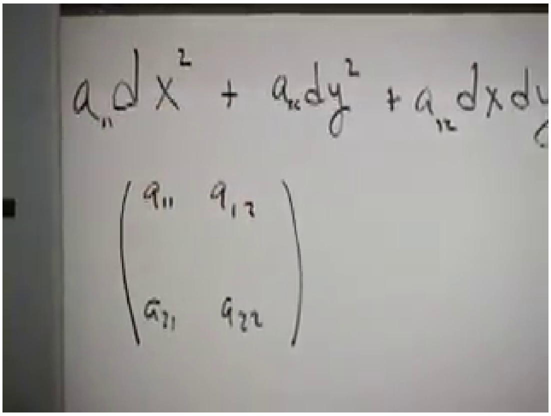


Figure 7.5: The quadratic differential form and its symmetric 2×2 coefficient matrix at the moment the matrix is identified as the metric.

Lecture frame: 00:24:07.

Uniform rectangular or oblique grids give constant coefficients. If the coordinate lines curve, bunch together, or spread apart, those coefficients become functions of position even when the underlying surface remains flat. The general spacetime notation is

$$ds^2 = g_{\mu\nu}(x)\, dx^\mu dx^\nu. \tag{7.10}$$

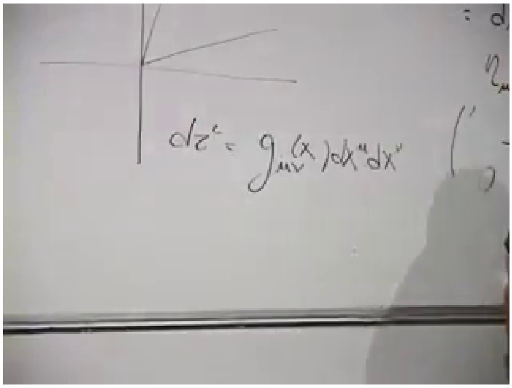


Figure 7.6: The general position-dependent metric after the coordinate grid is allowed to vary from point to point.

Lecture frame: 00:28:10.

Knowing $g_{\mu\nu}$ in a specified coordinate system determines local distances, times, and angles and thereby determines the geometry. The converse is not unique: one geometry has infinitely many coordinate descriptions, hence infinitely many arrays of coefficient functions related by coordinate transformations.

7.3 Coordinate complexity and intrinsic curvature

We must now distinguish a complicated metric from a curved space. Roll a sheet of paper into a cylinder without stretching it. Its appearance in the room changes, but all distances measured within the sheet remain the same. A creature confined to the paper cannot detect the bending by any intrinsic measurement. That is extrinsic bending, not the curvature meant here.

A hemisphere is different. It cannot be pressed flat without stretching, tearing, or compressing it. A flat map of Earth must distort either distance, area, angle, or some combination of them. This obstruction is intrinsic and can be detected without looking into a surrounding dimension.

Polar coordinates give the indispensable flat counterexample. On a plane, an infinitesimal radial displacement has length dr, while an angular displacement $d\theta$ at radius r has length $r\,d\theta$. Thus

$$ds^2 = dr^2 + r^2 d\theta^2, \qquad g_{\text{polar}} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}. \tag{7.11}$$

The coefficient r^2 varies across the plane, yet the plane is flat. The change of variables

$$x = r\cos\theta, \qquad y = r\sin\theta \tag{7.12}$$

returns the constant metric $ds^2 = dx^2 + dy^2$.

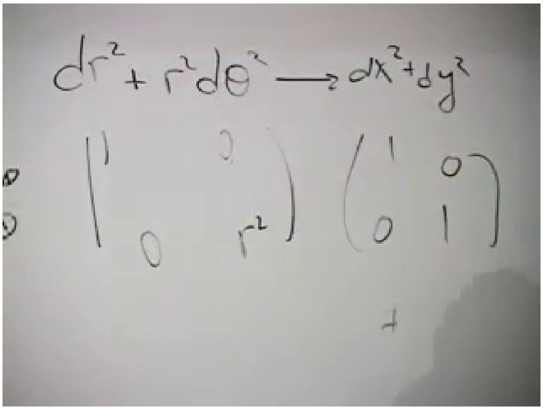


Figure 7.7: The polar metric and its return to the Cartesian metric. The position-dependent matrix on the left still describes a flat plane.

Lecture frame: 00:39:20.

Question from the audience. Should the angular distance in polar coordinates contain a sine?

Response. Not on the plane. There the arc length is $r\,d\theta$. A sine appears on the sphere, where a circle of latitude has radius $R\sin\theta$.

Take a sphere of radius R. Let θ be the polar angle measured from the north pole and ϕ the azimuthal angle. A meridional step has length $R\,d\theta$. At polar angle θ, the circle of latitude has radius $R\sin\theta$, so an azimuthal step has length $R\sin\theta\,d\phi$. Local Pythagoras gives

$$ds^2 = R^2\left(d\theta^2 + \sin^2\theta\,d\phi^2\right), \tag{7.13}$$

$$g_{\text{sphere}} = \begin{pmatrix} R^2 & 0 \\ 0 & R^2\sin^2\theta \end{pmatrix}. \tag{7.14}$$

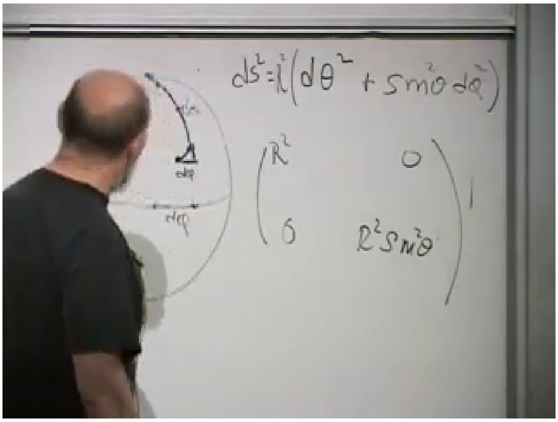


Figure 7.8: The sphere, its angular line element, and the corresponding metric matrix. The $\sin^2\theta$ factor records the shrinking circles of latitude.

Lecture frame: 00:45:15.

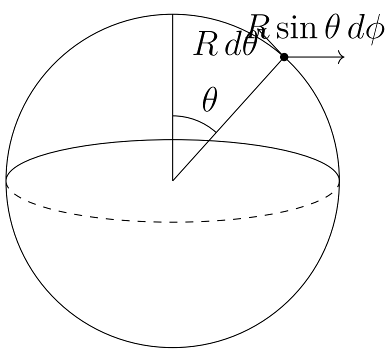


Figure 7.9: Clean reconstruction of the local differential legs used to obtain (7.13).

No global change of coordinates turns (7.13) into a constant Euclidean matrix. A two-dimensional observer can discover this intrinsically: a sufficiently large triangle made from geodesics has an angle sum greater than 180°. We have therefore reached the useful criterion. Variable coefficients do not by themselves imply curvature. Curvature is the obstruction to finding coordinates that make the metric constant throughout the region under consideration.

7.4 From curved spacetime to cosmology

General relativity permits spacetime itself to possess this intrinsic curvature. A position-dependent $g_{\mu\nu}$ may still be an artifact of coordinates, as in polar coordinates, but in a genuinely curved spacetime no coordinate transformation removes the variation globally.

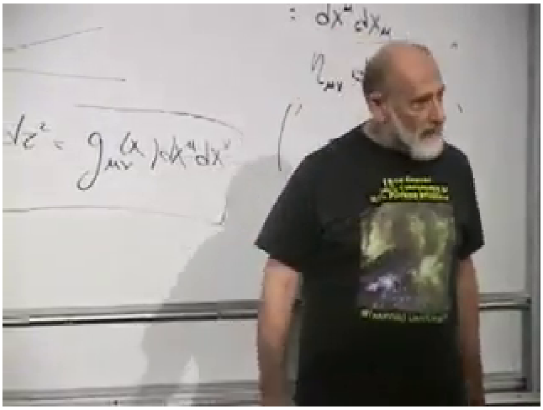


Figure 7.10: The transition from special coordinate examples to the general spacetime line element $ds^2 = g_{\mu\nu}(x)dx^\mu dx^\nu$.

Lecture frame: 00:48:40.

We will not attempt the full machinery of Einstein's equations. Instead we restrict attention to geometries appropriate to cosmology: space is taken to be homogeneous on sufficiently large scales, while the geometry may change with time. Homogeneous means that no location is distinguished from another. It does not mean flat: every point on a sphere is equivalent, yet the sphere is curved. It does not mean featureless on every scale either.

Question from the audience. How can the universe be homogeneous when galaxies form clusters, sheets, and filaments?

Response. The claim concerns coarse-grained scales. Stars, galaxies, and clusters are strongly inhomogeneous. When regions hundreds of millions of light-years across are compared, those structures average toward a uniform distribution. The cosmological principle is an empirical working hypothesis, not a logical necessity; sufficiently large systematic differences, including directional differences in the cosmic background, could disprove it.

The second large-scale assumption in this lecture is spatial flatness. At a fixed cosmic time, sufficiently large triangles appear Euclidean: their angles add to 180°. This is a statement about the geometry of three-dimensional space, not yet about the curvature of four-dimensional spacetime.

Question from the audience. What does simultaneity mean for points separated by cosmological distances, since their light reaches us at different times?

Response. We cannot place rods between distant galaxies or inspect them instantaneously. We receive signals later and reconstruct the geometry of the emitting events within a spacetime model. The procedure is indirect, but so are many geometric measurements in astronomy. Flatness means that the reconstructed large triangles show no detectable angular excess or deficit on the surveyed scale.

7.5 Comoving coordinates and the scale factor

Imagine a Cartesian grid painted onto the large-scale flow of galaxies. Each galaxy keeps its coordinate label as the grid expands. The label is not a distance in meters; it is an address. A time-dependent conversion factor turns coordinate separations into physical separations.

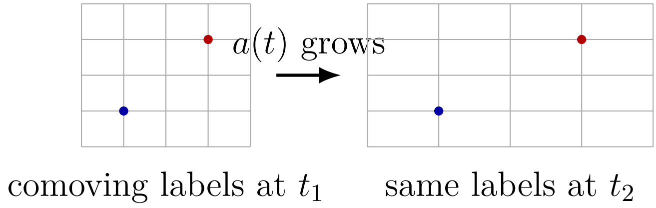


Figure 7.11: The comoving addresses stay fixed while the physical spacing of the coordinate grid grows.

Spatial homogeneity requires the same conversion factor everywhere at a given time. Spatial flatness allows the spatial part of the metric to be Cartesian. The resulting line element is

$$d\tau^2 = dt^2 - a^2(t)\left(dx^2 + dy^2 + dz^2\right). \qquad (7.15)$$

This is the spatially flat homogeneous cosmological metric in comoving coordinates. We may choose x, y, z dimensionless and give $a(t)$ units of length. Another convention can put the dimensions into the coordinates and make a dimensionless; only the product matters.

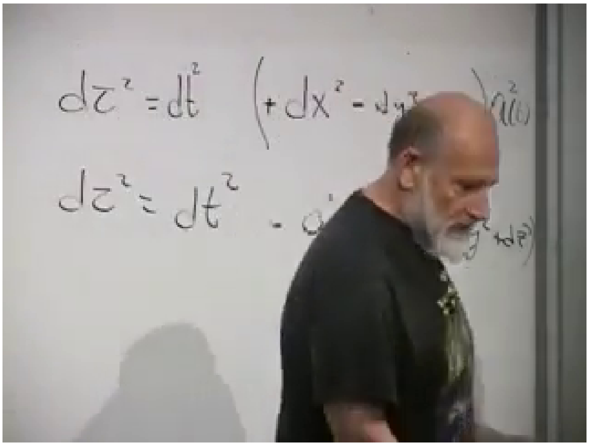


Figure 7.12: The spatially flat cosmological metric as completed on the board, with the common factor $a^2(t)$ multiplying all three spatial directions.

Lecture frame: 01:03:15.

If a were constant, we could absorb it by rescaling x, y, z, and (7.15) would be ordinary Minkowski spacetime in different units. Its time dependence is the nontrivial ingredient.

Question from the audience. If the coordinate grid expands, why do meter sticks, atoms, and galaxies not expand with it?

Response. The metric describes the average cosmic flow, not a force that compels every bound system to follow it. Electromagnetic forces hold a meter stick and an atom together; self-gravity holds a galaxy together. Those binding effects dominate the exceedingly weak background expansion on small scales. Nearby galaxies can also depart from the Hubble flow: Andromeda is pulled toward the Milky Way. Detailed studies of galactic binding show that visible matter alone is insufficient, one of the arguments for dark matter.

Question from the audience. Why is the scale factor not itself a force that pushes galaxies apart?

Response. It has the dimensions of length in the present convention, not force. Think of a cannonball launched upward. Once launched, its outward motion is inertial; gravity changes that motion by reducing its speed or pulling it back. Likewise, expansion is a state of motion. The dynamical question is how gravity and other sources change $\dot{a}$ and $\ddot{a}$, not what force is required to keep a moving.

The old expectation was that attractive gravity should decelerate the expansion. By the time of this lecture, astronomical evidence had shown that the recent expansion is instead accelerating, which means that the complete dynamics contains more than ordinary attractive matter. We postpone that issue until the geometrical setup is secure.

7.6 The Hubble law

Take two comoving galaxies whose fixed coordinate separation is Δx. At a common cosmic time their physical distance is

$$D(t) = a(t)\,\Delta x. \tag{7.16}$$

Differentiate while holding the comoving labels fixed:

$$\begin{aligned} V(t) = \dot{D}(t) &= \dot{a}(t)\,\Delta x \\ &= \frac{\dot{a}(t)}{a(t)} D(t). \end{aligned} \tag{7.17}$$

Define

$$H(t) = \frac{\dot{a}(t)}{a(t)}. \tag{7.18}$$

Then

$$V(t) = H(t)D(t). \tag{7.19}$$

At one cosmic time the same $H(t)$ applies everywhere; otherwise space would not be homogeneous. The recession speed is proportional to distance because every interval in the comoving grid is stretched by the same fractional rate.

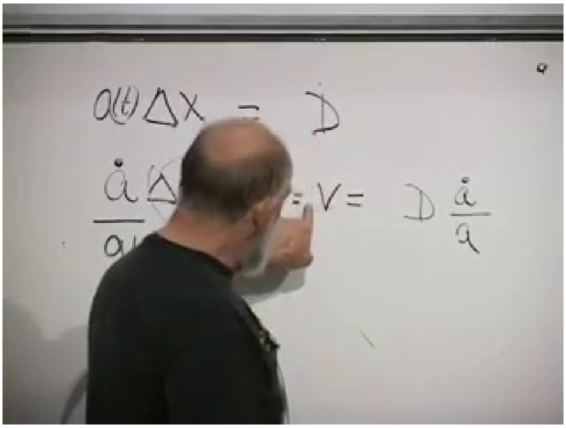


Figure 7.13: Physical distance $D = a\Delta x$ and its time derivative on the way to $V = (\dot{a}/a)D$.

Lecture frame: 01:16:45.

Question from the audience. For a distant galaxy, should the Hubble parameter be evaluated now or when the observed light was emitted?

Response. At the epoch being described. Light from a distant galaxy reports an earlier value of the expansion. A serious analysis must account for the time dependence of H; inserting today's value everywhere is only a local approximation.

Question from the audience. Where do the units reside in $D = a\Delta x$?

Response. A convenient choice is to treat the comoving label Δx as dimensionless and let a carry length. The opposite bookkeeping is possible. Either way, $H = \dot{a}/a$ has dimensions of inverse time.

7.7 Light in the expanding coordinates

Light follows a null path. Along such a path the proper-time interval vanishes:

$$d\tau^2 = 0. \tag{7.20}$$

Question from the audience. Why is the interval zero for light?

Response. That is the defining geometry of a light ray in relativity. Every local freely falling observer measures light to move at $c = 1$, so $dt^2 = d\ell^2$ locally and the spacetime interval vanishes. It does not mean that no coordinate time passes for the observer receiving the light.

For a ray moving only in the x-direction,

$$0 = dt^2 - a^2(t)\, dx^2. \tag{7.21}$$

Choosing the ray that moves toward increasing x,

$$dt = a(t)\, dx, \qquad dx = \frac{dt}{a(t)}, \qquad \frac{dx}{dt} = \frac{1}{a(t)}. \tag{7.22}$$

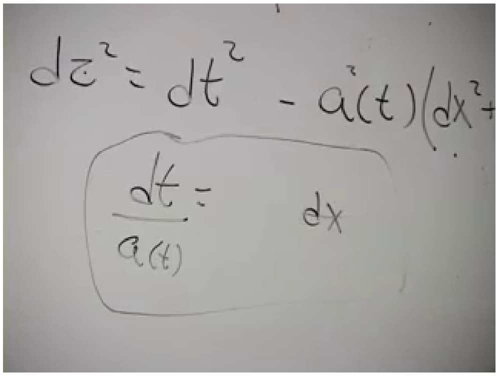


Figure 7.14: The null condition in the expanding metric and the resulting comoving light-ray equation $dx = dt/a(t)$.

Lecture frame: 01:23:45.

The coordinate speed dx/dt decreases as the scale factor grows. This does not mean that light locally slows down. One unit of comoving coordinate distance represents an ever larger physical distance, so the numerical coordinate speed changes while every local measurement still gives $c = 1$.

7.8 Why the time dependence matters

The central geometrical question is now unavoidable: how does $a(t)$ vary? The analogy is a projectile whose position is known at one instant. Its future trajectory requires an equation of motion. In cosmology, gravitational dynamics will supply the equation for $a(t)$.

Question from the audience. Why privilege the question whether $a(t)$ changes over the equally legitimate question whether, for example, the electron charge changes with time?

Response. There is no logical hierarchy. A measured change in the electron charge would be enormously important. We focus on $a(t)$ because observation already tells us that cosmic separations change, whereas the electron charge appears constant. Geometrically, a constant a could be removed by a change of spatial units and would leave flat spacetime; a time-dependent a cannot be removed that way.

Before measuring distant recession, one needs this geometrical framework to say what is being measured. At the crudest level, a source of known luminosity appears dimmer with distance because its light spreads over a larger area, while the displacement of known atomic spectral lines gives a redshift and hence a recession indicator. Hubble combined distance estimates with redshifts to uncover the proportionality (7.19), although the early numerical value of the coefficient was substantially wrong. Modern distance ladders and relativistic light-propagation calculations are much more elaborate, but the conceptual pair remains distance plus redshift.

7.9 The Hubble time and the age estimate

There is a simple estimate hidden in the Hubble law. Freeze a galaxy's present recession speed and run the motion backward uniformly. The time required to close its present distance would be

$$t = \frac{D}{V}. \tag{7.23}$$

Using $V = HD$,

$$t = \frac{D}{HD} = \frac{1}{H}. \tag{7.24}$$

The distance cancels, so every pair of comoving galaxies gives the same rough time. Susskind catches and corrects an easy verbal error here. The assumption is not that H remains constant. It is that the particular recession velocity is extrapolated backward as though it were constant.

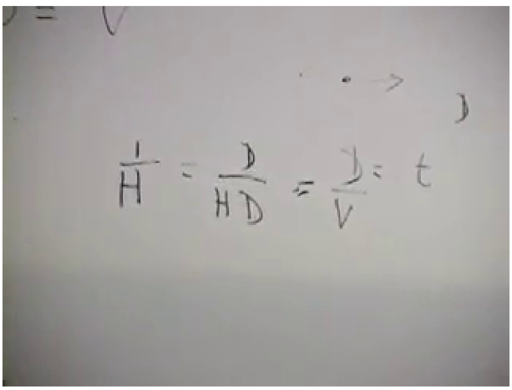


Figure 7.15: The Hubble-time estimate $H^{-1} = D/(HD) = D/V$, with distance canceling from the backward extrapolation.

Lecture frame: 01:33:45.

The inverse Hubble parameter is therefore a time scale, not automatically the exact age of the universe.

In a purely decelerating, matter-dominated picture, reverse the present velocity and let gravity pull the galaxies together. They meet sooner than they would at uniform velocity, so the present H^{-1} overestimates the age. The lecture quotes a refined age of about 13.5 billion years, close to the Hubble time but not identical to it.

Question from the audience. If there is no Hubble expansion inside our galaxy, what does it mean to run all galaxies backward until they collapse?

Response. The homogeneous description eventually ceases to resolve the real history of bound structures. Forward in time, small fluctuations grow gravitationally into galaxies and clusters. A literal reversed movie would require every microscopic motion to reverse and entropy to decrease; simply reversing the average Hubble flow does not undo structure formation. The backward extrapolation is a coarse cosmological estimate, not a film of every star retracing its path.

Question from the audience. Can an experiment distinguish an expanding intergalactic distance from a shrinking ruler?

Response. Only dimensionless ratios are observable. What is measured is that the distance between galaxies divided by an atomic or laboratory length standard increases. We conventionally hold the parameters of atomic and molecular physics fixed and say that the universe expands. One could choose the intergalactic separation as the standard and describe atoms, protons, and meter sticks as shrinking. The physics of the ratio would be unchanged.

That convention is natural because rods are made from atoms and their size is controlled by parameters that appear constant. A meter is ultimately a large number of microscopic lengths. Holding those microscopic standards fixed gives the familiar statement that $a(t)$ grows.

Question from the audience. Is (7.15) the only metric compatible with spatial homogeneity and flatness?

Response. Only up to coordinate transformations. The precise claim is that, under those assumptions and a suitable choice of cosmic time and comoving spatial coordinates, the metric can be brought to this form. Another coordinate system may make the same spacetime look quite different.

Question from the audience. When the backward separation reaches zero, the Hubble law also gives zero velocity. Does that prevent the galaxies from meeting?

Response. No. The H inferred today is not held fixed all the way back. In ordinary expanding solutions $H(t)$ was much larger at earlier times. The estimate H_0^{-1} is made by extending today's velocity uniformly only to obtain a scale; the real $a(t)$, $H(t)$, and velocities must be found from dynamics.

7.10 Newton's shell theorem and cosmological dynamics

To discover the dynamics, smear the galaxies into a homogeneous medium. Choose one galaxy as the origin and another with comoving label x. There is no physical privilege in this choice; homogeneity requires the final equation to be the same around every galaxy. The physical radius of the second galaxy is

$$R(t) = a(t)x. \tag{7.25}$$

At first, finding its gravitational acceleration seems to require summing the attraction of every galaxy in an enormous universe. Newton's shell theorem collapses the problem:

Proposition 7.1 (Newton's shell theorem)**.** *For an inverse-square force and a spherically symmetric mass distribution, matter outside a sphere through the test point contributes zero net force. The matter* *inside the sphere acts on the test point as though all enclosed mass were concentrated at the center.*

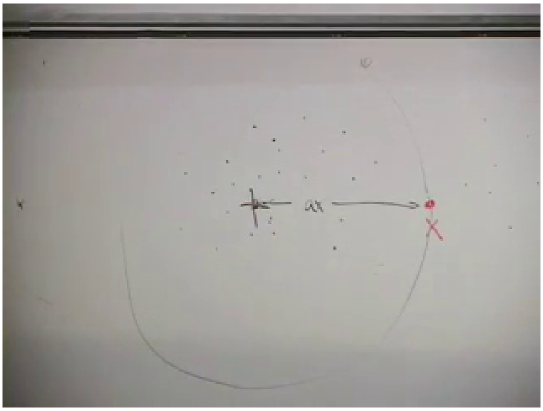


Figure 7.16: The homogeneous galaxy distribution, a test galaxy at comoving label x, and the sphere whose enclosed mass alone controls its radial motion. *Lecture frame: 01:50:05.*

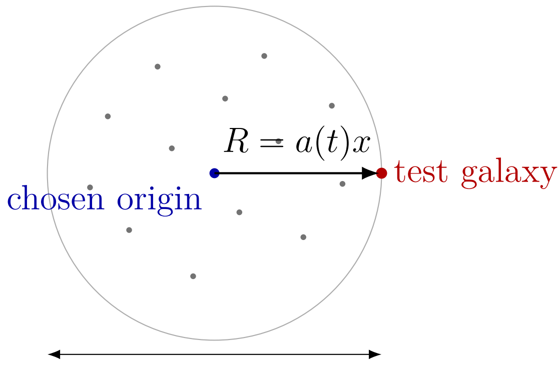


Figure 7.17: Clean reconstruction of the shell-theorem setup. The origin is a convenient choice, not a center of the universe.

The cosmological many-body problem has therefore become a familiar one-body problem: a projectile moving radially under the attraction of the mass enclosed inside its present sphere. Newton's equation will determine how $R = ax$, and hence $a(t)$, accelerates. The lecture deliberately stops before carrying out that calculation. The next meeting will derive the simplest matter-dominated cosmology rather than smuggling its answer into the geometrical setup.

7.11 Late questions: light, the Big Bang, and curvature

The formal lecture has ended, but the final questions expose several misleading pictures and are part of the physics rather than an appendix.

Question from the audience. If light from very distant objects was emitted when the universe was much smaller, why does it now appear to come from so far away? And if the early universe was compact, why is not all of it observable?

Response. One must solve the null-ray equation in the expanding coordinates. When $a(t)$ is very small, $dx/dt = 1/a(t)$ is very large: light crosses a large comoving interval while still moving locally at $c = 1$. As $a(t)$ grows, its comoving coordinate speed decreases. The precise emission point and the size of the observable region depend on the entire history of $a(t)$, so a numerical answer is postponed until a specific cosmology has been solved.

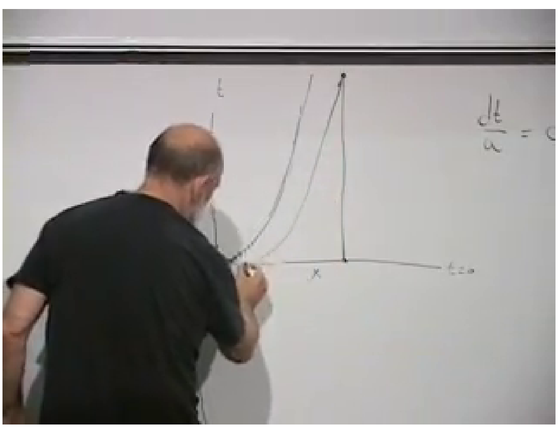


Figure 7.18: A light ray in the (x, t) plane: nearly horizontal in comoving coordinates when $a(t)$ is small, then progressively steeper as the scale factor grows.

Lecture frame: 01:56:20.

Integrating (7.22) makes the deferred calculation explicit:

$$\Delta x = \int_{t_e}^{t_0} \frac{dt}{a(t)}. \tag{7.26}$$

Whether the integral reaches arbitrarily large comoving distances as t_e approaches the beginning depends on the early-time form of $a(t)$.[1]

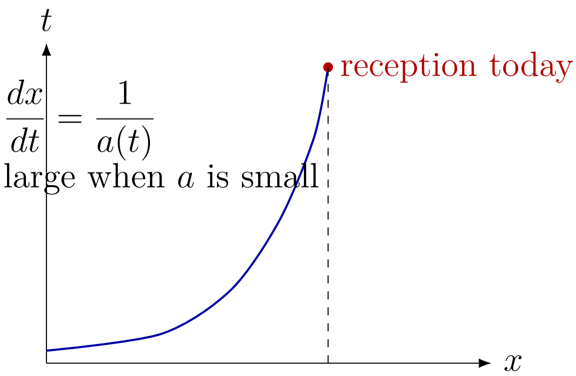


Figure 7.19: Clean version of the light-ray construction in comoving coordinates. The changing slope is a coordinate effect, not a changing local speed of light.

[1] Editorial clarification: Equation (7.26) is the integrated form of the board equation $dx = dt/a(t)$. It makes explicit the calculation that the lecture assigns to the next meeting without choosing a particular scale factor.

Question from the audience. Did the Big Bang eject matter from one point so that it should now lie on an expanding shell or ring?

Response. No. The model begins with a homogeneous, extremely dense space. Every large-scale separation grows; matter is not launched through pre-existing empty space from a preferred center. A balloon covered uniformly with dots is a useful two-dimensional analogy: as it expands, all dot separations grow while the surface remains uniformly populated. The space may even be infinite at every positive time; expansion does not imply that it began as a finite ball inside something else.

Question from the audience. Was gravity itself weaker or stronger during the expansion?

Response. The discussion keeps Newton's constant G fixed. The force nevertheless changes because the separations and enclosed densities change. Earlier, the same comoving galaxy was physically closer and the homogeneous matter was denser, so the gravitational deceleration was larger. This is precisely what the shell-theorem calculation is meant to quantify.

Finally, spatial curvature must not be confused with expansion. A small patch of Earth appears flat because its size is tiny compared with Earth's radius. By measuring sufficiently large triangles, one can either detect an angular excess or place a lower bound on the radius of curvature. Astronomy does the analogous experiment with enormous triangles. In the observational language used in the 2006 lecture, no curvature was detected across the visible region, so any curvature radius had to be substantially larger than that region. This does not prove an exactly flat universe; it bounds how curved the accessible part can be.

For a sphere, intrinsic curvature scales as R^{-2}. A growing sphere therefore becomes locally flatter. Likewise, if a homogeneous universe has nonzero spatial curvature, expansion increases its curvature radius and reduces the curvature visible in any fixed-size local patch. [2]

Question from the audience. If space is curved, must one component of the universe be contracting while another expands?

Response. No. Positive, zero, or negative spatial curvature describes the geometry of each cosmic-time slice. Expansion or contraction is controlled by the evolution of $a(t)$. A curved universe can expand; a flat universe can contract. Expansion may later reverse, just as an upward-moving projectile may fall back, but simultaneous contraction is not hidden inside curvature. The dynamical issue is whether the outward motion escapes the pull of the enclosed mass, falls back, or lies on the critical boundary.

[2] Editorial clarification: The R^{-2} scaling is the standard quantitative form of the lecture's statement that the numerical curvature decreases as a sphere grows.

That escape-velocity comparison is the last board-level idea of the lecture. Geometry has supplied the metric, the Hubble law, and the light-ray equation. Newton's theorem has reduced the dynamics to the mass inside one sphere. The next lecture begins exactly where this one should end: with the equation that determines $a(t)$, and with the question of whether the expansion escapes, recollapses, or sits on the knife edge.

CHAPTER 8

FRIEDMANN DYNAMICS: MATTER, RADIATION, AND VACUUM ENERGY

Overview. The kinematics of an expanding universe can be stated in one line: physical separations grow with a scale factor $a(t)$. The problem now is to determine that function. We shall recover the flat-space Friedmann equation from Newtonian energy conservation, solve it first for ordinary matter, and then see why radiation and vacuum energy lead to entirely different cosmic histories.

8.1 The geometry to be explained

At a fixed cosmic time, the large-scale universe appears homogeneous and isotropic. Galaxies, clusters, and superclusters are certainly real features, but on scales of hundreds of millions or billions of light-years their distribution becomes remarkably uniform. This empirical regularity is called the cosmological principle. The name does not make it a theorem; the force of the principle comes from observation.

There is a second empirical fact. Within present accuracy, large spatial triangles behave as Euclidean triangles. The comparison with surveying a football field is useful. If tightly stretched lines make triangles whose angles add to 180°, one has not proved that

Earth is globally flat. One has shown that Earth's curvature radius is large compared with the field. Cosmic surveying has the same logical form: space may have a very large curvature radius, but over the observable region it is well approximated as flat.

The reference geometry is Minkowski spacetime,

$$d\tau^2 = dt^2 - \left(dx^2 + dy^2 + dz^2\right). \tag{8.1}$$

The bracket is simply the Euclidean distance element in three dimensions. In general relativity, by contrast, the coefficients of the coordinate differentials are fields. They may vary with position and time, and mixed terms may also occur. These metric coefficients are the relativistic successors of the Newtonian gravitational field.

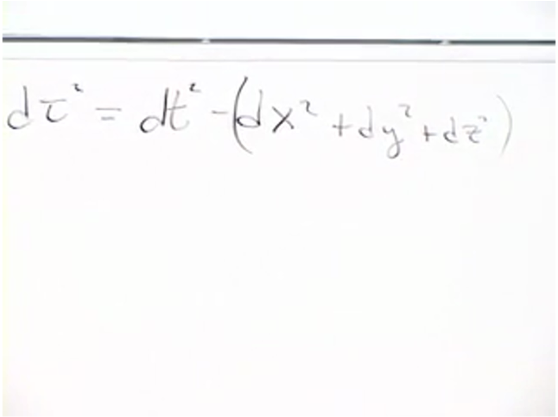


Figure 8.1: The Minkowski proper-time element written at the beginning of the lecture. Source frame at 00:01:50.

For a spatially flat expanding universe, homogeneity and isotropy leave one time-dependent scale factor:

$$d\tau^2 = dt^2 - a^2(t)\left(dx^2 + dy^2 + dz^2\right). \qquad (8.2)$$

We use units in which $c = 1$, so time and length have the same dimensions. There are two consistent conventions. One may give x, y, z dimensions of length and take a dimensionless, or treat x, y, z as dimensionless labels and let a carry the dimensions of length. The lecture uses the second convention because it makes the physical meaning of $a(t)$ visible.

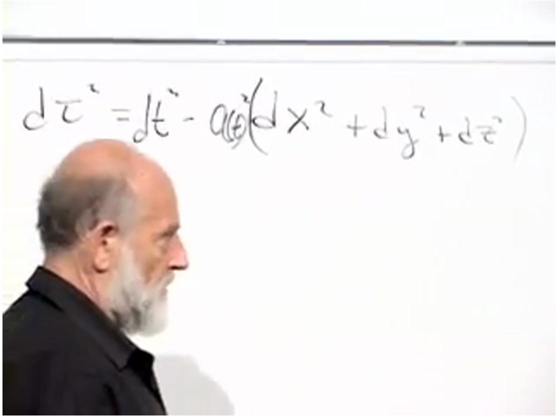


Figure 8.2: The flat expanding metric, with the scale factor multiplying the Euclidean spatial line element. Source frame at 00:09:12.

Imagine coordinates painted on a rubber sheet. The coordinate marks remain attached to the sheet while the physical distance between them grows. Thus a coordinate difference such as "from $x = 0$ to $x = 1$" is only a pair of labels. The distance represented by that interval is $a(t)$. Finding the equation that governs this scale factor is the dynamical problem of the lecture.

8.2 Closed, flat, and open spatial geometries

Flatness is an observational simplification, not a consequence of homogeneity. There are three homogeneous and isotropic spatial geometries: positive, zero, and negative curvature. To understand the positive case, it helps to be precise about dimension.

The locus

$$X_1^2 + X_2^2 = R^2 \tag{8.3}$$

is a circle, or one-sphere S^1. Its filled interior is a disk, also called a two-ball. Adding one embedding coordinate gives

$$X_1^2 + X_2^2 + X_3^2 = R^2, \tag{8.4}$$

the ordinary two-sphere S^2. Adding a fourth embedding coordinate gives

$$X_1^2 + X_2^2 + X_3^2 + X_4^2 = R^2, \tag{8.5}$$

a three-sphere S^3, which is itself a three-dimensional space. We cannot picture its four-dimensional embedding, but the intrinsic geometry does not need that embedding in order to exist.

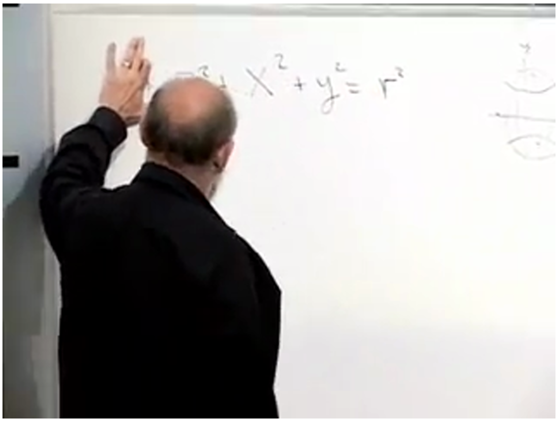


Figure 8.3: The embedding equation used to introduce the three-sphere. Source frame at 00:16:40.

Every point of an ideal sphere is geometrically equivalent to every other, and every direction at a point is equivalent. A sphere is therefore both homogeneous and isotropic. If its radius is multiplied by $a(t)$, the whole space expands or contracts without selecting a center on the sphere. Tracing such a universe backward does not place all matter at one special location in an exterior void. The entire space becomes small, so every pair of points becomes close. The Big Bang is not an explosion from one point of space.

On a positively curved sphere, geodesic triangles have angle sum greater than 180°. Flat triangles have angle sum exactly 180°. In a homogeneous negatively curved geometry, represented imperfectly by a saddle, the angle sum is less than 180°. These intrinsic tests distinguish the three cases.

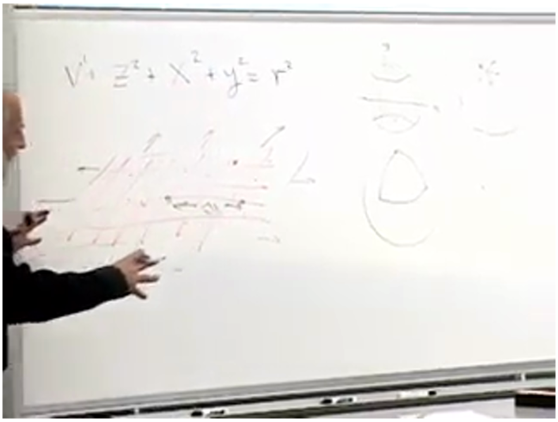


Figure 8.4: Positive, zero, and negative curvature sketched through their characteristic triangle shapes. Source frame at 00:37:30.

A perfectly flat local metric does not by itself prove that space is topologically infinite. Compact flat topologies are mathematically possible, although the lecture adopts the simplest infinite Euclidean model. [1]

8.3 Comoving labels and Hubble's law

Return now to the expanding flat sheet and draw a coordinate grid on it. In comoving coordinates, the grid expands with the large-scale matter flow. A galaxy, or more accurately a sufficiently coarse-grained cluster of galaxies, keeps an approximately fixed coordinate label even though its physical distance from another comoving object changes.

[1]Editorial clarification: This global-topology qualification is not needed for any of the subsequent local dynamics.

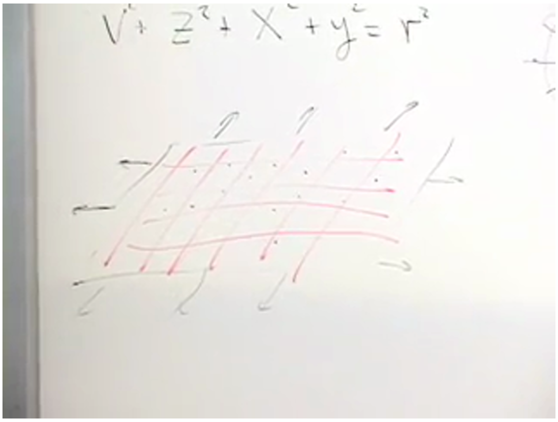


Figure 8.5: A coordinate mesh carried along by an expanding spatial sheet. Source frame at 00:22:20.

For two objects separated along the x-axis by a fixed comoving interval Δx, their physical distance is

$$D(t) = a(t)\,\Delta x. \tag{8.6}$$

Differentiating while holding Δx fixed gives

$$v(t) = \dot{a}(t)\,\Delta x = \frac{\dot{a}(t)}{a(t)} D(t). \tag{8.7}$$

Defining

$$H(t) \equiv \frac{\dot{a}(t)}{a(t)}, \tag{8.8}$$

we obtain Hubble's law,

$$v = H(t) D. \tag{8.9}$$

The word "constant" in "Hubble constant" means constant across space at a specified cosmic time. It does not imply constancy through cosmic history.

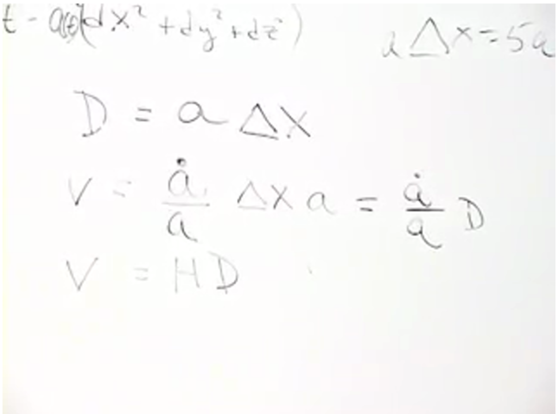


Figure 8.6: The board derivation $D = a\Delta x$, $v = (\dot{a}/a)D = HD$. Source frame at 00:26:40.

Question from the audience. If a distant galaxy is seen at an earlier time, should its inferred Hubble parameter really equal the value measured nearby today?

Response. Only when the lookback time is small compared with the age of the universe. Over modest distances, $H(t)$ changes little enough that one number is useful. At truly cosmological distances we see earlier epochs, and the time dependence of a and H must be included. That is one reason we need a dynamical equation for the scale factor.

8.4 Null rays in an expanding coordinate system

Question from the audience. Does the scale factor affect the propagation of light between distant galaxies?

Response. A light ray follows a null trajectory, so $d\tau = 0$. For a ray moving along the positive x-axis, $dy = dz = 0$, and the expanding metric gives

$$0 = dt^2 - a^2(t)dx^2, \qquad dt = a(t)dx, \qquad \frac{dx}{dt} = \frac{1}{a(t)}. \tag{8.10}$$

The coordinate speed decreases as $a(t)$ grows, but the local physical speed of light remains $c = 1$. A fixed coordinate interval represents an increasing physical distance.

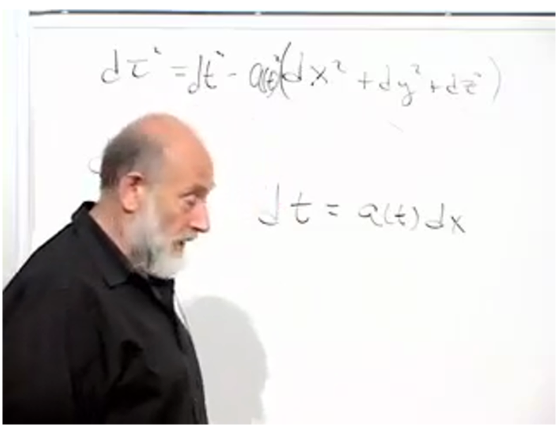


Figure 8.7: The null condition and $dt = a(t)dx$ for radial light propagation. Source frame at 00:30:54.

One can choose coordinates in which this same fact sounds like "light slows down farther away," but then lengths and clocks acquire correspondingly awkward descriptions. The expanding-space language is cleaner because it preserves the local statement that every freely falling observer measures the same speed of light.

Question from the audience. Is there evidence that local light propagation and atomic physics are the same in distant galaxies?

Response. Only dimensionless comparisons are operationally meaningful. Choose an atomic size as a unit of length and an atomic or nuclear period as a unit of time, then ask how many such periods light requires to cross the chosen length. Spectral observations support the statement that these dimensionless ratios are the same in distant regions. The evidence is indirect, but it is precisely the kind of comparison by which one tests the uniformity of local laws.

When the observable universe was extremely small, light could cross a given physical region quickly in the corresponding light-crossing unit. This does not mean that c was different. It means that the region itself was small.

8.5 A Newtonian sphere inside the universe

We now turn from kinematics to dynamics. Place ourselves at the origin and draw a sphere large enough to contain many galaxies and clusters. Select a galaxy on its boundary. In a homogeneous universe the mass distribution is spherically symmetric around us, and Newton's shell theorem gives two crucial facts:

1. all spherically distributed matter outside the chosen radius exerts zero net Newtonian force on the boundary galaxy;
2. all matter inside the sphere acts, for this radial motion, as if it were concentrated at the center.

We use a frame in which we are at rest, so velocities and accelerations in the following argument are measured relative to us.

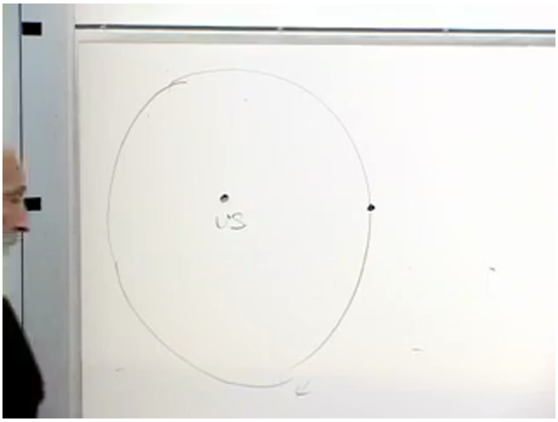


Figure 8.8: The shell-theorem construction: us at the center and a test galaxy on the boundary of a large homogeneous sphere. Source frame at 00:41:10.

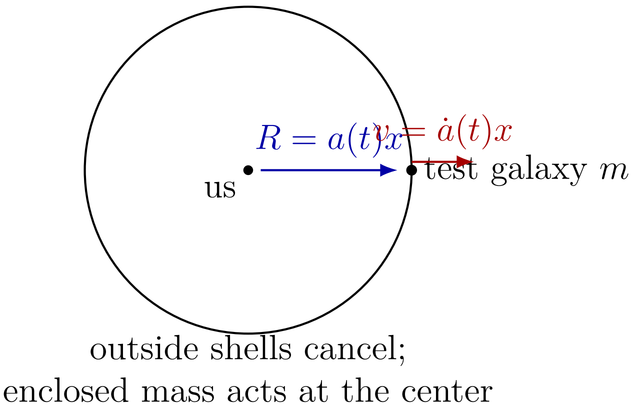


Figure 8.9: Clean reconstruction of the Newtonian construction.

If the galaxy has fixed comoving coordinate x, its physical radius and velocity are

$$R(t) = xa(t), \qquad v(t) = x\dot{a}(t) = H(t)R(t). \tag{8.11}$$

Let M be the mass enclosed by the sphere and m the mass of the test galaxy. The Newtonian energy is

$$E = \frac{1}{2}mv^2 - \frac{GMm}{R}. \tag{8.12}$$

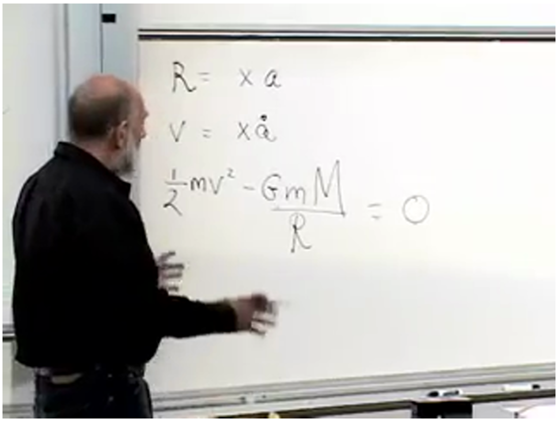


Figure 8.10: The complete board sequence $R = xa$, $v = x\dot{a}$, and the Newtonian energy equation specialized to $E = 0$. Source frame at 00:48:40.

The energy may be positive, zero, or negative. Start at the boundary between escape and recollapse, $E = 0$, for which

$$\frac{1}{2}v^2 = \frac{GM}{R}. \tag{8.13}$$

Question from the audience. What does zero total energy mean for the receding galaxy?

Response. It is exactly the escape-velocity case. Positive energy means motion above escape velocity; negative energy means motion below escape velocity and, in the matter-only Newtonian picture, eventual return. We begin at the boundary between those possibilities and return to the other signs after deriving the flat equation.

Question from the audience. Does the enclosed mass M vary as the sphere expands?

Response. No. The sphere is comoving, so the same particles remain inside it. Its physical volume grows and its density falls, but its total enclosed mass remains fixed. This distinction between M and $\rho(t)$ is essential.

For a homogeneous density $\rho(t)$,

$$M = \frac{4\pi}{3}\rho(t)R^3. \tag{8.14}$$

Substitution into (8.13) gives

$$\frac{1}{2}v^2 = \frac{G}{R}\left(\frac{4\pi}{3}\rho R^3\right). \tag{8.15}$$

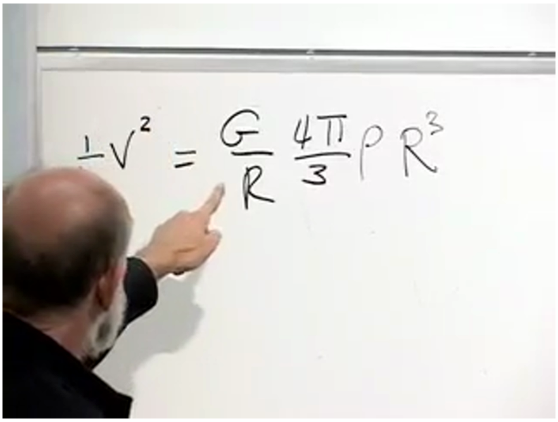


Figure 8.11: The enclosed-mass substitution in the zero-energy equation. Source frame at 00:54:16.060.

Using $v = HR$, we find

$$\frac{1}{2}H^2R^2 = \frac{4\pi G}{3}\rho R^2, \tag{8.16}$$

$$H^2 = \frac{8\pi G}{3}\rho. \tag{8.17}$$

Thus the zero-energy, spatially flat expansion law is

$$\boxed{\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho(t)}. \tag{8.18}$$

The cancellation of R^2 is not accidental. A homogeneous universe must give the same $H(t)$ no matter which comoving galaxy was chosen as the test object.

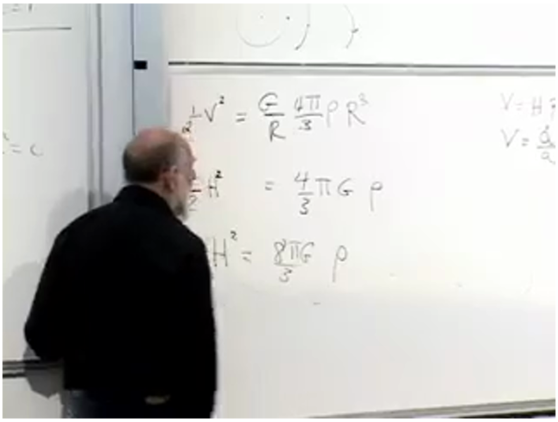


Figure 8.12: The cancellation of the radius and the resulting flat Friedmann relation on the board. Source frame at 00:57:30.

Question from the audience. Is there evidence for the zero-energy relation?

Response. Both sides can be inferred observationally: H from recession data and ρ by accounting for stars, gas, dark matter, and vacuum energy. The lecture-era comparison found the two sides consistent to about percent accuracy. The qualification matters: all forms of energy must be included, and the later vacuum-energy discussion changes the naive escape-velocity interpretation.

In units $c = 1$, mass density and energy density have the same units. The symbol ρ will therefore denote the total energy density appropriate to the component under discussion.

8.6 Ordinary matter and the two-thirds law

Equation (8.18) is not closed until we know how $\rho(t)$ changes. First fill a comoving rectangular box with slowly moving, nonrelativistic particles. Its coordinate dimensions are fixed, while its physical volume grows as

$$V(t) = a^3(t)\,\Delta x\,\Delta y\,\Delta z. \tag{8.19}$$

The particle number and rest mass in the box remain fixed, so

$$\rho_{\mathrm{m}}(t) = \frac{C_{\mathrm{m}}}{a^3(t)}. \tag{8.20}$$

Normalizing to the present epoch t_0,

$$\frac{\rho_{\mathrm{m}}(t)}{\rho_{\mathrm{m},0}} = \left(\frac{a_0}{a(t)}\right)^3. \tag{8.21}$$

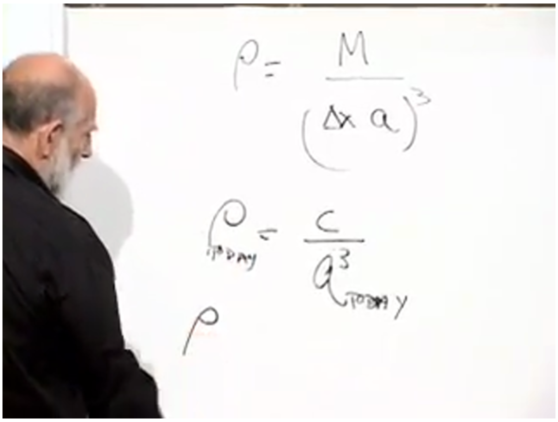


Figure 8.13: Matter density, its a^{-3} dilution, and present-day normalization. Source frame at 01:04:47.940.

Question from the audience. What is the rough present energy density of the universe, and can ordinary atoms alone provide it?

Response. A count of visible baryonic matter gives only part of the answer. Dark matter and dark energy must also be represented as equivalent mass-energy. The lecture quotes a rough scale ranging from an atom to a few dozen hydrogen-mass equivalents per cubic meter, emphasizing order of magnitude rather than a precision inventory.

Insert (8.20) into (8.18) and absorb all time-independent factors into a positive constant C:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{C}{a^3}, \qquad \dot{a}^2 = \frac{C}{a}. \tag{8.22}$$

There are two transparent ways to solve it. A power-law trial $a = Kt^p$ requires

$$p - 1 = -\frac{p}{2}, \qquad p = \frac{2}{3}. \tag{8.23}$$

Equivalently, take the expanding square-root branch and integrate:

$$a^{1/2}\frac{da}{dt} = \sqrt{C}, \qquad \int a^{1/2} da = \sqrt{C} \int dt, \tag{8.24}$$

so, after choosing the Big Bang origin of time,

$$a^{3/2} \propto t, \qquad \boxed{a(t) \propto t^{2/3}}. \tag{8.25}$$

The live derivation briefly loses a factor while matching coefficients, then returns to the integral and verifies the exponent. The notes retain the check but display only the corrected algebra.

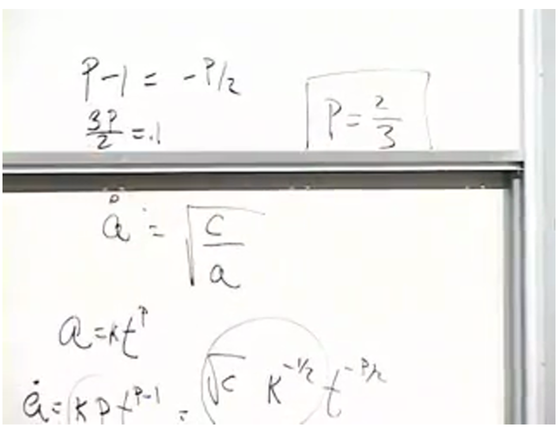


Figure 8.14: The power-law trial and the boxed exponent $p = 2/3$. Source frame at 01:13:05.

Differentiating the solution gives

$$H(t) = \frac{\dot{a}}{a} = \frac{2}{3t}. \tag{8.26}$$

The Hubble parameter was larger in the past. Consequently, when looking far enough away to see appreciable lookback time, one should not extend the local linear relation using today's H_0 unchanged.

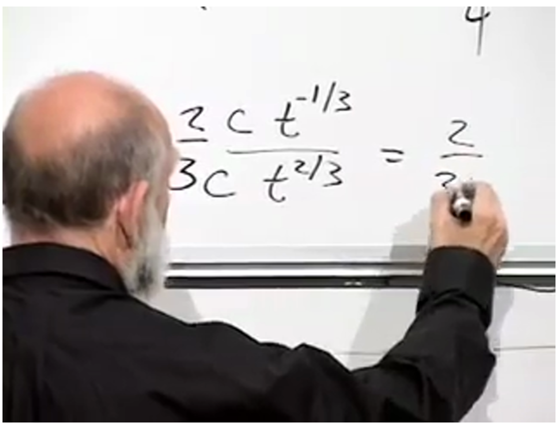


Figure 8.15: Taking $\dot{a}/a$ for $a \propto t^{2/3}$ gives $H = 2/(3t)$. Source frame at 01:20:42.

8.7 Energy sign and spatial curvature

Newtonian reasoning has carried us through the flat zero-energy case. General relativity adds a fact that cannot be obtained from the shell theorem alone: the sign of the Newtonian-style specific energy corresponds to the sign of the spatial curvature.

Total energy	Spatial curvature	Matter-only fate
positive	negative (open)	expands forever
zero	zero (flat)	approaches escape boundary
negative	positive (closed)	expands, turns, recollapses

The last column assumes no vacuum energy. In the projectile analogy, positive energy means above escape velocity, zero means exactly at escape velocity, and negative means below escape velocity. Einstein's equations identify these three cases with open, flat, and closed spatial geometry respectively.

8.8 Photons and the one-half law

The matter law $\rho \propto a^{-3}$ assumes that the energy of each particle is approximately fixed while the number density falls. Photons violate that assumption. For a photon,

$$E = h\nu = \frac{hc}{\lambda}. \tag{8.27}$$

Expansion stretches its wavelength in proportion to the scale factor,

$$\lambda \propto a, \qquad E \propto a^{-1}. \tag{8.28}$$

The photon number density still falls as a^{-3}, and each photon's energy contributes one additional factor of a^{-1}. Therefore

$$\boxed{\rho_{\mathrm{r}}(t) \propto a^{-4}(t)}. \tag{8.29}$$

The guitar-string analogy captures the physical picture. Pluck a string and then lengthen its vibrating portion: the wavelength lengthens and the tone drops. A photon mode in an expanding comoving box behaves similarly. Running the picture backward, compression shortens wavelengths and raises photon energies.

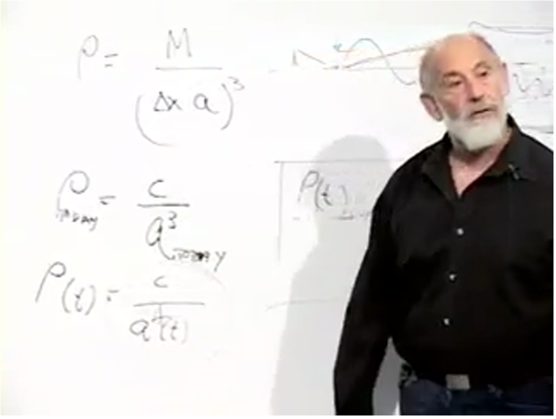


Figure 8.16: The matter-density relation extended on the board to the radiation scaling $\rho_{\mathrm{r}} \propto a^{-4}$. Source frame at 01:29:20.

Using $\rho_{\mathrm{r}} \propto a^{-4}$ in the flat Friedmann equation gives

$$\left(\frac{\dot a}{a}\right)^2 = \frac{C_{\mathrm{r}}}{a^4}, \qquad \dot a \propto a^{-1}, \tag{8.30}$$

and hence

$$\boxed{a(t) \propto t^{1/2}}. \tag{8.31}$$

This grows more slowly than $t^{2/3}$. Far enough into the past, radiation must dominate ordinary matter because a^{-4} rises faster than a^{-3} as a decreases. The lecture places the matter-radiation crossover around the epoch when the scale factor was roughly a thousand times smaller than today. Before that crossover the universe was radiation dominated; after it, for a long interval, it was matter dominated.

8.9 Vacuum energy and exponential expansion

Vacuum energy has the opposite scaling behavior. Its density is a property of empty space and does not dilute as the universe expands:

$$\rho_\Lambda = \text{constant}. \tag{8.32}$$

Ordinary matter and radiation become denser toward the past, while this term does not. It was therefore less important in the early universe and becomes more important as the other components dilute. The lecture uses the then-current estimate that vacuum energy supplies about 70% of today's cosmic energy density.

With constant ρ_Λ, equation (8.18) gives a constant expansion rate

$$H_\Lambda = \sqrt{\frac{8\pi G}{3}\rho_\Lambda}, \qquad \frac{\dot{a}}{a} = H_\Lambda. \tag{8.33}$$

Therefore

$$\boxed{a(t) = a(t_i)e^{H_\Lambda(t-t_i)}}. \tag{8.34}$$

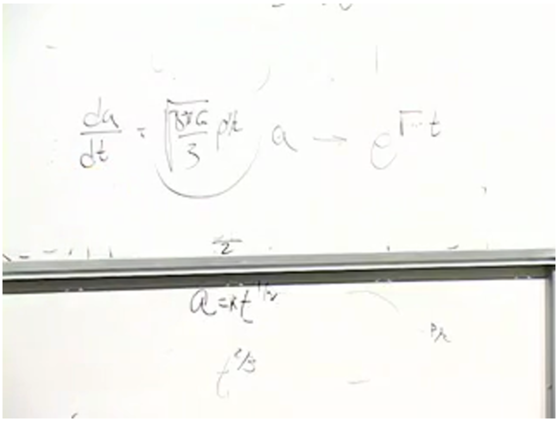


Figure 8.17: The constant-density equation leading to exponential expansion. Source frame at 01:35:48.

At the observed scale quoted in the lecture, the universe doubles in linear size over roughly ten to twelve billion years. The origin of this small but nonzero vacuum density remains unexplained; its existence and its effect on the expansion are empirical facts.

Question from the audience. Is vacuum energy the same thing as dark energy in this discussion?

Response. Yes. Here "dark energy" denotes a constant vacuum-energy density, equivalently a cosmological constant. More general time-dependent dark-energy models can be imagined, but they are not part of this lecture's calculation.

8.10 How an expanding universe redshifts light

Question from the audience. How can photons know that we chose expanding comoving coordinates and therefore lower their energy?

Response. They do not respond to our labels. They propagate through a geometry whose physical separations change. A wave train occupying a region that doubles in physical size acquires a doubled wavelength. The comoving coordinates merely describe this stretching economically; they do not cause it.

Question from the audience. Where do the background photons reaching us today come from?

Response. Looking farther away means looking earlier. Eventually we see the epoch when the cosmic material was an ionized plasma, opaque because photons scattered repeatedly from free charged particles. The boundary from which photons could last travel freely to us appears as a sphere surrounding the observer: the surface of last scattering.

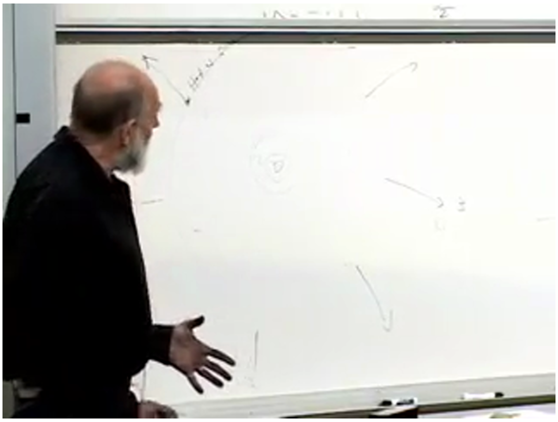


Figure 8.18: The observer-centered sketch of the surface of last scattering and increasingly distant shells. Source frame at 01:42:05.

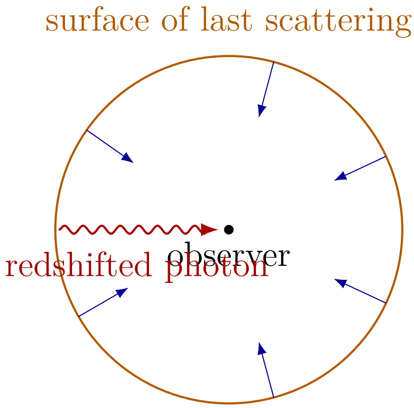


Figure 8.19: Clean reconstruction of the last-scattering picture.

The radiation was emitted with a spectrum characteristic of a hot plasma and reaches us with wavelengths roughly a thousand times longer. In the lecture's Doppler language, the distant emitting surface recedes rapidly and its radiation is strongly redshifted. In the geometric language, the wave is stretched during propagation. Both pictures encode the same relation,

$$1 + z = \frac{\lambda_0}{\lambda_{\rm em}} = \frac{a_0}{a_{\rm em}}. \tag{8.35}$$

[2]

At an earlier observing epoch, the last-scattering sphere would have been closer in the corresponding picture, and the observed background photons would have been less redshifted and therefore more energetic. As cosmic time advances, photons reaching a comoving observer from that early plasma have progressively longer wavelengths.

[2] Editorial clarification: Equation (8.35) is the standard quantitative form of the lecture's stretching and Doppler descriptions.

8.11 Where does a redshifted photon's energy go?

Question from the audience. A photon is emitted with energy $h\nu_{\mathrm{em}}$ and arrives redshifted with less energy. What happened to the difference?

Response. The Newtonian derivation began as an energy equation. Its cosmological form relates the expansion rate to the material energy density, so matter energy and expansion cannot be treated as separately conserved accounts. The lecture describes the lost photon energy as entering the kinetic bookkeeping of expansion. A piston filled with radiation makes the analogy concrete: radiation pressure pushes the piston outward, the photon wavelengths increase, and their lost energy becomes mechanical work and kinetic energy of the piston.

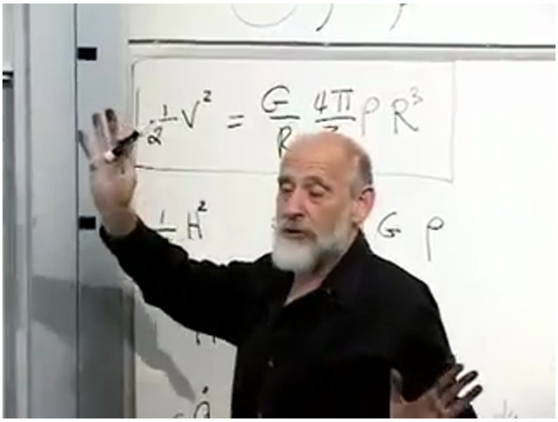


Figure 8.20: The board places the boxed Newtonian energy equation above the corresponding Friedmann equation. Source frame at 01:44:42.

Question from the audience. What couples the radiation energy to the expansion?

Response. Pressure. In a cylinder with a movable piston, photons transfer momentum to the wall. As the piston moves, the radiation does work; its wavelength grows and its energy falls. The homogeneous expanding universe has no literal outer piston, but the $p\,dV$ analogy explains why the energy density of radiation falls faster than number dilution alone.

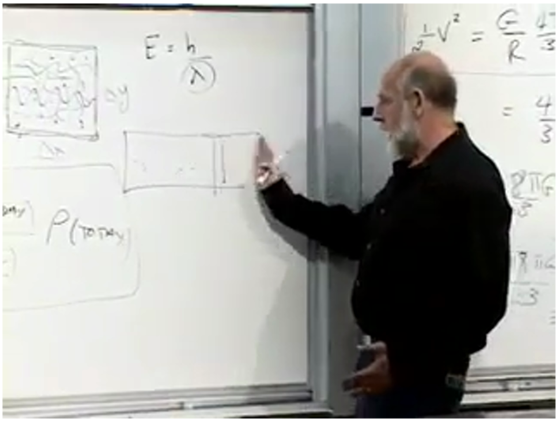


Figure 8.21: The movable photon-filled piston used to visualize pressure work. Source frame at 01:46:40.

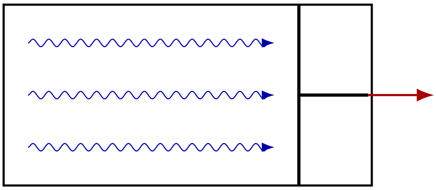


Figure 8.22: Clean reconstruction of the radiation-piston analogy.

Energy in general relativity requires care. In a generic expanding spacetime there need not be a unique conserved global energy analogous to the energy of an isolated Newtonian system. Local stress-energy conservation still holds, and for a homogeneous fluid it yields the continuity equation

$$\dot{\rho} + 3H(\rho + p) = 0, \tag{8.36}$$

which contains the pressure-work term behind the matter and radiation scaling laws.[3]

Vacuum energy makes the puzzle especially sharp. Since its density stays constant while volume grows, the total vacuum energy in a comoving region increases. Its negative pressure is part of the gravitational source and drives accelerated expansion. One should therefore not demand separate conservation of the naive "matter energy in a comoving box" without including the geometry and pressure work.

[3]Editorial clarification: Equation (8.36) is the precise local general-relativistic statement underlying the lecture's deliberately Newtonian energy-bookkeeping analogy.

8.12 Why the scale factor can be removed from the time term

Question from the audience. Why does the scale factor multiply the spatial part of the metric but not the time part?

Response. That is a choice of time coordinate. Suppose one begins in a conformal-time coordinate t with

$$d\tau^2 = A^2(t)\left(dt^2 - dx^2\right). \tag{8.37}$$

Define a new cosmic time T by

$$dT = A(t)\,dt. \tag{8.38}$$

Then

$$d\tau^2 = dT^2 - A^2\big(t(T)\big)dx^2. \tag{8.39}$$

The common factor in front of the time differential has disappeared, while the physical scale factor remains in the spatial term.

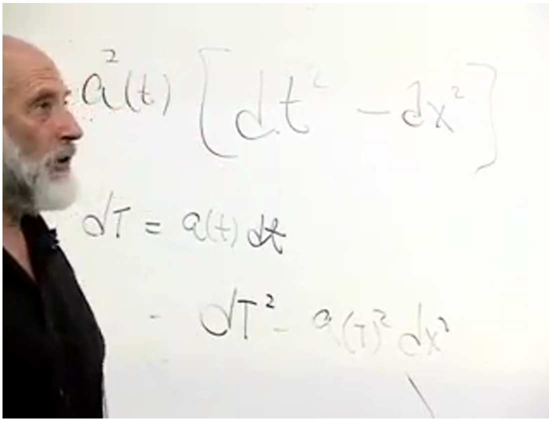


Figure 8.23: The time redefinition $dT = A(t)dt$ that converts conformal time to cosmic time. Source frame at 01:51:05.

8.13 Curvature, vacuum energy, and cosmic fate

Question from the audience. If the universe were positively curved, should we already see galaxies coming toward us?

Response. Not necessarily. A projectile launched below escape velocity first travels outward, reaches a turning point, and only later falls back. We observe the universe during an expanding phase. Without vacuum energy, a positively curved matter universe would eventually recollapse. A sufficiently large positive vacuum energy prevents that turnaround, and the observed value appears large enough to make the long-term expansion exponential regardless of a small spatial curvature.

The exponential future is not an energetic paradise.

Stars eventually burn out, matter and radiation densities fall exponentially, and distant regions pass beyond causal contact. Vacuum energy remains while ordinary structure becomes progressively colder and more dilute.

Question from the audience. Looking farther into the past, are there more photons than ordinary particles?

Response. The photon-to-baryon number ratio is already enormous, roughly 10^{10} photons per baryon in the lecture's estimate, and both numbers in a comoving volume remain approximately fixed. What changes toward the past is mainly the energy per photon. Photon energies rise as a^{-1}, while nonrelativistic rest masses do not. Radiation therefore dominates the energy density at early times even though the number ratio itself need not change.

The central lesson is that geometry alone does not determine the cosmic history. The flat Friedmann equation supplies the relation between expansion and energy density, but the material content supplies the time dependence. Nonrelativistic matter gives $a \propto t^{2/3}$, radiation gives $a \propto t^{1/2}$, and constant vacuum energy gives exponential expansion. Those three laws organize the successive eras of the universe and explain why the same scale factor controls distances, redshifts, and the changing Hubble parameter.